Financing Southeast Asia's Blue Economy

Development Assistance and Beyond

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DEVELOPMENT ASSISTANCE AND BEYOND

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Foreword

Comprising eleven countries (Brunei Darussalam, Cambodia, Indonesia, Lao People's Democratic Republic, Malaysia, Myanmar, the Philippines, Singapore, Thailand, Timor-Leste and Viet Nam), the Southeast Asia region has experienced rapid economic growth, emerging as the world's fourth-largest economic bloc in 2024. Important to this trajectory is the region's blue economy – the economic activities tied to coastal, marine and freshwater resources. Even as it presents opportunities, the region's blue economy faces mounting pressures and transitional risks. A range of stressors - including overfishing, marine pollution, habitat destruction and climate change - has undermined the resilience of Southeast Asia's blue ecosystems. At the same time, the global shift toward low-carbon economies poses transition risks, as stricter climate policies and changing market preferences may reduce demand for carbon-intensive marine activities and expose certain sectors to economic disruption. Addressing these threats and transition risks is, therefore, essential for the near and long-term outlook of Southeast Asia's blue economy.

Effective financial capital allocation is needed to navigate these opportunities and challenges, but the region faces a financing gap of nearly USD 2.1 trillion through 2030. Cognisant of this, the Association of Southeast Asian Nations (ASEAN) – which brings together all eleven Southeast Asian countries, with Timor-Leste joining in 2025 – has elevated the blue economy on the regional policy agenda, with member states pursuing a co-ordinated approach since the 2021 ASEAN Leaders' Declaration on the Blue Economy. A key milestone was the adoption of the ASEAN Blue Economy Framework in 2023. An overarching policy blueprint, the Framework provides a structured articulation of the region's blue economy priorities and elaborates on key considerations for operationalising them.

In particular, the Framework identifies "sustainable investment and finance" as an enabler of blue economy development in Southeast Asia, acknowledging that adequate and well-targeted financial resources are necessary for the implementation of the Framework's strategic priorities. Overall, there is a strong focus on the need to reduce reliance on development assistance and instead shift towards a longer-term approach to blue economy finance and investment. This report – a landscape study of finance and investment for the blue economy in Southeast Asia – aims to support ASEAN and its member states in this endeavour, helping illustrate opportunities and challenges in diversifying away from development assistance as a source of blue economy finance.

This report starts with a review of key trends in the blue economy in Southeast Asia (Chapter 1). It then delves into the current landscape of official development assistance (Chapter 2) – the rationale being that unpacking the precise nature of finance diversification requires understanding what Southeast Asia is diversifying from. Next, it analyses the other sources of blue economy finance and investment to highlight not only their use cases, but also the barriers that the region faces in accessing them (Chapter 3). It then shifts to the solutions to these barriers – innovative mechanisms and instruments that may unlock other sources of capital – detailing case studies and lessons learned (Chapter 4). It concludes with a synthesis of findings, providing recommendations for the national and regional level, as well as for development partners (Chapter 5).

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Shashwat Koirala (Economist) led the co-ordination and drafting of the report, with contributions from Gabriele Crisfoaro (Junior Policy Analyst). The work was supervised by Jens Sedemund (Head of the Climate, Biodiversity and Sustainable Ocean Unit) with strategic guidance from Eva Beuselinck (Head of the Policies and Networks Division) – both within the OECD's Development Co-operation Directorate. Strategic steer in stakeholder engagement was provided by colleagues from the OECD's Global Relations and Co-operation Directorate: Alexander Bohmer (Head of the Southeast Asia Division), Max Bulakovskiy (Head of the Project Implementation Unit), Hans Koger (Junior Policy Analyst) and Marta Bertanzetti (Junior Policy Analyst). Harsh Desai (Acting Head of the Statistical Collections and Dissemination Unit, Development Co-operation Directorate) provided guidance on generating forecasts of official development assistance for the ocean economy. Meral Gedik provided editorial support and prepared the report for publication.

The report was informed by discussions and insights gathered through engagement with stakeholders involved in developing and implementing the ASEAN Blue Economy Framework and its subsequent implementation plan. Two milestones, in particular, offered opportunities for engagement and information-gathering: the Intersessional Meeting of the ASEAN Co-ordinating Task Force on Blue Economy (August 2025) and the ASEAN Blue Economy Forum (September 2025). The author is grateful for the partnership of the ASEAN Secretariat – Kanchana Wanichkorn, Arief Rizky Bakhtiar, Soon Hong Kwan and Latifahaida Abdul Latif – and the Economic Research Institute for ASEAN and East Asia – Intan M. Ramli – in facilitating engagement with these fora, as well as for their substantive input during the development of the report. Thanks are also due to Michelle Voyer of the Australian National Centre for Ocean Resources & Security and Kelly Hoareau of the Blue Economy Cooperative Research Centre and the blue economy focal points of ASEAN member states for their inputs and insights.

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Abbreviations and acronyms

ADB	Asian Development Bank
ACMF	ASEAN Capital Markets Forum
ASEAN	Association of Southeast Asian Nations
CCRIF SPC	Caribbean Catastrophe Risk Insurance Facility Segregated Portfolio Company
CO_2	Carbon dioxide
COAST	Caribbean Oceans and Aquaculture Sustainability Facility
DAC	Development Assistance Committee
DFC	United States International Development Finance Corporation
DFNS	Debt-for-nature swap
EEZ	Exclusive economic zone
ESG	Environmental, social and governance
EPI	Environmental Performance Index
FDI	Foreign direct investment
FTE	Full-time equivalent
GDP	Gross domestic product
GSSS	Green, social, sustainability and sustainability-linked
GVA	Gross value added
IBRD	International Bank for Reconstruction and Development
IDA	International Development Association
IDB	Inter-American Development Bank
IFC	International Finance Corporation
IUU	Illegal, unreported and unregulated (fishing)
Lao PDR	Lao People's Democratic Republic
LDC	Least developed country
LIC	Low-income country
LMIC	Lower middle-income country
MDB	Multilateral development bank
MPA	Marine protected area
MSP	Marine spatial planning
NDC	Nationally Determined Contribution
NGO	Non-governmental organisation
OCR	Ordinary Capital Resources
ODA	Official development assistance
ODF	Official development finance
OECD	Organisation for Economic Co-operation and Development
OOF	Other official flows
PEFA	Public Expenditure and Financial Accountability
PES	Payment for ecosystem services
SDG	Sustainable Development Goal
SLB	Sustainability-linked bonds
TCFD	Task Force on Climate-Related Financial Disclosure
TFCCA	Tropical Forest and Coral Reef Conservation Act
THB	Thai baht
UMIC	Upper middle-income country
UNEP FI	United Nations Environment Programme Finance Initiative
USD	United States dollar
VAT	Value-added tax

Executive summary

Southeast Asia's blue economy is both a major growth engine and a source of mounting risk. Together with Oceania, Southeast Asia's share of the global ocean economy has grown from 7.4% in 1995 to 10.7% in 2019. Yet the region faces structural productivity gaps, heavy reliance on carbon-intensive activities and growing environmental pressures that erode marine and coastal natural capital. Without a shift toward more sustainable, higher-value blue activities and better protection of critical ecosystems, the region risks stranded assets, lost livelihoods and foregone opportunities in emerging sectors.

Development assistance currently plays a crucial, if relatively small, role in financing the blue economy in Southeast Asia. Ocean-related official development assistance (ODA) is well-placed to support essential public goods such as data, capacity building, governance reforms and early-stage project preparation that are difficult to finance through commercial capital, helping countries begin to address overfishing, pollution, ecosystem degradation and climate risks. However, ODA flows to the blue economy in the region are constrained and exhibit problematic patterns in scale, allocation and delivery. Ocean-related ODA to ASEAN averaged just USD 680 million annually in 2022-2023, and early projections suggest this could decline further as donor budgets tighten. Slow disbursement, sectoral concentration and possible mismatch with needs all weaken ODA's ability to fund ocean priorities while unlocking resources from broader sources.

On the latter, Southeast Asia faces uneven availability of public and private capital to support blue economy priorities. The region confronts a blue economy finance gap of nearly USD 2.1 trillion through 2030, yet macroeconomic headwinds, fiscal space constraints, relatively low tax to gross domestic product ratios and weaknesses in public financial management reduce governments' capacity to invest in coastal resilience, ecosystem restoration and sustainable ocean industries at the scale required. On the private side, constraints such as concentration of international credit and equity in the region's advanced economies and shallow domestic capital markets in some countries, as well as limited financial inclusion impede adequate and effective blue economy finance and investment. At the same time, limited data, weak use of integrated ocean policies and underdeveloped financial policy levers further limit the flow of the available capital into blue economy priorities.

Blending different sources of capital can conceivably ease capital constraints, but has not been a panacea for the region. In fact, even at its peak, the total private finance mobilised by ODA for ocean activities and land-based ocean-relevant activities in Southeast Asia totalled only USD 128.4 million, and much of this was concentrated in a few countries and blue economy sectors.

Meanwhile, innovative instruments offer opportunities to navigate constraints in capital availability and flow to Southeast Asia's blue economy, although their uptake across the region remains uneven. Blue use-of-proceeds bonds are the most established mechanism in the region, adapting conventional debt structures to channel capital towards ocean-related projects. Debt-for-nature swaps can simultaneously ease fiscal pressures and generate funding for marine conservation, yet their use in Southeast Asia remains marginal. Blue carbon mechanisms present opportunities to monetise ecosystem services and generate revenue streams for coastal protection, but adoption in the region remains patchy. Parametric insurance products can reduce investment risks through predefined, trigger-based payouts; however, Southeast Asia lacks precedents despite successful implementation in blue economy contexts elsewhere.

Moreover, the viability of innovative blue finance instruments is not guaranteed, underscoring the continued importance of development support even as these mechanisms seek to reduce dependence on it. The same economic fundamentals that constrain conventional capital flows – including macroeconomic stability, project-level returns and creditworthiness – also shape the feasibility of innovative instruments. In sub-investment-grade countries, blue bonds and debt-for-nature swaps frequently require credit enhancements from development finance institutions to secure commercially viable terms. Parametric insurance products often depend on blended finance arrangements to offset otherwise prohibitive premiums in highly climate-vulnerable settings. Blue carbon projects may also struggle to achieve financial viability from carbon revenues alone, necessitating complementary income streams to support long-run sustainability. Beyond financial constraints, limited implementation capacity – including a lack of bankable project pipelines, inadequate governance frameworks (e.g. marine spatial planning) or weak data systems – can create additional bottlenecks. Critically, additionality remains contested; instruments may simply refinance existing commitments or redirect obligated payments rather than genuinely mobilising new resources or improving development and environmental outcomes.

Bearing these factors in mind, tackling the blue finance gap in Southeast Asia demands a co-ordinated response across national governments, regional institutions and development partners:

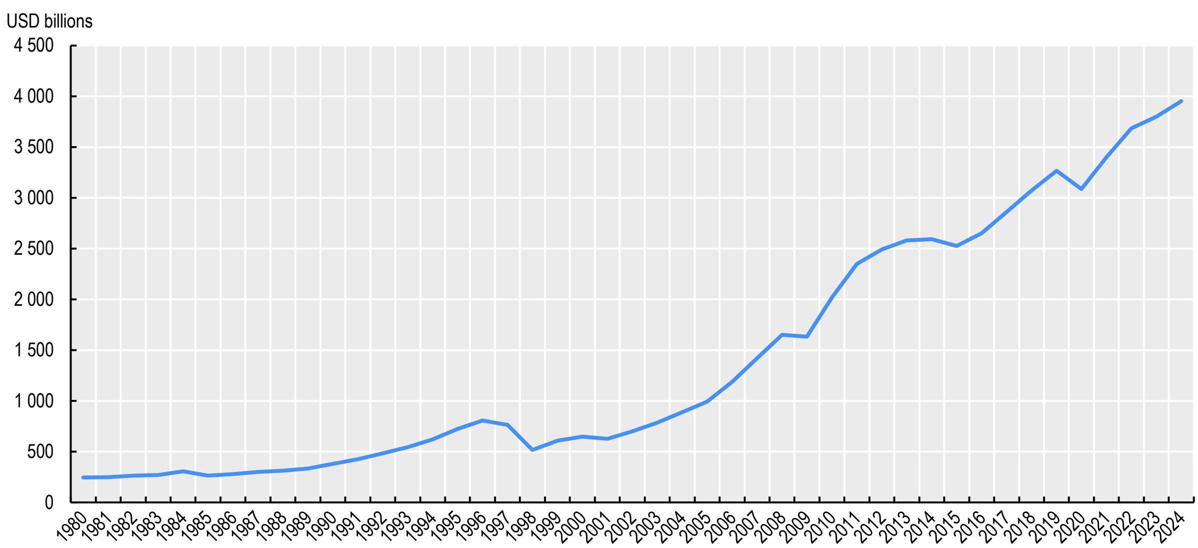
- **National level:** Strengthen policy and fiscal foundations for blue finance by developing integrated blue economy strategies, leveraging existing tools and systematically integrating blue priorities into aid co-ordination.
- **Regional level:** Enhance regional co-ordination under ASEAN by reinforcing the institutional architecture for blue finance, mainstreaming blue economy objectives into regional integration efforts and exploring the case for a regional blue finance facility.
- **Development partners:** Maximise the value of development support by directing funding to activities systematically undervalued by markets, using blended and other innovative instruments only where they demonstrably deliver additionality and pursuing co-benefits between blue economy outcomes and wider development priorities.

1 Blue economy trends and pressures in Southeast Asian countries

Southeast Asia has emerged as the world's fourth-largest economic bloc, with its blue economy – economic activities tied to coastal, marine and freshwater resources – playing an important role in this trajectory. Yet, mounting pressures threaten the long-term performance of these sectors. This chapter analyses the key trends and challenges facing Southeast Asia's blue economy, examining both the environmental and economic forces at play. It then explores recent regional policy developments, particularly the Association of Southeast Asian Nations (ASEAN) Blue Economy Framework adopted in 2023, and assesses how these policy commitments create the impetus for mobilising long-term finance and investment.

Over the past 25 years, Southeast Asia's gross domestic product (GDP) surged from under USD 650 billion in 2000 to nearly USD 4 trillion in 2024 (see Figure 1.1). Differences in growth between sectors have reshaped the region's economic structure, with services accounting for 53.9% of GDP in 2023, while agriculture and industry declined to 30.3% and 15.8%, respectively (ASEAN, 2024[1]). Economic activities in Southeast Asian countries have increasingly moved up the value chain, driven by global shifts like decarbonisation and digitalisation. The region is poised to become one of the fastest-growing data centre markets and has developed strengths in advanced sectors such as semiconductors, electric vehicles, e-commerce and fintech (OECD, 2023[2]).

Figure 1.1. Southeast Asia's GDP over time



Source: IMF (2025[3]), *World Economic Outlook*, https://www.imf.org/external/datamapper/PPPSH@WEO/SEQ.

Southeast Asia's unique geographical features make the blue economy (see Box 1.1) a key part of its economic trajectory. Mostly consisting of coastal nations, the region has some of the longest coastlines in the world: Indonesia and the Philippines, for instance, have the third and sixth longest in the world (World Population Review, 2025[4]). The exclusive economic zones of Southeast Asian countries constitute an important share of their territories, often exceeding the size of their terrestrial area – for instance, in the case of Malaysia (Mohamed Rashidi et al., 2021[5]).

The region is also a global hub for marine, coastal and freshwater biodiversity. It contains almost 34% of the world's coral reefs, covering nearly 100 000 square kilometres, which are home to more than 600 of the 800 known reef-building coral species (Burke and Selig, 2002[6]). Its biological richness extends to vital coastal ecosystems, with the area also hosting 51 of the world's 70 mangrove species and 23 of the 50 known seagrass species[1] (Burke and Selig, 2002[6]). The region – and in particular the Coral Triangle[2] – harbours some of the world's highest marine fish richness,[3] hosting over 4 000 marine fish species (Limmon et al., 2020[7]). Biodiversity in Southeast Asia's freshwater ecosystems is also exceptionally high. For instance, the region accounts for more than 30% of all known freshwater fish species globally (Patricio et al., 2023[8]).

These endowments promote economic welfare and livelihoods while providing ecosystem services indispensable for human well-being. The paragraphs below detail key economic, environmental and policy trends in Southeast Asia's blue economy.

Box 1.1. What is the blue economy?

There is no universally agreed definition of the blue economy. The term is often used interchangeably with others such as "maritime economy", "(sustainable) ocean economy" or "sustainable blue economy", though their meanings are not always consistent across contexts.

OECD (2016[9]) puts forth a definition and measurement framework for the ocean economy, conceptualising it as the sum of two components: i) market-based industries that use, operate on or benefit from proximity to marine and coastal resources and ii) non-market ecosystem services provided by the ocean. This framework does not explicitly include freshwater resources, which also contribute to human well-being in both measurable and non-measurable ways and are often considered part of the blue economy, especially when they are ecologically or economically linked to marine environments (Blue Water Intelligence, 2025[10]).

Another area of debate is whether the blue economy is inherently "sustainable", particularly given the environmental pressures exerted by nature-dependent economic activities on the ecosystems they rely upon. This is complicated by the fact that the concept of sustainability itself is contested. “Weak” sustainability assumes natural capital to be replaceable economic capital, while “strong” sustainability treats them as complementary but not interchangeable (le Gouvello and Simard, 2024[11]). This distinction fundamentally shapes which activities can be included under the umbrella of a blue economy.

ASEAN’s definition of the blue economy encompasses coastal, marine and freshwater resources, and emphasises the importance of leveraging these resources in a sustainable and inclusive manner (ASEAN, 2023[12]).

1.1. The blue economy is prominent in Southeast Asia, but it faces structural and transition challenges

Existing estimates suggest that ocean and inland water resources contribute significantly to Southeast Asian economies – reaching up to 30% of GDP in certain Southeast Asian countries[4] (ERIA, 2023[13]). However, a comparable and complete picture of the size and composition in the region remains elusive due to data gaps. The OECD's work on ocean economy measurement provides a glimpse into the economic contribution of different ocean economic activities (see Table 1.1) – which are a part of the blue economy.

Table 1.1. List of ocean economic activities, both established and emerging

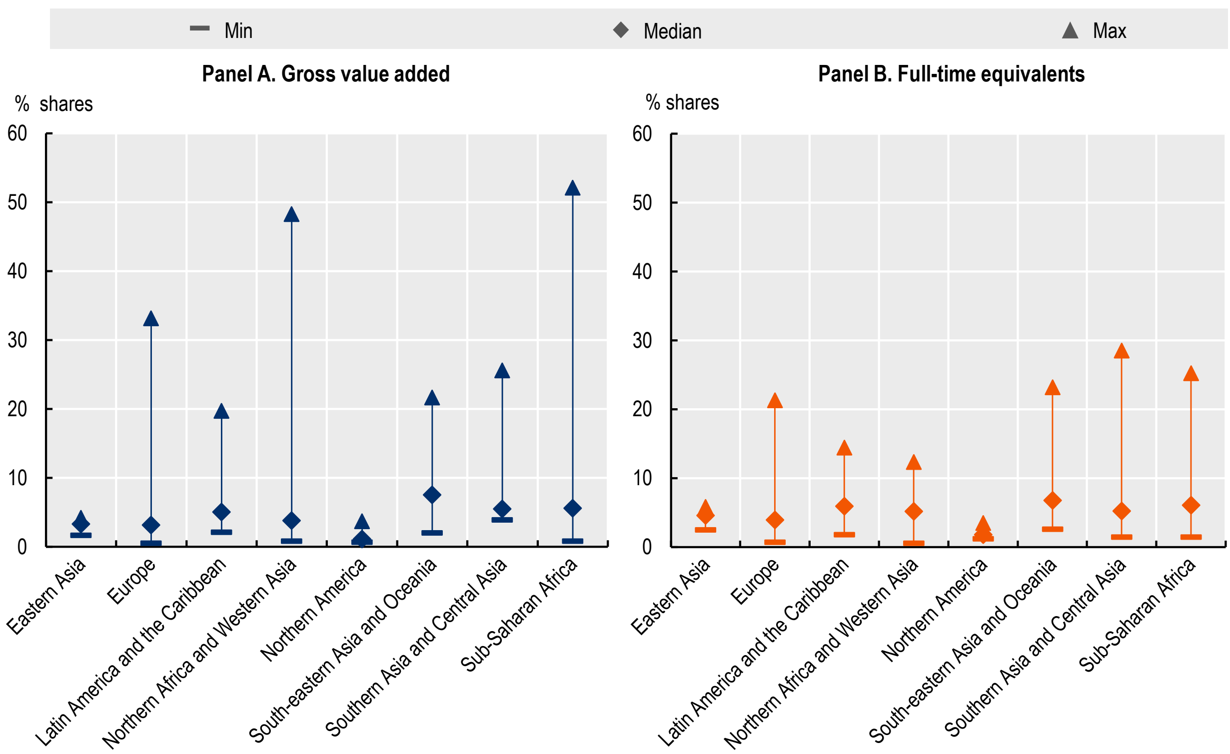
ESTABLISHED	EMERGING
Capture fisheries and seafood processing	Marine aquaculture
Shipping and ports	Deep-water and ultra-deep-water oil and gas
Shipbuilding and repair	Offshore wind energy
Offshore oil and gas (shallow water)	Ocean renewable energy
Marine manufacturing and construction	Marine and seabed mining
Maritime and coastal tourism	Maritime safety and surveillance
Marine business services	Marine biotechnology
Marine research and development and education	High-tech marine products and services
Dredging	

Source: OECD (2016[9]), *The Ocean Economy in 2030*, https://doi.org/10.1787/9789264251724-en.

The data – available only for the wider Southeast Asia and Oceania region, rather than Southeast Asia specifically – show that the region has one of the highest relative contributions of the ocean economy globally. As shown in Figure 1.2, ocean economic activities in Southeast Asia and Oceania account for a median of 7.5% of total Gross Value Added (GVA) and 6.8% of full time equivalent (FTE) employment, demonstrating that the ocean economy plays a major role in both output and jobs. Both figures are well above those seen in most other regions. For comparison, median GVA shares are around 3.3% in Eastern Asia, 3.1% in Europe and just 1.1% in Northern America.

There is, however, substantial heterogeneity within the region. The share of the ocean economy ranges from 2.0% to 21.7% in GVA terms and 2.6% to 23.2% in FTE terms. Compared to other world regions, this variation is relatively high (see Figure 1.2).

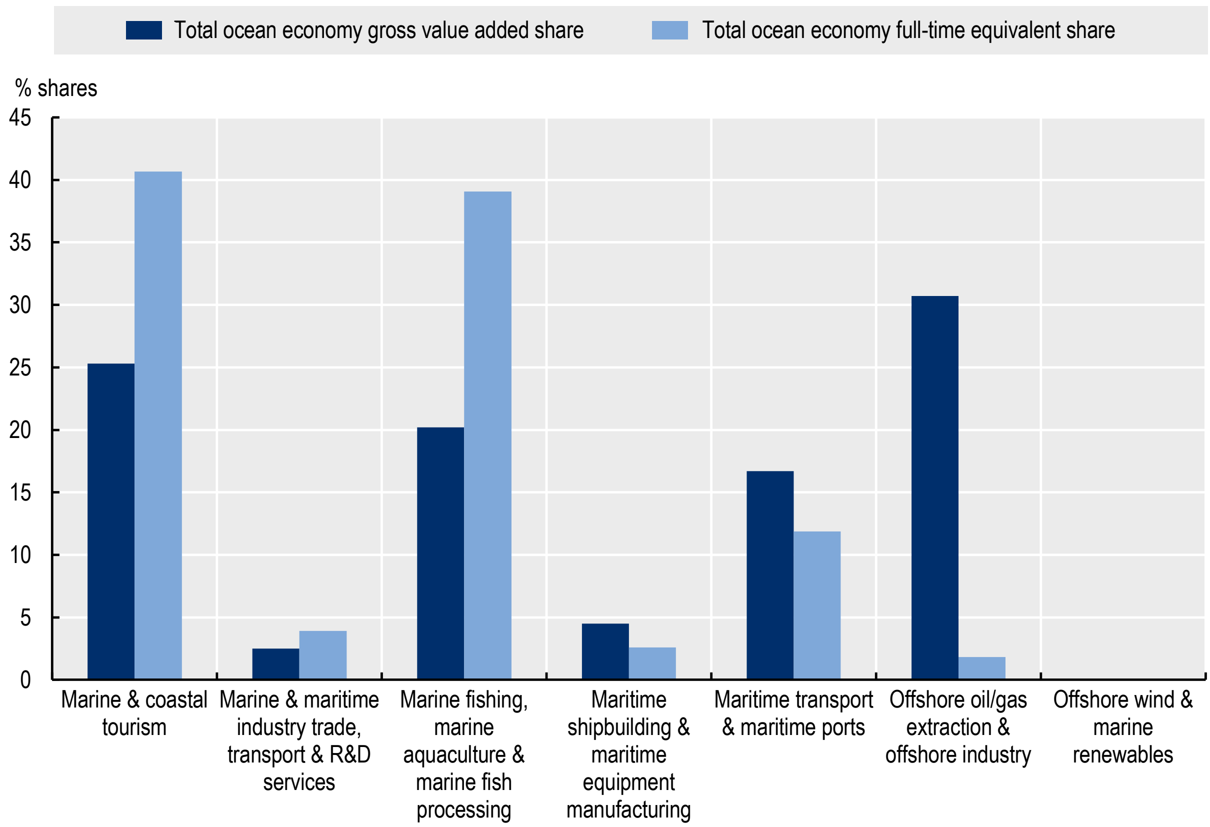
Figure 1.2. Contribution of ocean economy to the overall economy



Source: OECD (2025[14]), *The Ocean Economy to 2050*, https://doi.org/10.1787/a9096fb1-en.

Over the past 25 years, Southeast Asia and Oceania's ocean economy has evolved into a diverse mix of sectors, each contributing in different ways (see Figure 1.3). Some sectors are central to employment, notably marine fishing and aquaculture[5] and coastal tourism, which together accounted for nearly 80% of FTE jobs in 2020. Others, like offshore oil and gas extraction and offshore industry, are dominant in terms of output, contributing over 30% of gross value added (GVA) while employing less than 2% of the workforce (OECD, 2025[14]). A few sectors, such as maritime transport and ports, play a dual role, contributing meaningfully to both employment and output.

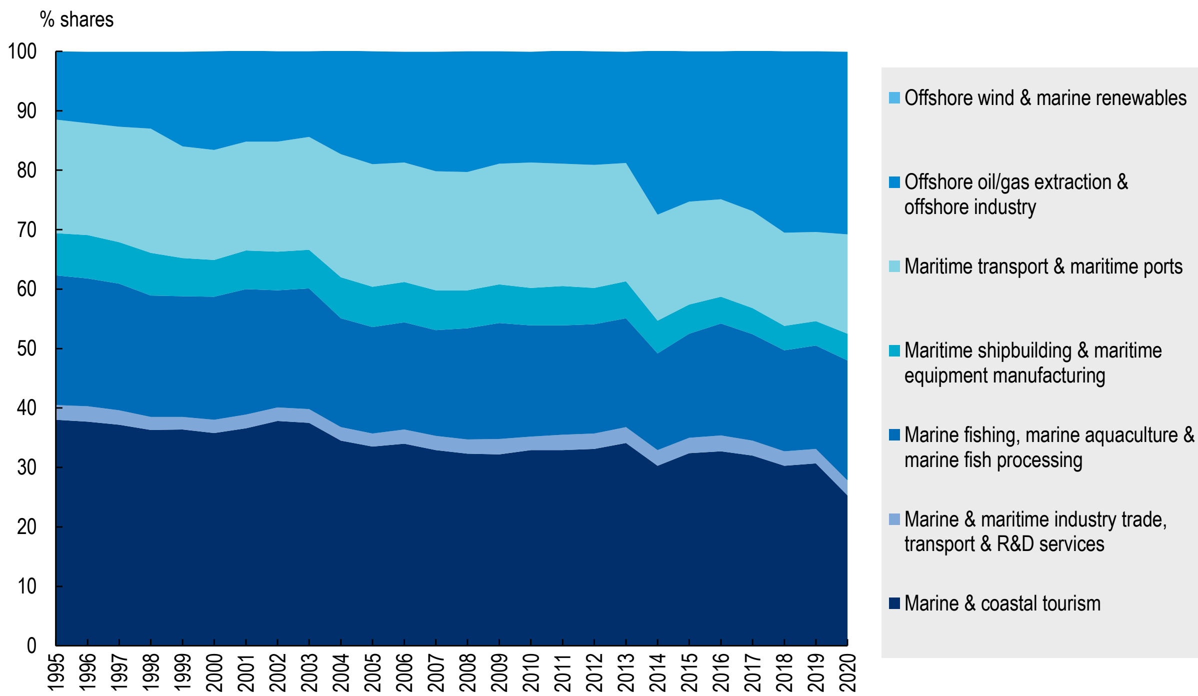
Figure 1.3. Sectoral composition of Southeast Asia and Oceania's ocean economy in 2020



Source: OECD (2025[14]), *The Ocean Economy to 2050*, https://doi.org/10.1787/a9096fb1-en.

The rapid growth of offshore oil and gas and associated industries is striking and has overtaken all other sectors in terms of GVA since the mid-1990s (see Figure 1.4). While this growth has boosted economic output, it also introduces transition risks. As global energy systems shift toward decarbonisation, the region faces potential exposure to stranded assets and volatility in fossil fuel markets. At the same time, the negligible presence of offshore wind and marine renewables (OECD, 2025[14]) highlights a missed opportunity to diversify the ocean economy and align with climate goals.

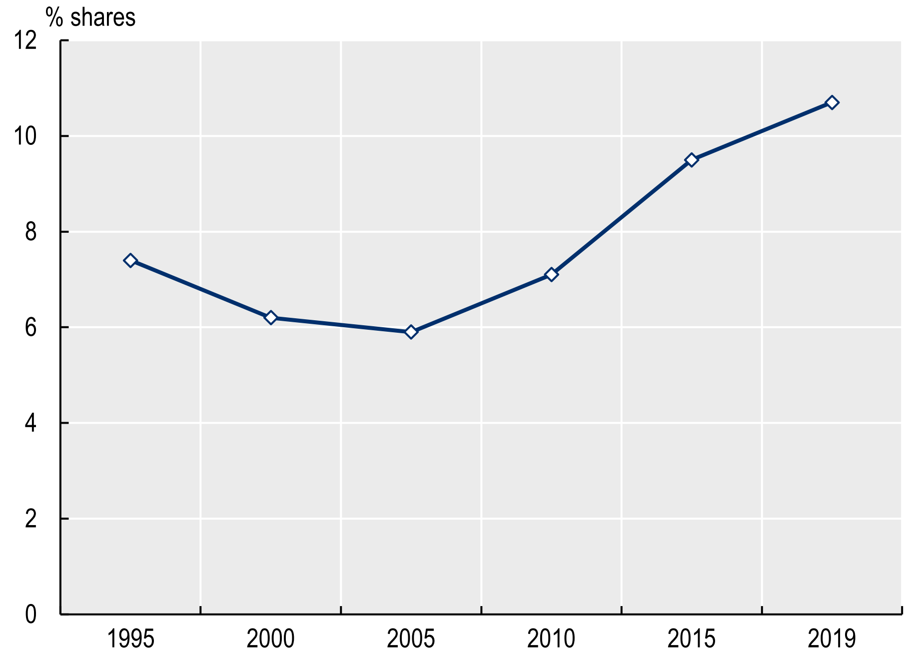
Figure 1.4. Composition of Southeast Asia and Oceania's ocean economy, in terms of gross value added, over time



Source: OECD (2025[14]), *The Ocean Economy to 2050*, https://doi.org/10.1787/a9096fb1-en.

The contribution of Southeast Asia and Oceania to the global ocean economy has also grown noticeably. The region's share of global ocean economy GVA grew from 7.4% in 1995 to 10.7% in 2019 (see Figure 1.5) while its share of global ocean economy employment grew from 17.8% to 18.4% over the same period. The OECD projects that, should historical patterns be repeated into the future, the global ocean economy may more than double in value, from approximately USD 2.6 trillion in 2020 to nearly USD 5.1 trillion by 2050 (OECD, 2025[14]). This potential surge, combined with the emergence of Southeast Asia as a major player in the global ocean economy, offers the region major opportunities for economic development in the medium and long term.

Figure 1.5. Southeast Asia and Oceania's share of the global ocean economy, in terms gross value added



Source: OECD (2025[14]), *The Ocean Economy to 2050*, https://doi.org/10.1787/a9096fb1-en.

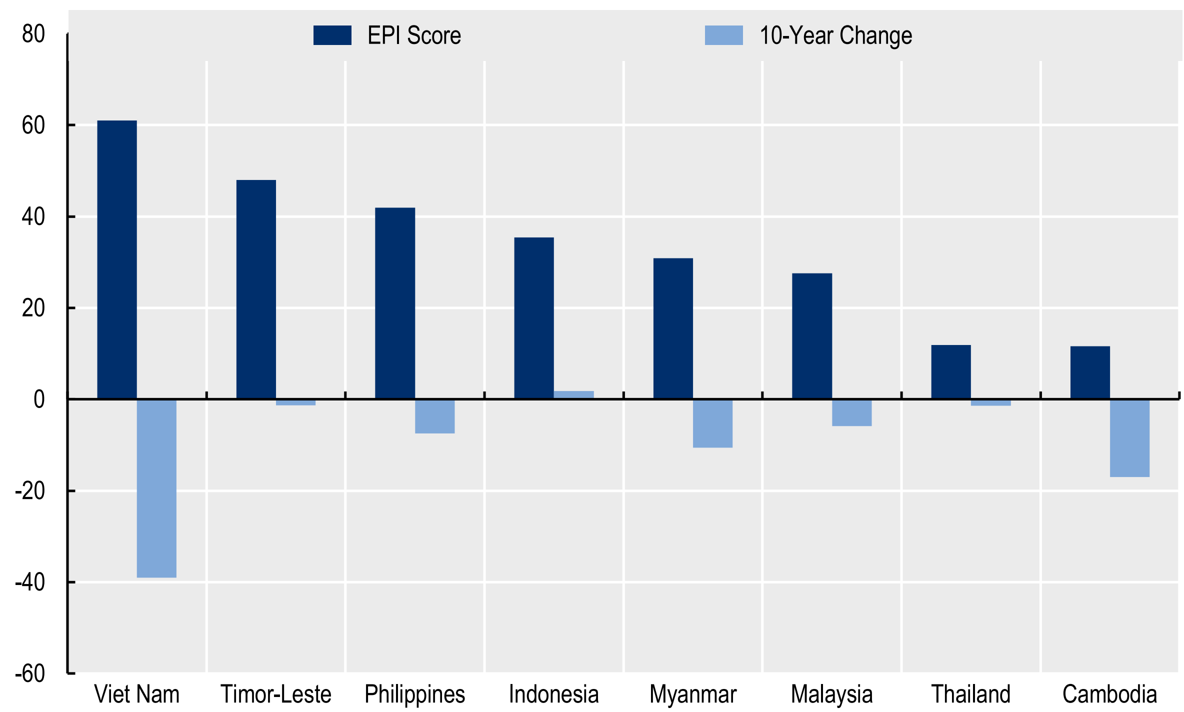
But tackling structural challenges is essential to leveraging the blue economy for sustainable and resilient development. While Southeast Asia and Oceania employ approximately 18% of the global ocean economy workers, they only generate 10.7% of the global ocean economy production (in GVA terms) (OECD, 2025[14]). Comparatively, Europe accounts for 6.1% of the global ocean economy FTE employment but generates 20.1% of global GVA – reflecting higher levels of productivity per worker (OECD, 2025[14]). Europe's higher productivity reflects its more technology-driven blue economy, with greater emphasis on high-value activities such as marine biotechnology, advanced maritime services and digitalised port operations (BlueInvest, 2023[15]). This suggests yet-untapped opportunity in Southeast Asia to make productivity gains in the blue economy – whether through diversifying into emerging sectors or adding value along the production chain of established sectors.

1.2. Environmental degradation threatens blue ecosystems and the economic activities they sustain

Beyond the transition risks associated with low-carbon shift, Southeast Asia is vulnerable to an array of environmental pressures. These pressures – ranging from overfishing and habitat destruction to pollution and climate-related hazards – are partly linked to the expansion of the blue economy. For sectors such as wild-capture fisheries and dive-based marine tourism, which can collapse when ecosystems degrade, these environmental pressures pose a direct threat to economic viability. They also indirectly affect sectors such as ports and shipping, which depend on healthy ecosystems for services like coastal protection. More generally, environmental stressors can undermine macroeconomic performance, such as through weaker aggregate productivity or deteriorating sovereign creditworthiness (Zenios, 2022[16]), as recent empirical work on natural capital depletion (Managi et al., 2024[17]) or climate-related financial risks suggests (Carè, 2023[18]).

Unsustainable fishing practices[6] represent one of the most pressing threats to the region's blue economy. In 2022, most Southeast Asian countries showed poor fish stock health according to Yale's Environmental Performance Index (EPI) (see Figure 1.6). Scores for nations like Thailand, Cambodia, Malaysia and Myanmar are especially low, indicating that much of their catch comes from overexploited or collapsed stocks. Only Viet Nam had a moderate score, though it declined sharply over the past decade, signalling worsening overfishing pressures (Yale Center for Environmental Law & Policy, 2024[19]). The prevalence of illegal, unreported and unregulated (IUU) fishing exacerbates pressures on fish stock health and causes economic losses. The Philippines incurs annual losses of approximately USD 1 billion attributable to IUU fishing, while Cambodia's fisheries sector has been subject to European Union trade restrictions since 2014 due to inadequate controls over illegal fishing activities (ERIA, 2023[13]).

Figure 1.6. Fish stock status, according to the Yale Environmental Performance Index

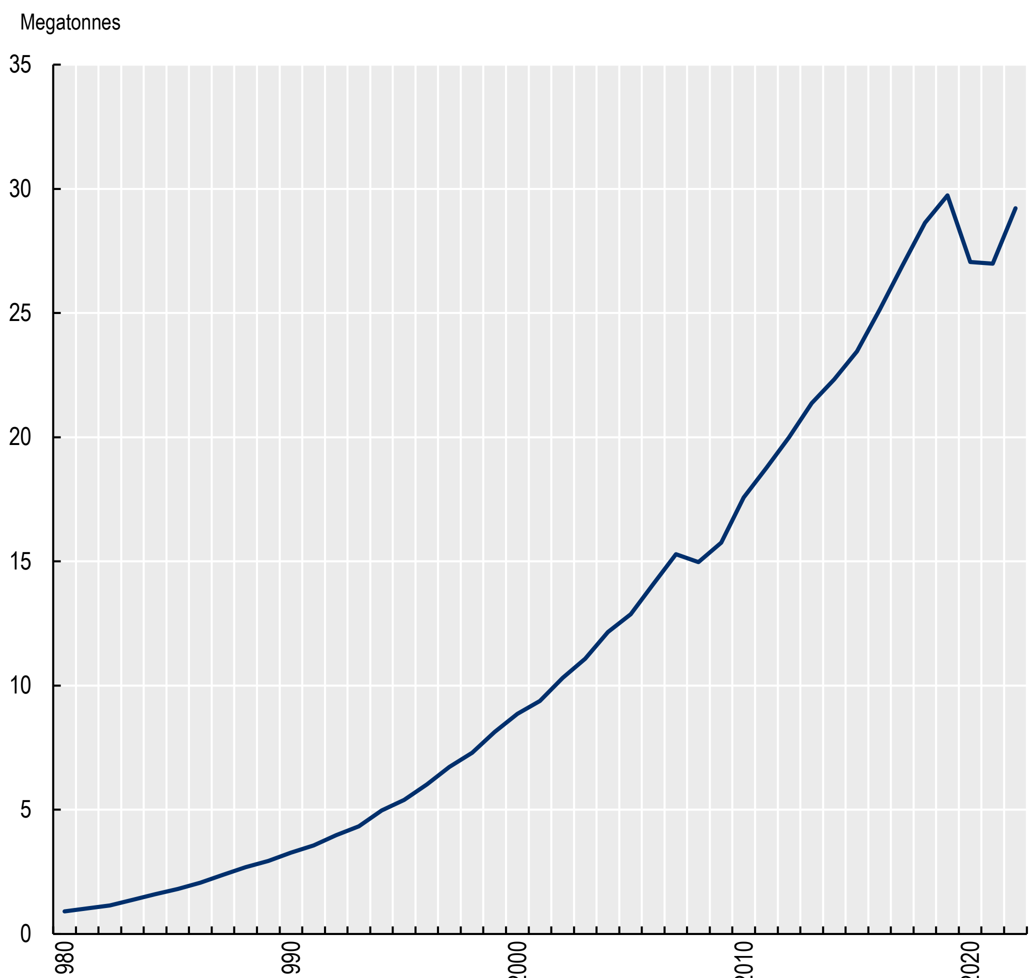


Note: Fish stock status is the percentage of a country's total catch that comes from overexploited or collapsed stocks, considering all fish stocks within a country's exclusive economic zone. A score of 100 indicates that none of a country's fish catch come from stocks that are overexploited or collapsed, and a score of 0 indicates worst performance. Data not available for Singapore, Brunei Darussalam and Lao People's Democratic Republic.
Source: Yale Center for Environmental Law & Policy (2024[19]), *2024 Environmental Performance Index*, https://epi.yale.edu/epi-results/2022/component/fss.

The degradation of critical coastal habitats represents a fundamental threat to the ecological foundation of the blue economy. Mangrove ecosystems, which provide essential services such as coastal protection, carbon sequestration, nursery habitats for commercially important species (e.g. Asian seabass, mullets), and support for aquaculture productivity, have undergone severe decline across the region. Between 1989 and 2017, Cambodia lost 42% of its mangrove coverage, primarily due to conversion for salt production, charcoal manufacture and aquaculture operations. Indonesia, despite possessing 3.31 million hectares of mangroves – representing 20% of the global total – faces annual losses ranging from 6 200 to 52 000 hectares (ERIA, 2023[13]). Nearly half of these removals are attributed to aquaculture development. Coral reef systems face similarly severe pressures, with approximately one-third of Indonesia's reefs classified in poor condition. Projections indicate that over 80% of Indonesian reefs are likely to experience coral bleaching in five out of every ten years during the 2030s, driven by destructive fishing methods, pollution from agricultural and urban sources and climate-related thermal stress (ERIA, 2023[13]).

Pollution – and in particular, marine plastic pollution – has emerged as both an environmental crisis and an economic liability for Southeast Asian nations. OECD projections show that plastic use has skyrocketed in the region (see Figure 1.7), but waste management infrastructure has not kept pace (OECD, 2025[20]). As a result, the OECD estimates that ASEAN could account for nearly 35% of global plastic leakage by 2050 (OECD, 2025[20]), with coastal and riverine areas bearing the bulk of the burden. The economic repercussions of plastic pollution are substantial: direct costs across the Asia-Pacific region exceed USD 10.8 billion annually, with Indonesia alone accounting for over USD 450 million in documented impacts – calculations that do not incorporate remediation expenses or indirect ecosystem damage (ERIA, 2023[13]).

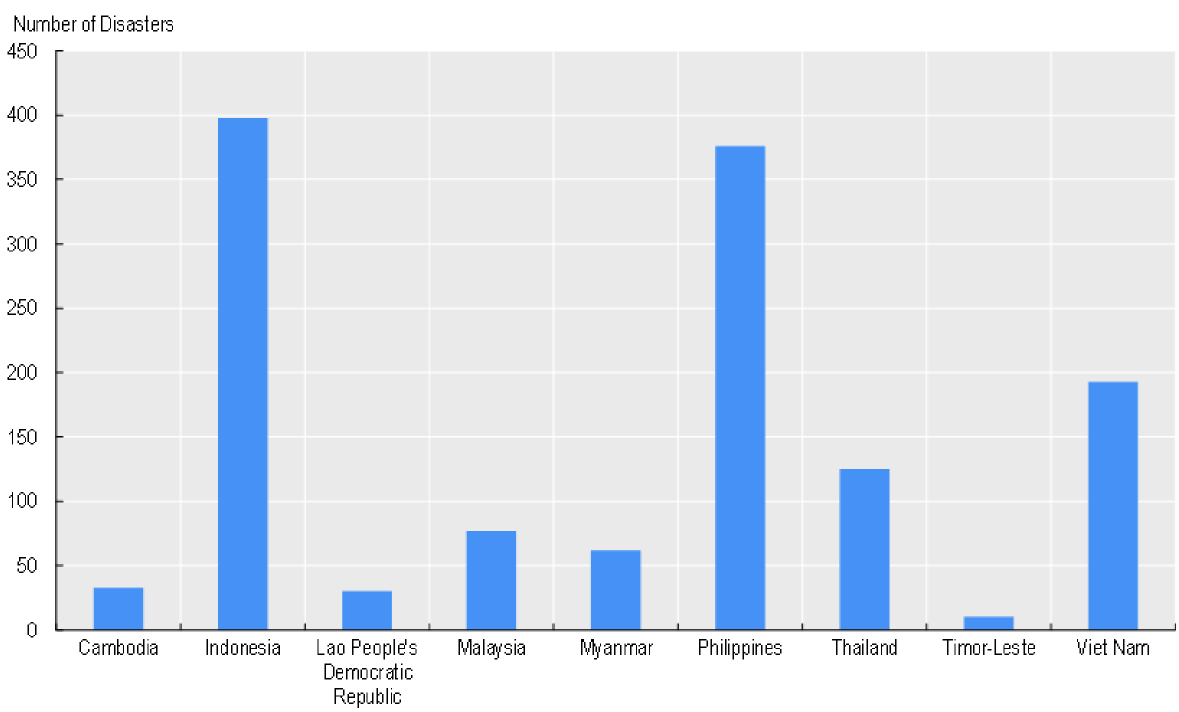
Figure 1.7. Plastic use in ASEAN countries



Source: OECD (2025[20]), *Regional Plastics Outlook for Southeast and East Asia*, https://doi.org/10.1787/5a8ff43c-en.

Climate change functions as a threat multiplier, intensifying existing environmental pressures while introducing new vulnerabilities across blue economy sectors. Observational data from the Philippines demonstrate that sea levels in the Philippine Sea have risen approximately 12 centimetres over the past two decades, a rate three times greater than the global average (ERIA, 2023[13]). Concurrently, research establishes linkages between elevated sea-surface temperatures and increased typhoon frequency and intensity, creating compounding risks for coastal populations and infrastructure, especially in the most disaster-prone countries in the region (see Figure 1.8). These climatic shifts undermine the resilience of critical coastal ecosystems, including mangroves, seagrass beds and coral reefs, which buffer coasts from storm surges and erosion and provide nursery habitats for different marine species. The challenge is compounded by institutional weaknesses, including underutilisation of disaster risk reduction and management funding, inadequate integration of coastal spatial planning with infrastructure development and insufficient investment in ecosystem-based adaptation strategies (ERIA, 2023[13]).

Figure 1.8. Number of climatological, geophysical, hydrological and meteorological natural disasters, 2000-2025



Note: Does not include biological (e.g. epidemics) and extra-terrestrial (e.g. geomagnetic storms) disasters; no recorded disasters in Singapore and Brunei Darussalam
Source: EM-DAT (2025[21]), *The International Disaster Database*, https://www.emdat.be/.

Collectively, these environmental impacts carry profound long-term implications. Viewing the blue economy through a wealth framework – which accounts for the asset value of marine fish stocks, blue carbon ecosystems and other natural capital (World Bank, 2024[22]) – reveals how ocean degradation represents not merely a flow problem (reduced annual output) but a stock problem (erosion of long-term wealth). Marine fish stocks and blue carbon ecosystems (e.g. mangroves, seagrasses, salt marshes) represent substantial economic assets whose depletion imposes intergenerational costs that conventional GDP accounting fails to capture (Bertram et al., 2021[23]; World Bank, 2024[22]).

1.3. ASEAN's regional policy response is constrained by insufficient long-term finance and investment

Integrated blue economy management – which accounts for the interlinkages between socioeconomic activities and natural ecosystems – can help Southeast Asia strike a balance between economic development and environmental protection. For instance, decarbonisation pathways for Southeast Asia necessitate not only the scale-up of renewable energy and electrification but also a shift away from unabated coal-fired power plants – the largest current source of emissions in the region (IEA, 2023[24]). Ocean-based sources of energy, including offshore renewables, are a prime candidate to fill that void. In fact, as the fastest-growing sector globally (31% growth between 2000 and 2020) (OECD, 2025[14]), offshore wind and marine renewables also hold one of the keys to economic development.[7] Meanwhile, efforts to conserve and restore coastal and marine ecosystems represent a boon not only to marine biodiversity but also to climate change mitigation – given the role of these ecosystems as a carbon sink (see Chapter 4 for a more detailed discussion on blue carbon) – and climate change adaptation – given the natural protection they offer against natural hazards (OECD, 2025[25]).

The Association of Southeast Asian Nations (ASEAN) – as an intergovernmental organisation that facilitates diplomatic engagement, economic integration and cultural exchange among its member states – has sought to promote such an integrated approach at regional level. Since the 2021 ASEAN Leaders' Declaration on the Blue Economy, member states have pursued a co-ordinated regional approach to the blue economy[8] (ASEAN, 2021[26]). The Declaration recognises oceans and seas as critical to the region's development and highlights the blue economy's relevance across all three ASEAN pillars. It provides a shared definition and framework for the blue economy, offering a common basis for regional co-operation, and calls for co-ordinated action through capacity development, resource mobilisation and policy implementation.

The Declaration was followed by the launch of the ASEAN Blue Economy Framework in 2023, which establishes a concrete structure for advancing a sustainable, inclusive and resilient blue economy. The Framework is organised around three core components: guiding principles, strategic objectives and enabling mechanisms (Figure 1.9). The guiding principles – value creation, inclusivity and sustainability – set the foundational norms for the region's blue economy. Strategic objectives translate these principles into priority action areas for member states, while enablers identify the mechanisms and conditions necessary to implement these strategies effectively. To facilitate the implementation of the Framework, ASEAN has created a Co-ordinating Taskforce on Blue Economy, which comprises blue economy focal points from ASEAN member states. It meets periodically to co-ordinate next steps for the Framework, including developing a regional blue economy implementation plan.

Figure 1.9. The ASEAN Blue Economy Framework

Source: ASEAN (2023[12]), *ASEAN Blue Economy Framework*, https://asean.org/wp-content/uploads/2023/09/ASEAN-Blue-Economy-Framework.pdf.

Operationalising the integrated vision of the ASEAN Blue Economy Framework requires adequate and effective use of financial resources. But Southeast Asia confronts a USD 2.1 trillion blue economy financing gap through 2030 (see Table 1.2). There is notable heterogeneity across sectors, with the largest gaps in capital-intensive sectors. Resilient Ports has the largest gap at USD 1 510 000 million, followed by Marine Offshore Wind Renewable Energy (USD 151 800 million) and Solid Waste Management (USD 145 000 million). Wastewater Management (USD 97 000 million) and Non-point Source Pollution Management (USD 87 100 million) show mid-range gaps, while Fishing (USD 200 million), Marine and River Ecosystems (USD 44 million) and Fisheries (USD -27 million) indicate lower or negative funding needs. While the blue economy's performance and sustainability depend on more than finance alone – and a zero or negative gap does not mean a sector is without problems – financing gaps remain a key bottleneck to achieving ASEAN's blue economy vision.

Table 1.2. Blue economy finance gap to meet the Sustainable Development Goals by 2030 in the Asia Pacific

USD millions

Themes	Segments	Pacific	Southeast Asia	South Asia
Ecosystem and Natural Resource Management	Marine & River Ecosystems	121	44	51
	Fishing	22	200	2 001
	Fisheries	14	-27	28
	Seafood Processing & Distributions	930	4 775	3 429
	Aquaculture & Algalculture	1 001	9 782	9 843
Pollution Control	Solid Waste Management	80 500	145 000	47 400
	Resource Efficiency and Circular Economy	9 000	58 600	69 900
	Non-point Source Pollution Management	26 300	87 100	463 800
	Wastewater Management	13 900	97 000	150 400
Sustainable Coastal and Marine Development	Coastal and Marine Tourism	1 776	1 396	773
	Coastal Resilience	1 700	3 600	6 340
	Resilient Ports	881 000	1 510 000	1 280 000
	Green Ports	8 858	63 641	22 086
	Green Shipping	2 171	5 649	5 532
	Marine Offshore Wind Renewable Energy	25 200	151 800	235 900
	Marine Tidal, Wave, Geothermal Renewable Energy	100	500	800
Total		1 052 593	2 139 060	2 298 283

Source: ADB/UNEP/UNDP (2022[27]), *Financing the Blue* Economy*: Investments in Sustainable Blue Small–Medium Enterprises and Projects in Asia and the Pacific*, https://doi.org/10.22617/tcs220281-2.

The ASEAN Blue Economy Framework itself recognises the importance of sustainable finance and investment, listing it as one of the "blue enablers" (ASEAN, 2023[12]). It calls for financing solutions that can support Southeast Asia's transition to a blue economy over the long term. In particular, the Framework highlights the need to reduce reliance on development assistance – typically the most concessional form of capital for developing countries – and instead mobilise a broader range of financial sources and instruments to realise the region's blue economy ambitions (ASEAN, 2023[12]).

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Notes

[1] The most widespread seagrass (marine angiosperm) species in Southeast Asia is Thalassia hemprichii. Other common species include Cymodocea serrulata and Cymodocea rotundata (Fortes et al., 2018[31]).

[2] The Coral Triangle is a biodiversity hotspot, often known as the "Nursery of the Seas" (Coral Triangle Initiative, 2026[33]).

[3] Southeast Asia's marine biodiversity also encompasses marine mammals and reptiles: for instance its marine mammal fauna comprises about 30 species – a quarter of the world's marine mammal species (PERRIN, 2002[28]).

[4] ERIA (2023[13]) suggests that Thailand's blue economy accounts for roughly 30% of GDP and employs 25% of the country's workforce.

[5] This underscores the importance of Southeast Asia's endowment of commercially important fish species (e.g. yellowfin tuna).

[6] These include practices like bottom trawling – which drags heavy nets across the seabed and destroys habitats (Olsgard et al., 2008[29]) – and dynamite fishing – which uses explosives to kill or stun fish and in the process destroys reef structures (Hampton-Smith, Bower and Mika, 2021[30])

[7] When well-governed, offshore wind farms can also function as an effective conservation measure since they entail certain spatial restrictions (Fitkov-Norris, Witt and Simmons, 2025[32]).

[8] This builds on regional policy responses on specific ocean issues. For instance, all ASEAN Member States endorsed collective commitments such as the Bangkok Declaration on Combating Marine Debris in the ASEAN Region (ASEAN, 2019[34]) and ASEAN Framework of Action on Marine Debris (ASEAN, 2021[35]).

2 Development assistance: A crucial but constrained basis for blue economy finance

The Association of Southeast Asian Nations (ASEAN) Blue Economy Framework envisions a future less dependent on development assistance, but any diversification strategy requires understanding the current baseline. This chapter examines ocean-related official development assistance (ODA) to Southeast Asia: its scale, sectoral allocation, geographic distribution and delivery modalities. It subsequently explores the implications of budgetary pressures on ocean-related ODA to Southeast Asia. Understanding these patterns is critical to assessing what gaps alternative capital sources ought to fill.

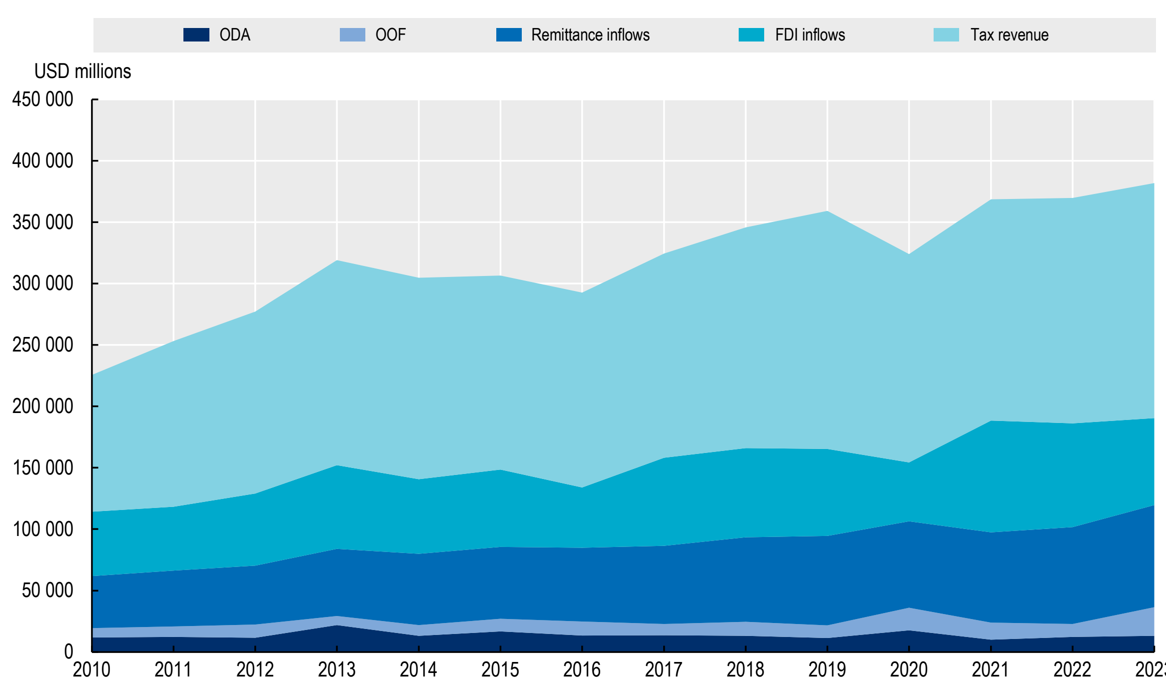
2.1. Development assistance is vital despite its small share of the overall finance mix

Official development finance (ODF) – comprising both ODA and other official flows (OOF)[1] – represents a small but strategically important component of Southeast Asia's financing landscape. Between 2010 and 2023, total ODF increased from USD 19.4 billion to USD 36.5 billion (see Figure 2.1). ODA rose from USD 11.8 billion to USD 13.1 billion over the same period, having reached a peak of USD 21.8 billion in 2013. OOF volumes also expanded, growing from USD 7.7 billion in 2010 to USD 10.5 billion in 2022, before spiking to USD 23.4 billion in 2023.

ODF forms part of the region's financing mix – typically defined as the combination of other external finance inflows (i.e. foreign direct investment and remittances) and tax revenues (Cattaneo and Piemonté, 2021[1]). Despite its growth, ODF remains a relatively small component of the broader financing landscape (see Figure 2.1). Between 2010 and 2023, ODF accounted for roughly 6–11% of total external inflows plus tax revenues, with the peak in 2020 (11%). In 2023, ODF represented just under 10% of the overall financing mix.

The small relative volumes mask the importance of development finance – particularly ODA – for Southeast Asia and its blue economy. First, most Southeast Asian countries rely to some extent on development finance, making it a common source of funding across the region. This reliance is especially pronounced among the region's lower-income countries, which not only benefit from the most concessional forms of development finance but also depend on it more heavily than upper-middle-income countries. Second, the grant-based and highly concessional nature of development finance such as ODA makes it an irreplaceable catalytic instrument: it not only addresses funding shortfalls, but also strengthens the underlying conditions for other sources and types of capital to flow, including into the blue economy.

Figure 2.1. ODA as a share of the overall financing mix of Southeast Asian countries



Source: OECD (2025[2]), *Creditor Reporting System*, https://data-explorer.oecd.org/; World Bank (2025[3]), *Data Bank*, https://databank.worldbank.org/.

2.1.1. Most ASEAN countries access development finance, with lower-income countries receiving the most concessional terms

Access to development finance – from both bilateral and multilateral sources – depends on a range of country-level factors, such as income level and creditworthiness, which reflect different dimensions of a country's development (Cattaneo and Piemonté, 2021[1]). These factors are typically formalised into eligibility criteria used by international organisations and multilateral development banks, determining not only whether a country can access finance, but also the terms on which it is offered. Eligibility for ODA and access to the different lending windows of multilateral development banks are the most salient of these criteria[2] (see Box 2.1). ODA eligibility allows countries to receive grant support and highly concessional loans, although the minimum level of concessionality varies by income group (OECD, 2024[4]). Meanwhile, multilateral development banks (MDB) operate multiple lending windows: the lowest-income countries qualify for highly concessional loans similar to ODA, while others receive preferential but near-market-rate finance.

Box 2.1. Eligibility for and access to development finance

ODA eligibility

Eligibility for official development assistance is governed by the OECD's Development Assistance Committee (DAC). For countries to be ODA-eligible, they must be classified as low or middle income according to their gross national income per capita. Countries that achieve high-income status for three consecutive years graduate from the list of eligible ODA recipients.

ODA concessionality is also influenced by a country's income level. The "softness" of an ODA loan is measured by its grant element – the percentage by which the present value of the loan falls short of its face value, due to features such as a below-market interest rate, a long grace period, or a long maturity. For a loan to be considered ODA eligible, it must have a grant element of at least:

- 45% in the case of bilateral loans to the official sector of least developed countries (LDC) and other low-income countries (LICs), calculated at a discount rate of 9%
- 15% in the case of bilateral loans to the official sector of lower middle-income countries (LMIC), calculated at a discount rate of 7%
- 10% in the case of bilateral loans to the official sector of upper middle-income countries (UMIC), calculated at a discount rate of 6%
- 10% in the case of loans to multilateral institutions and international non-governmental organisations (NGO), calculated at a discount rate of 5% for global institutions and multilateral development banks and 6% for other organisations and international NGOs.

Multilateral windows

MDBs structure their financing through differentiated lending windows that align with countries' income levels and borrowing capacity. Concessional windows provide grants or highly concessional loans to the lowest income and most vulnerable countries, while non-concessional windows offer loans on preferential but closer-to-market terms to countries with stronger creditworthiness (Cattaneo and Piemonté, 2021[1]).

For example, the Asian Development Bank (ADB) allocates concessional assistance to lower-income and more financially vulnerable member states (Group A), while more creditworthy or middle-income members typically borrow through its Ordinary Capital Resources (OCR) (Group C). Some countries receive a mix of concessional and OCR financing (Group B).

Source: OECD (2024[4]), Converged Statistical Reporting Directives for the Creditor Reporting System (CRS) and the Annual DAC Questionnaire, DCD/DAC(2024)40/FINAL/en/pdf; ADB (2025[5]), Classification and Graduation of Developing Member Countries, https://www.adb.org/sites/default/files/institutional-document/31483/om-a1.pdf.

Nearly all ASEAN countries are ODA-eligible and have access to the different MDB lending windows, making development finance a widely used tool across the region. Nine of the region's eleven countries – Cambodia, Indonesia, Lao People's Democratic Republic (PDR), Malaysia, Myanmar, the Philippines, Thailand, Timor-Leste and Viet Nam – are eligible for official development assistance (OECD, 2025[6]). These countries also have access to various multilateral lending windows, such as the World Bank's concessional (International Development Association) and non-concessional (International Bank for Reconstruction and Development) arms, as well as the Asian Development Bank's concessional and OCR windows (see Table 2.1). Only Brunei Darussalam and Singapore are ODA-ineligible and fall outside the scope of MDB lending.

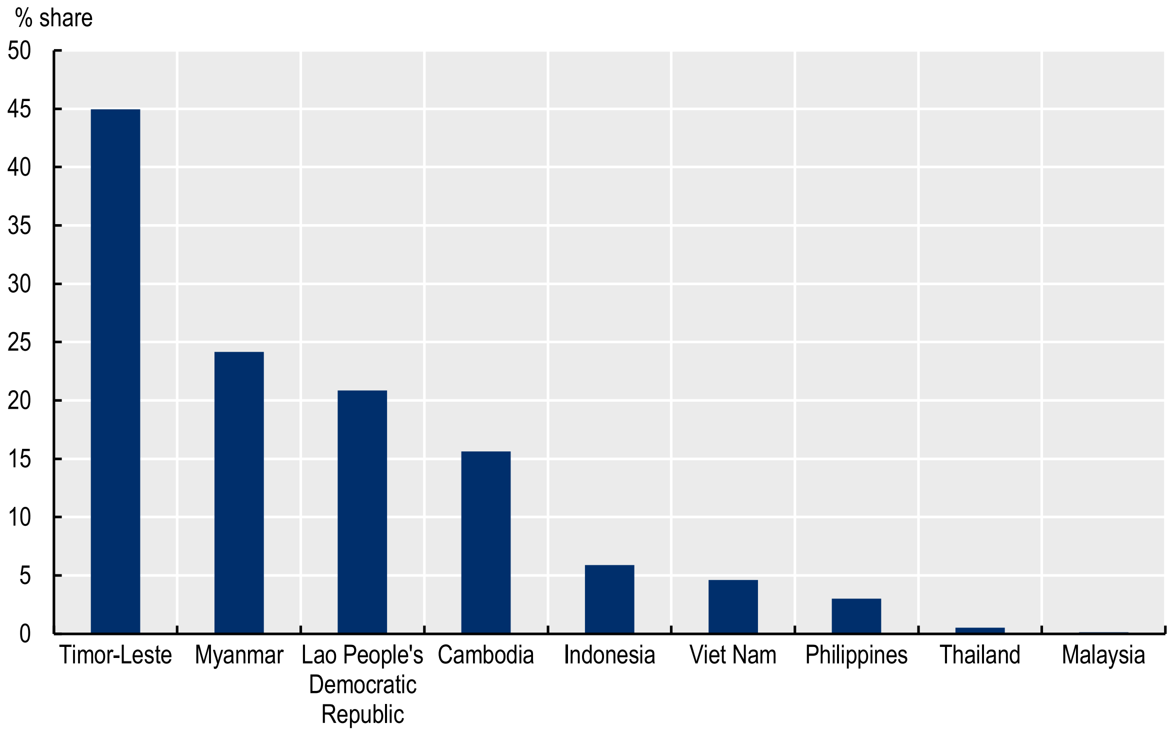
Table 2.1. Eligibility of Southeast Asian countries for development finance

Countries	ODA eligible	World Bank Lending Group	ADB Country Group
Brunei Darussalam	No		
Cambodia	Yes (LDC)	IDA	Group A (Concessional assistance-only)
Indonesia	Yes (UMIC)	IBRD	Group C (Regular OCR-only)
Lao PDR	Yes (LDC)	IDA	Group A (Concessional assistance-only)
Malaysia	Yes (UMIC)	IBRD	Group C (Regular OCR-only)
Myanmar	Yes (LDC)	IDA	Group A (Concessional assistance-only)
Philippines	Yes (LMIC)	IBRD	Group C (Regular OCR-only)
Singapore	No		
Thailand	Yes (UMIC)	IBRD	Group C (Regular OCR-only)
Timor-Leste	Yes (LDC)	Blend	Group B (OCR Blend)
Viet Nam	Yes (LMIC)	IBRD	Group C (Regular OCR-only)

Note: IDA refers to International Development Association. IBRD refers to International Bank for Reconstruction and Development
Source: OECD (n.d.[7]), DAC List of ODA Recipients: Effective for reporting on 2024 and 2025 flows, https://www.oecd.org/content/dam/oecd/en/topics/policy-sub-issues/oda-eligibility-and-conditions/DAC-List-of-ODA-Recipients-for-reporting-2024-25-flows.pdf; ADB (2025[5]), Classification and Graduation of Developing Member Countries, https://www.adb.org/sites/default/files/institutional-document/31483/om-a1.pdf; World Bank (2025[8]), World Bank Country and Lending Groups, https://datahelpdesk.worldbank.org/knowledgebase/articles/906519-world-bank-country-and-lending-groups.

Development finance is particularly important for the region's lower-income countries, which benefit from more concessional loans due to both higher concessionality thresholds for ODA and access to concessional MDB lending. For example, three of the region's least developed countries – Cambodia, Lao PDR and Myanmar – can access the World Bank's International Development Association, which provides interest-free loans and grants, and are classified as Group A countries by the ADB, making them priorities for concessional lending (see Table 2.1). The other LDC in the region, Timor-Leste, receives a blend of concessional and non-concessional lending from both the World Bank and the ADB (see Table 2.1). This importance is reflected in the extent to which ASEAN LDCs rely on ODA: between 2010 and 2023, ODA accounted for 45 % of the financing mix for Timor-Leste, 24 % for Myanmar, 21 % for Lao PDR and 16 % for Cambodia (Figure 2.2). By contrast, ODA represents less than 1 % of the financing mix in countries such as Malaysia and Thailand, with Indonesia, the Philippines and Viet Nam falling between these extremes at 5–6 %. This concentration of support in lower-income countries makes ODA a critical tool for reducing the development divide within ASEAN – a key priority for regional cohesion under the ASEAN Blue Economy Framework.

Figure 2.2. ODA as a share of the financial inflows and tax revenues across ODA-eligible Southeast Asian countries



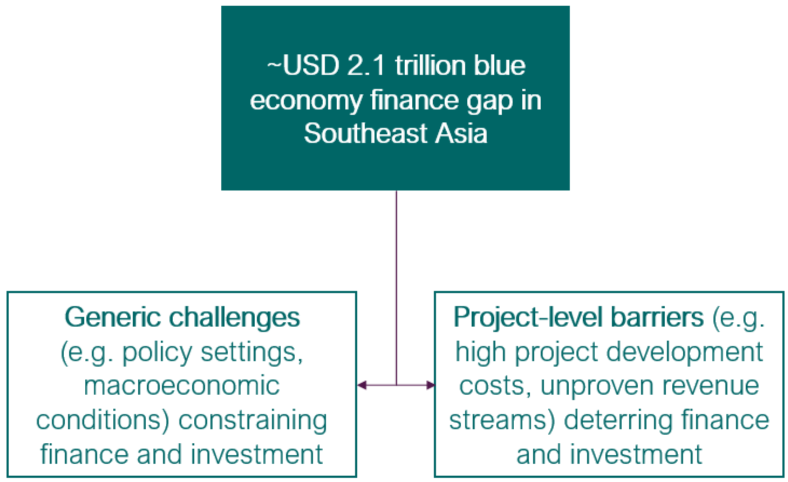
Source: OECD (2025[2]), *Creditor Reporting System,* https://data-explorer.oecd.org/; World Bank (2025[3]), *Data Bank,* https://databank.worldbank.org/.

2.1.2. Development finance can help overcome the different barriers to blue economy finance and investment

Development finance is indispensable for advancing sustainable development priorities. It can help fund activities that would otherwise be under- or unfunded, such as public goods and basic services – whether healthcare, education, climate change adaptation or social protection. Moreover, when crises occur – from natural disasters to pandemics – development finance functions as a safety net, cushioning contractions in other sources of capital (UNCTAD, 2025[9]). Beyond this direct funding role, development finance also serves as a catalyst for mobilising both private and public resources at significantly larger scales. Blended finance structures can reduce investment risk and mobilise private investment: in 2023, various leveraging mechanisms mobilised nearly USD 70 billion in private finance (OECD, 2025[10]). Development finance can also strengthen the enabling conditions – such as infrastructure and institutional quality – improving the overall investment climate. Meanwhile, support to enhance public finance management can catalyse domestic resource mobilisation and bolster fiscal capacity (Chami, Darkey and Williams, 2021[11]).

This carries important value in the blue economy, where its concessional nature – especially of ODA – and focus on social returns make it well suited to addressing barriers to finance and investment. These barriers operate at two levels: generic constraints, which limit overall capital availability, and project-specific challenges that deter individual investments (see Figure 2.3).

Figure 2.3. Barriers to finance and investment for the blue economy



Source: Adapted from OECD (2025[12]), *Promoting Sustainable Ocean Economies: Guidance for Development Co-operation*, https://doi.org/10.1787/72055d7f-en.

Generic constraints significantly shape blue economy finance and investment (OECD, 2025[12]). Limited fiscal space and high public debt, particularly when debt service takes up a large share of budget revenue, are well-known constraints in developing countries (Isgut, 2025[13]) (see Chapter 3 for a discussion of the fiscal constraints facing Southeast Asian countries). Recent overlapping shocks – including the COVID-19 pandemic – have further pressured domestic budgets and limited the availability, affordability and accessibility of external finance, as well as exposing borrowers to foreign exchange risks (United Nations, 2024r[14]). For the blue economy, this means that governments have less room to finance capital-intensive, long-term investments. Insufficient regulatory and policy frameworks also deter necessary investment: in particular, policies to strengthen sustainable marine resource management, establish clear community rights and incentivise sustainable ocean enterprises remain nascent (OECD, 2025[12]). Simultaneously, systemic capacity shortfalls – at both institutional and individual levels – reduce both investment readiness and the ability to absorb available capital.

At the project level, multiple constraints can directly impede capital deployment in the blue economy (OECD, 2025[12]). High development costs for activities such as offshore renewables mean that multi-year grant support is often required to achieve investment readiness, yet grant financing remains limited and short term. The ticket sizes for small-scale fisheries projects mean that due diligence costs typically exceed the deal size – deterring investor interest (OECD, 2025[12]). Nature-based solutions (e.g. mangrove restoration for coastal resilience) frequently lack clear revenue profiles due to insufficient performance data, while ocean-focused small and medium-sized enterprises often fail standard creditworthiness assessments. Meanwhile, the longer-term time horizons required for blue economy investments, such as coastal protection infrastructure or large-scale ecosystem restoration, to generate financial returns conflict with investor preferences for shorter time frames for a return on investment (OECD, 2025[12]).

These macroeconomic and project-specific barriers are interrelated. Limited fiscal space reduces the public co-financing needed to de-risk projects; weak regulatory environments heighten project-level uncertainty and deter private investment, reinforcing a cycle of underinvestment in the region's blue economy; and systemic capacity constraints are compounded by the specialised technical expertise often required in blue economy activities (for example, marine biotechnology).

These macroeconomic and project-specific barriers help explain why development finance remains vital for Southeast Asia's blue economy. It occupies parts of the financing landscape that other sources typically avoid – funding public goods such as marine data systems, ecosystem restoration and regulatory reforms, and supporting countries and communities with limited market access – while also catalysing other forms of capital by de-risking projects and strengthening foundations. In practice, concessional resources and ODA often absorb early-stage and policy risks, cover project preparation and technical assistance costs, and provide guarantees that make blue economy investments acceptable to domestic and international investors. This same catalytic function underpins many of the instruments discussed later in the report. Blue bonds, debt-for-nature swaps, blue carbon markets and parametric insurance, examined in Chapter 4, have typically relied on development partners for their design, piloting and scaling, whether through credit enhancements or capacity-building.

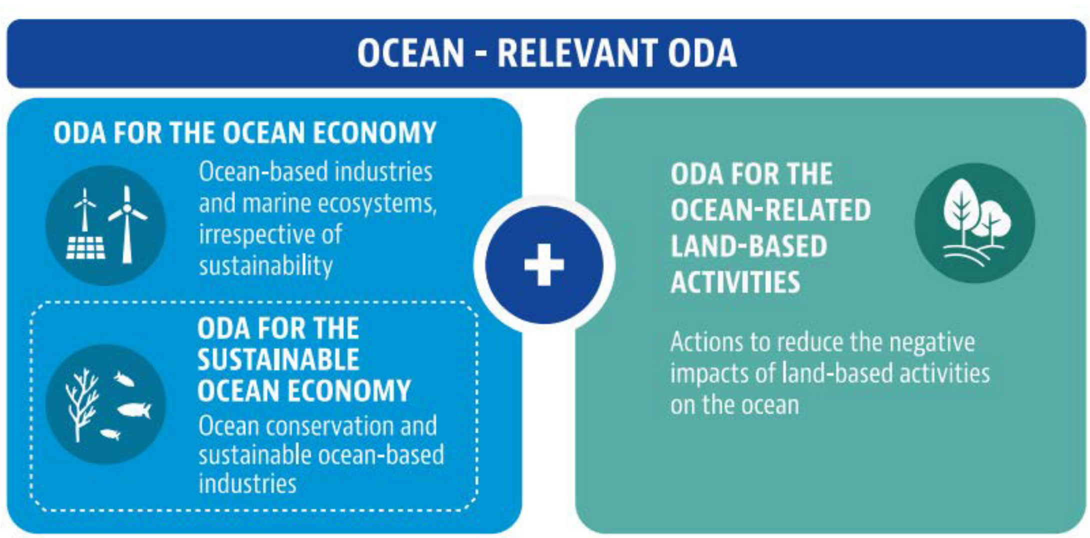
2.2. Ocean-related ODA to Southeast Asia shows problematic patterns in scale, allocation and delivery

The OECD's framework for tracking development finance for the ocean economy[3] centres on ODA (see Figure 2.4). Data on non-ODA development finance (e.g. OOF) are not available. In particular, the OECD tracks:

- **ODA for the ocean economy encompasses support** to all ocean economy activities, regardless of sustainability.
    - A subset, **ODA for the sustainable ocean economy**, captures only activities promoting sustainable resource use, management and environmental protection.
- The third indicator, **ODA for land-based activities that reduce harm to the ocean**, recognises that land-based activities spill over into marine and coastal environments.

For the purposes of this report, the sum of ODA for the ocean economy and ODA for land-based activities that reduce harm to the ocean will be referred to as **ocean-related ODA**. Meanwhile, the sum of ODA for the sustainable ocean economy and ODA for land-based activities that reduce harm on the ocean will be referred to as **sustainable ocean-related ODA**.

Figure 2.4. Framework for measuring ocean-related ODA



Source: OECD (2020[15]), *Sustainable Ocean for All: Harnessing the Benefits of Sustainable Ocean Economies for Developing Countries*, https://doi.org/10.1787/bede6513-en.

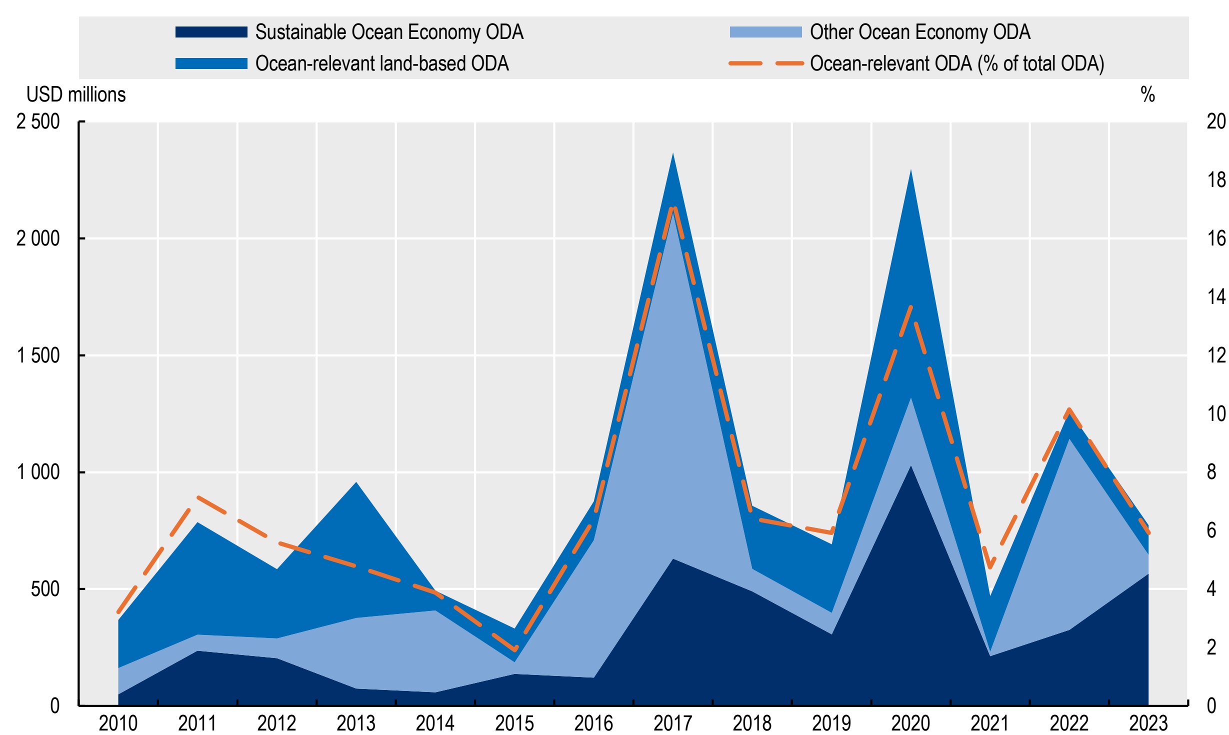
An analysis of ocean-related and sustainable-related ODA to Southeast Asia shows that there are several challenges. High year-over-year volatility – driven by concentration in large infrastructure projects – creates unpredictability. Significant gaps persist between commitments and actual disbursements, also driven by the skew of ocean-related ODA commitments towards large projects. Allocations do not always align with countries' ocean and ODA dependencies, while heavy reliance on loans rather than grants and dominance of project-type interventions can limit strategic coherence.

2.2.1. Ocean-related ODA has increased but exhibits high year-over-year volatility driven by large individual transport projects

Ocean-related ODA to ASEAN grew substantially between 2010 and 2023. It increased from USD 367.7 million in 2010 to USD 773.7 million in 2023 – more than doubling and outpacing the 1.67-fold increase in ocean-related ODA to all developing country recipients. Between 2010 and 2023, ocean-related ODA's share of total ODA to ASEAN oscillated between 6% and 14% – noticeably larger than the 2% global average, suggesting ocean priorities feature more prominently in ASEAN ODA allocations.

However, extreme year-over-year volatility also characterises ocean-related ODA to ASEAN. Between 2010 and 2023, commitments ranged from USD 331.3 million (2015) to roughly USD 2.4 billion (2017). Despite the generally increasing trend, spike years are followed by pronounced dips. After ocean-related ODA crossed USD 2 billion in 2017 and 2020, marked decreases followed: -64% from 2017 to 2018 and -80% from 2020 to 2021.

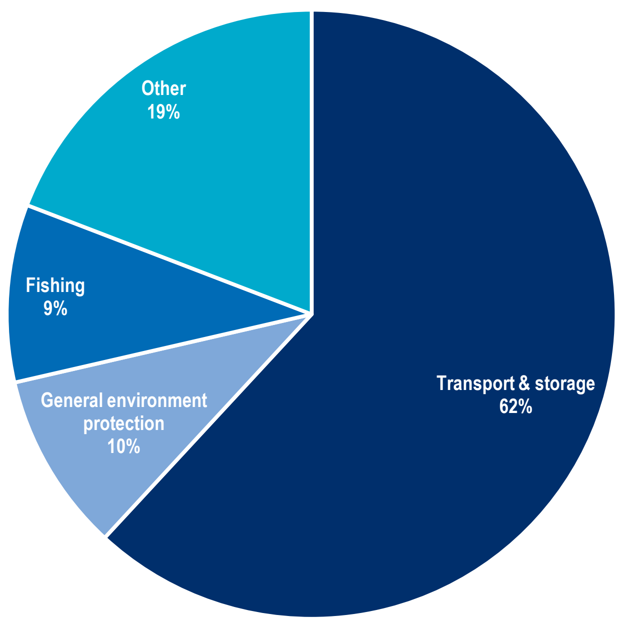
Figure 2.5. Ocean-related ODA over time to ASEAN



Source: Authors' calculations based on OECD (2025[16]), *Data Platform on Development Finance for the Sustainable Ocean Economy*, https://oecd-main.shinyapps.io/ocean/.

This volatility stems primarily from a small number of large ocean economy projects that dominate ocean-related ODA to ASEAN, most of which are in transport and storage, typically single large port upgrading and expansion projects. In 2022-2023, transport and storage accounted for USD 1.3 billion – over 60% of ocean-related ODA to ASEAN – while the next three largest sectors together received only 27% (Figure 2.6). While this concentration in transport and storage aligns with the region's largest blue economy financing gap and the sector's high capital intensity, it also introduces substantial volatility. At the same time, the near absence of investment in other areas of need – especially in emerging, equally capital-intensive sectors such as marine renewable energy – represents a potential missed opportunity. Moreover, although the financing gap is useful for identifying underfunded areas, excessive reliance on it risks skewing resources towards capital-intensive activities at the expense of lower-cost but essential functions for the blue economy. Taken together, the pronounced concentration of ODA in the transport sector raises important questions about the balance and long-term resilience of blue economy development in Southeast Asia.

Figure 2.6. Sectoral distribution of ocean-related ODA to ASEAN 2022-2023



Source: Authors' calculations based on OECD (2025[16]), *Data Platform on Development Finance for the Sustainable Ocean Economy*, https://oecd-main.shinyapps.io/ocean/.

Over the 2010-2023 period, only 34% of ocean-related ODA to Southeast Asia supported the sustainable ocean economy. Including ODA for land-based activities that reduce ocean harm increases this share to 66%. This implies, nonetheless, that over 30% of ocean-related ODA projects do not explicitly target environmental sustainability.

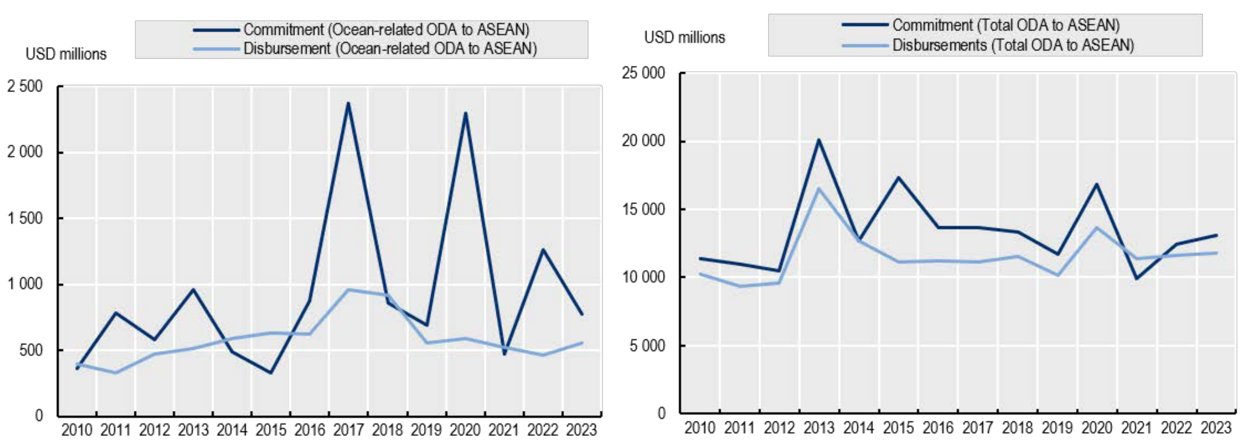
While sustainable ocean economy ODA has grown substantially – from USD 49.1 million in 2010 to USD 565.5 million in 2023, with its share of ocean-related ODA rising from 13% to 73% – this apparent progress masks considerable year-to-year volatility driven by the transport sector's dominance. In years with major ocean-related ODA spikes – 2017, 2020 and 2022 – sustainable ocean economy ODA accounted for only 27%, 45% and 26% of total ocean-related ODA respectively. This pattern reflects how large transport infrastructure projects, particularly port expansion and upgrading, drive overall commitments upward while often lacking explicit sustainability criteria. The result is an unpredictable sustainability trajectory determined by project-specific funding cycles rather than strategic prioritisation of environmental objectives, making it difficult to assess whether ocean-related ODA is genuinely becoming more sustainable or simply experiencing cyclical fluctuations tied to infrastructure investment timing.

2.2.2. The dominance of large individual projects creates delivery and modality challenges

Ocean-related ODA commitments to ASEAN consistently exceed disbursements, and by a wider margin than ODA to ASEAN overall. Over 2010-2023, commitments exceeded disbursements by approximately USD 5 billion, meaning 38% of committed funds remained undisbursed (Figure 2.7). This exceeds the 14% undisbursed of overall ODA to ASEAN over the same period. The most striking feature is the pattern of large commitment spikes, particularly in 2017, 2020 and 2022, that are not matched by corresponding disbursements in the years that follow. These spikes correspond to years when transport sector commitments peaked, particularly for port-related projects.

Figure 2.7. Commitments versus disbursements

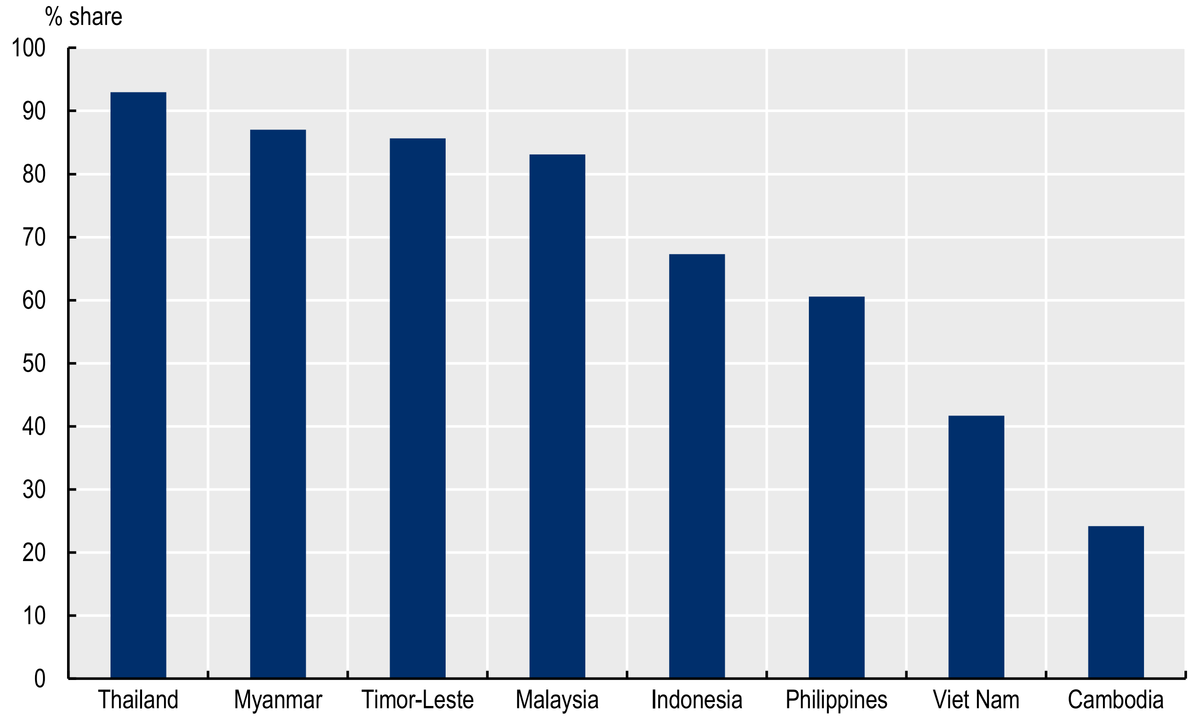
LHS: Ocean-related ODA to ASEAN; RHS: Total ODA to ASEAN



Source: Authors' calculations based on OECD (2025[16]), *Data Platform on Development Finance for the Sustainable Ocean Economy*, https://oecd-main.shinyapps.io/ocean/.

The mismatch remains salient when evaluating individual projects and countries. In a sample of ocean-related ODA projects with consistent reporting between 2010-2013, only 50% of committed funds had been disbursed by end-2023, compared to 66% for overall ODA. Strong heterogeneity exists across recipients: Thailand has received over 90% of committed funds whereas Cambodia has received less than 25% (see Figure 2.8).

Figure 2.8. Share of commitments from 2010-2013 disbursed as of 2023

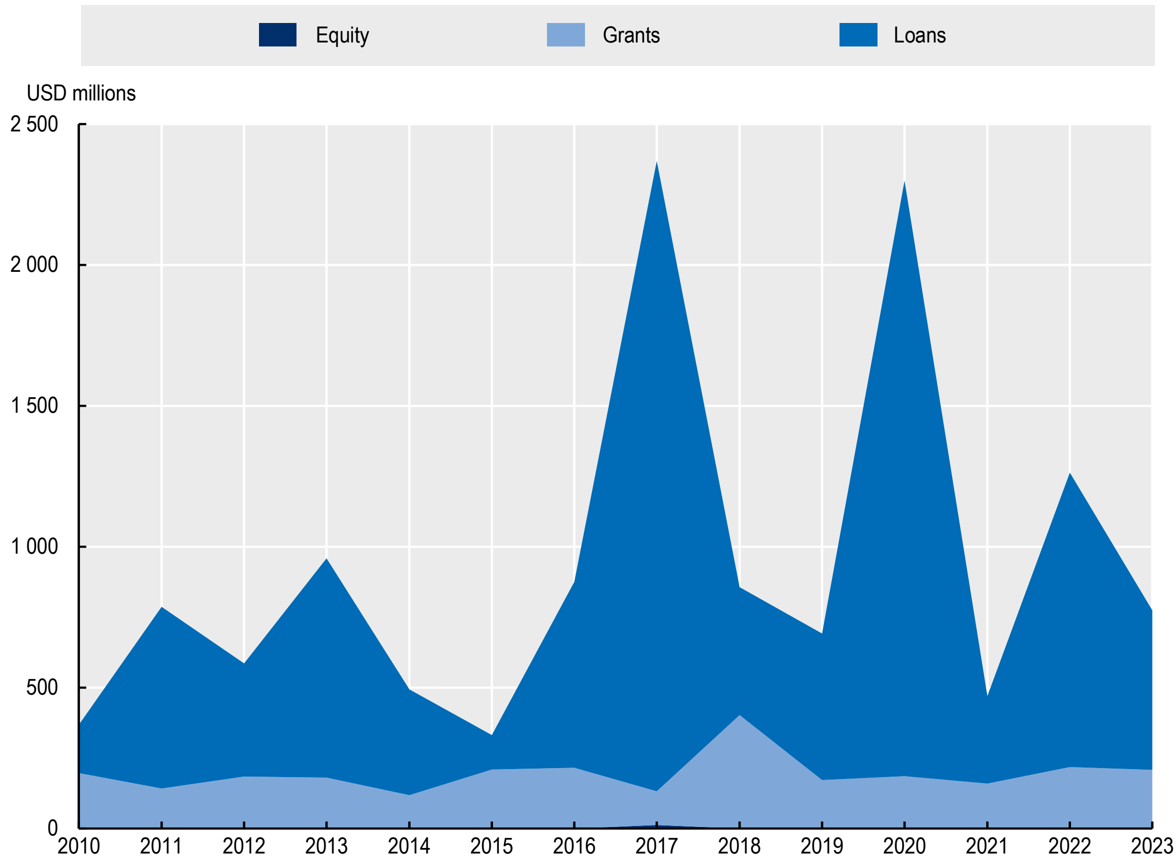


Note: Sample of projects with continuous reporting (i.e. same CRSID used between 2010 and 2023).
Source: Authors' calculations based on OECD (2025[16]), *Data Platform on Development Finance for the Sustainable Ocean Economy*, https://oecd-main.shinyapps.io/ocean/.

Virtually all ocean-related ODA to ASEAN takes the form of project-type interventions. Over 2010-2023, project-type interventions never dropped below 85% of total ocean-related ODA, often exceeding 90%. This contrasts sharply with overall ODA to ASEAN, where project-type interventions constitute just 45-47% of activities over the same period. This echoes the significant skew of ocean-related ODA towards singular large projects, typically in the transport sector. Excessive reliance on project-type interventions, at the expense of other modalities like budget support or technical assistance, can represent an ad hoc, fragmented approach to development co-operation, generating high transaction costs for recipient countries (e.g. multiple reporting and accounting requirements) and potentially undermining long-term development progress due to lack of strategic coherence across projects.

Loans account for nearly 80% of ocean-related ODA over 2010-2023 (Figure 2.9), significantly exceeding loan shares in overall ODA to ASEAN. This reflects large infrastructure investment projects prevalent in ocean-related ODA, particularly in the transport sector, which are typically loan-financed. Even in other ocean-related sectors like fisheries and water supply and sanitation, loans predominate. Environmental protection is the only exception, being entirely grant-funded.

Figure 2.9. Loans and grants over time in ocean-relevant ODA



Source: Authors' calculations based on OECD (2025[16]), *Data Platform on Development Finance for the Sustainable Ocean Economy*, https://oecd-main.shinyapps.io/ocean/.

2.2.3. ODA allocations do not consistently reflect countries' ocean and aid dependencies

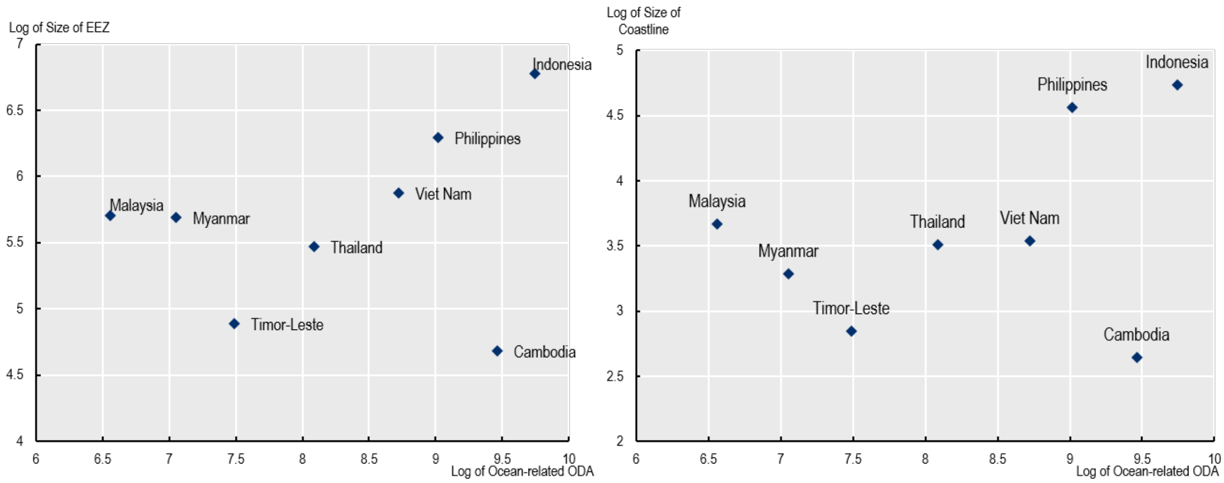
The largest recipients of ocean-related ODA to Southeast Asia have shifted over time. Before 2017, Viet Nam was the largest recipient. Indonesia emerged as the largest in 2017. Recently, Cambodia has become prominent. For these recipients, ocean-related ODA accounts for a significant share of total ODA. In 2022-2023, ocean-related ODA represented nearly 19% and 13% of Indonesia and Cambodia's total ODA respectively.

Examining relationships between ocean-related ODA allocations and two types of dependency provides insight into allocation patterns: ocean dependency (indicated by exclusive economic zone (EEZ) size and coastline length, proxying reliance on ocean resources) and ODA dependency (indicated by ODA's share of the financing mix, proxying reliance on development assistance).

Some alignment exists between ocean-related ODA allocations and ocean dependency as proxied by EEZ and coastline size (Figure 2.10). Indonesia, with the largest EEZ and coastline, receives the largest ocean-related ODA. However, notable exceptions exist. Myanmar receives relatively small amounts despite considerable EEZ and coastline – particularly striking given that Cambodia, with smaller EEZ and coastline, receives the second-largest amount. While this likely reflects competing priorities in Myanmar, where development assistance focuses on conflict and fragility challenges, other countries including the Philippines, Timor-Leste and Viet Nam also receive smaller amounts despite having larger EEZ and coastlines than Cambodia.

Figure 2.10. Ocean-related ODA versus size of EEZ and length of coastline

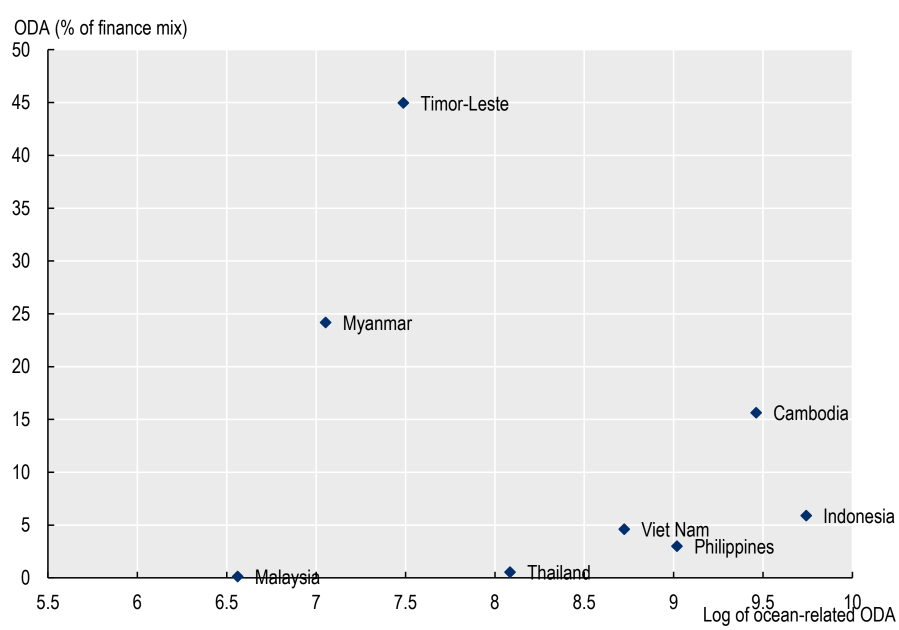
LHS: EEZ vs. ocean-related ODA; RHS: Coastline vs. ocean-related ODA



Source: Authors' calculations based on OECD (2025[16]), *Data Platform on Development Finance for the Sustainable Ocean Economy*, https://oecd-main.shinyapps.io/ocean/.

The relationship between ocean-related ODA and overall ODA dependency shows no clear pattern (Figure 2.11). Malaysia, receiving the least ocean-related ODA, is also least reliant on ODA generally, suggesting alignment. This also holds for Thailand, which receives below-median ocean-related ODA and is not ODA-reliant. However, Myanmar and Timor-Leste present contrasting cases: despite heavy ODA reliance (24% and 45% of their financing mix respectively), both receive relatively small amounts of ocean-related ODA, trailing countries like Indonesia, the Philippines and Viet Nam that rely far less on ODA overall. This pattern suggests that ODA dependency alone does not determine ocean-related allocations.

Figure 2.11. Ocean-related ODA versus share of ODA in overall financing mix



Source: Authors' calculations based on OECD (2025[16]), *Data Platform on Development Finance for the Sustainable Ocean Economy*, https://oecd-main.shinyapps.io/ocean; and OECD (2025[2]), *Creditor Reporting System*, https://data-explorer.oecd.org/.

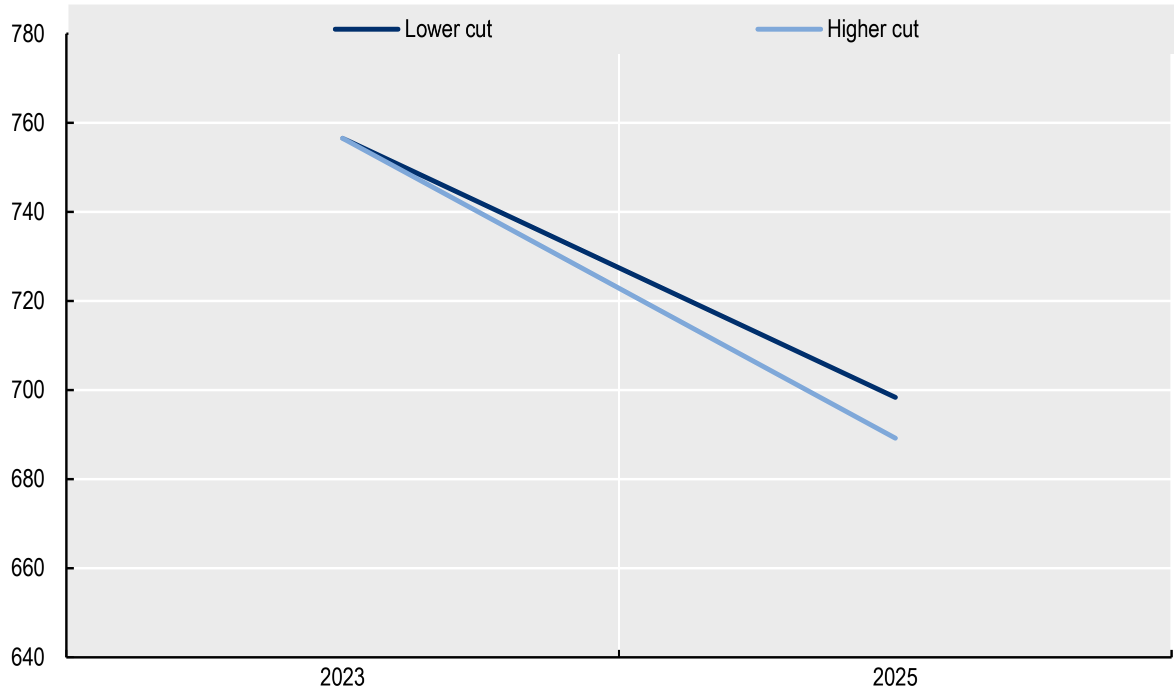
2.3. Declining aid budgets increase the urgency of mobilising alternative capital sources

Bilateral official development assistance is under significant strain. After falling by 9% in 2024, ODA is projected to decline by a further 9–17% in 2025 (OECD, 2025[17]). This sustained contraction stems from announced cuts by eleven OECD DAC members, including a historic development: four major providers – France, Germany, the United Kingdom and the United States – are cutting aid simultaneously for two consecutive years, the first time this has occurred in nearly three decades. These four countries have collectively accounted for close to two-thirds of total ODA over the past decade, with the United States alone representing one quarter. If current trends continue, ODA is projected to fall back to 2020 levels by 2027, threatening to reverse recent gains at a time when global needs are rising (OECD, 2025[17]).

Ocean-related ODA to Southeast Asia is vulnerable to these broader pressures, given the composition of bilateral providers supporting the region's ocean priorities. Three of the four OECD DAC members announcing cuts (France, the United States and Germany) were among the top ten providers (both bilateral and multilateral) of ocean-related ODA (OECD, 2025[16]). Their relative importance varies across countries: Germany dominates ocean-related ODA to Thailand and Timor-Leste, while France is a major provider to Viet Nam (OECD, 2025[16]). These countries therefore face pronounced exposure to cuts in ocean-related ODA.

Indeed, simulations based on announced aid reductions suggest that ocean-related ODA flows to ASEAN could contract in 2025. In 2023, Development Assistance Committee members provided approximately USD 756.5 million in ocean-related ODA to ASEAN member states. Under a low-cut scenario, this volume is projected to decline to USD 698.4 million. Under a high-cut scenario, funding could fall further to USD 689.2 million, representing an overall decline of roughly 8–9% (Figure 2.12).

Figure 2.12. Early projections of DAC members ocean-related ODA to Southeast Asia



Note: The key distinction between these two projections lies in how United States' Official Development Assistance is calculated. In the first projection ("lower cut"), United States' foreign aid through United States Agency for International Development is estimated to decrease by 38% compared to 2024 levels, drawing on conservative estimates from the Center for Global Development's analysis of funding reductions for the 2024-2025 fiscal year. The second projection ("higher cut") anticipates a more dramatic 82% reduction in United States Agency for International Development funding from 2024 levels, corresponding to the United States' Secretary of State's March 2025 announcement regarding the cancellation of 82% of the agency's programmes.

Source: Authors' calculation based on OECD (2025[16]), *Data Platform on Development Finance for the Sustainable Ocean Economy*, https://oecd-main.shinyapps.io/ocean; and OECD (2025[17]), *Cuts in official development assistance: OECD projections for 2025 and the near term,* https://doi.org/10.1787/8c530629-en.

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Notes

[1] ODA consists of grants and sufficiently concessional loans (see Box 2.1 for a more detailed explanation of concessionality) provided by official agencies to promote economic development and welfare in developing countries (OECD, 2024[4]). OOF includes non-concessional official lending that does not meet ODA criteria but still comes from official sources (OECD, 2024[4]).

[2] There is overlap in the underlying factors that dictate a country's ODA eligibility and its access to different windows of multilateral development banks. But they are governed by different institutional processes and affect access to different types and sources of development finance.

[3] While these indicators do not capture the full spectrum of blue economy activities in Southeast Asia, given that they do not include freshwater resources, they help assess the overall landscape. For more detailed information on the methodology, see OECD (2020[15]) and OECD (2025[16]).

3 Diversifying blue economy finance: Opportunities and constraints across alternative sources

Official development assistance remains important for Southeast Asia's blue economy, but its limitations are increasingly apparent. Yet alternative sources – domestic public finance, international private capital and domestic private capital – are not straightforward substitutes. Each operates under different constraints, targets distinct activities and serves separate functions within the Association of Southeast Asian Nations (ASEAN) Blue Economy Framework. This chapter examines how these sources align with the Framework's strategic pillars, identifies what limits their availability and flow in Southeast Asia and assesses which policy and institutional gaps prevent effective capital deployment. Understanding these dynamics is critical to developing realistic strategies for blue economy financing.

3.1. Different capital types serve distinct purposes, but their flow to ASEAN's blue economy is poorly tracked

3.1.1. Each capital type – international private, domestic private, and domestic public – plays a distinct role within the ASEAN Blue Economy Framework

Financial capital is not fungible. Depending on the provider of this capital (e.g. public versus private), the associated requirements (e.g. commercial return versus public value) and its use cases will vary. This means that domestic public, international private and domestic private sources of capital can support the implementation of the ASEAN Blue Economy Framework in distinct ways (see Figure 3.1).

Figure 3.1. Different capital types mapped to the strategic pillars of the ASEAN Blue Economy Framework

EXAMPLES OF SUPPORT THE ABEF'S THREE STRATEGIC PILLARS

	Conservation and Management	Science, Technology and Innovation	Priority Sectors
International Public (e.g., official development assistance)	Funds foundational efforts e.g. grants for conservation	Supports technology transfer, e.g. to close gaps in novel blue economy sectors	De-risks private investment in sectors like aquaculture and offshore renewable
Domestic Public (e.g., fiscal instruments)	Funds environmental management (e.g. monitoring IUU fishing)	Encourages research & innovation in blue sectors (e.g. fiscal incentives)	Funds foundational blue economy infrastructure
International Private (e.g., FDI, private equity, venture capital)	Supports nature-based solutions (namely to meet ESG commitments)	Invests in commercial ventures that develop technologies in blue economy sectors	Provides capital for major infrastructure investments (e.g. for coastal tourism)
Domestic Private (e.g., local retail banking)	Finances smaller-scale conservation/eco-tourism efforts	Invests in/lends to small enterprises that are developing innovative "blue" solutions	Funds local businesses, SMEs in traditional sectors like fisheries

Source: Authors' creation based on ASEAN (2023[1]), *ASEAN Blue Economy Framework*, https://asean.org/wp-content/uploads/2023/09/ASEAN-Blue-Economy-Framework.pdf.

Domestic public finance channels government revenues towards public priorities. This source targets public goods that markets tend to underprovide: ports, coastal protection, marine research, fisheries enforcement, and environmental management. Because it does not require commercial returns, domestic public finance suits activities with high social and environmental value but limited commercial potential – though deployment depends on fiscal capacity and policy priorities. Within the Framework, domestic public finance could support environmental management under Conservation and Management, foundational infrastructure for Priority Sectors, and fiscal incentives for innovation. Where deployed effectively, it creates governance structures and regulatory frameworks enabling other investments – though this requires adequate fiscal space and political commitment.

International private capital flows from institutional investors, commercial banks and corporations seeking financial returns (see Box 3.1 for two exceptions). This capital targets larger profit-generating activities: shipping, offshore renewables, capital-intensive aquaculture, and coastal tourism infrastructure. Projects must generate revenues sufficient to justify costs and risks – though flow depends on risk perceptions and enabling environments. Across the Framework's pillars, international private capital could back nature-based solutions with environmental, social and governance returns (Conservation and Management), blue economy technology ventures (Science, Technology and Innovation) and major infrastructure like ports and offshore renewables (Priority Sectors). International capital can deliver the scale that domestic sources cannot match alone, though its profit focus means public goods and uncertain ventures may be ignored without public sector intervention.

Box 3.1. Non-profit seeking private capital: Remittances and philanthropic contributions

While technically private, philanthropic funding is largely concerned with generating social and environmental return. It is mainly grant based and, in the case of the blue economy, can essentially function similarly to public sources of finance, although the areas of focus are typically set by philanthropy leadership (e.g. board of trustees) in line with the organisation's mission. Another exception is remittances, which are private transfers between or within households. Their objective and use case are highly variable, but there is evidence that remittances can improve well-being at the household level. Although there has been some momentum to incentivise the use of remittances for collective purposes, such as the green transition, their inherent household-to-household nature makes directing them in a concerted and coherent way a challenge.

Source: OECD (2021[2]), Private Philanthropy for Development – Second Edition: Data for Action, https://doi.org/10.1787/cdf37f1e-en; OECD (2025[3]), DAC Blended Finance Guidance 2025, https://doi.org/10.1787/e4a13d2c-en; OECD (2005[4]), Migration, Remittances and Development, https://doi.org/10.1787/9789264013896-en;

Domestic private capital comes from local commercial banks, institutional investors, corporations and microfinance institutions. Like international capital, it generally seeks economic returns, though often with greater willingness to accept smaller investments. This source could support all Framework pillars at smaller scales: modest eco-tourism and conservation (Conservation and Management), locally adapted solutions from small enterprises (Science, Technology and Innovation), and local businesses operating in traditional sectors (Priority Sectors). Given the Framework's inclusivity emphasis, domestic private capital could be particularly valuable – potentially reaching communities and enterprises international capital skips due to small scale or perceived risk, though this depends heavily on local financial market development.

3.1.2. Visibility of the flows from different capital sources into the blue economy in Southeast Asia remains limited

Tracking capital flows is essential to assuring the adequacy and impact of finance and investment for the blue economy. Without knowing how much capital is available, where it comes from, and where it goes, it is difficult to identify redundancies, address financing gaps, or seize missed opportunities. Such information gaps can mean that capital suitable for – and even intended for – the sustainable blue economy may be diverted elsewhere, co-opted to facilitate further exploitation of ocean spaces, or be concentrated in limited geographies and sectors. Beyond international public finance (see Chapter 2), there is limited transparency on other sources of finance and investment for the blue economy in Southeast Asia – meaning that it is unclear what flows actually exist and whether they are optimally deployed.

In the case of domestic public finance, there is available information on fiscal instruments relevant to the blue economy in Southeast Asia. In particular, the OECD Policy Instruments for the Environment database provides detailed data on ocean-related policy mechanisms employed by Southeast Asian countries – covering an array of domestic revenue-generating tools (e.g. taxes) and government expenditures (e.g. subsidies) (OECD, 2025[5]). Table 3.1 shows the number and types of these instruments used across different Southeast Asian countries. Fees and taxes dominate the ocean-related environmental policy mix in Southeast Asian countries, highlighting a skew towards mechanisms that mobilise domestic revenues while advancing environmental objectives. In contrast, policy instruments such as payment for ecosystem service schemes are present in negligible numbers.

Table 3.1. Count of ocean-relevant economic environmental policy instruments in Southeast Asian countries

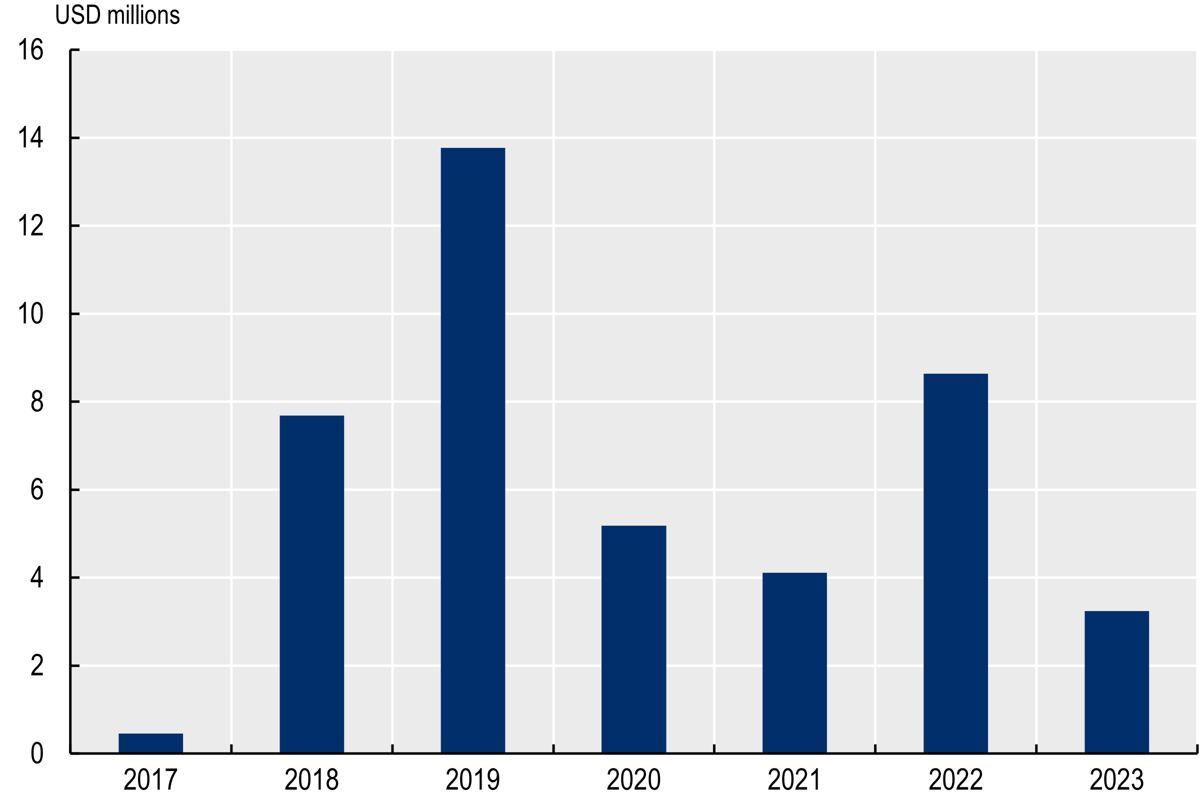
Countries	Fee (including charges)	Payment for ecosystem service	Per physical unit (recurrent)	Tax	Tax credit	User rights
Brunei Darussalam	20	0	0	2	0	0
Cambodia	6	0	0	1	0	0
Indonesia	10	1	1	2	1	0
Lao People's Democratic Republic (PDR)	2	0	0	1	0	0
Malaysia	11	0	0	2	0	0
Myanmar	12	0	0	1	0	0
Philippines	8	0	0	5	0	0
Singapore	18	0	0	3	0	0
Thailand	11	0	0	3	0	0
Viet Nam	6	0	0	0	0	1

Note: Timor-Leste is not included due to unavailable data.
Source: OECD (2025[5]), Policy Instruments for the Environment Database, http://oe.cd/pinedatabase (accessed in June 2025).

However, a more comprehensive accounting of domestic public finance – especially of expenditures – for the blue economy is missing (OECD, 2025[6]). This is a global challenge. Many developing countries and emerging markets do not systematically track how much they spend on the blue economy. As a result, data on public expenditures for the blue economy are far less advanced than those for climate and biodiversity, which have benefited from early co-ordination on international good practice (Siverwood-Cope and Ling, 2021[7]). A full review of opportunities to generate public revenue from beneficiaries of blue natural capital is also often not common across countries (World Bank, 2021[8]). In the case of Southeast Asia, work on enhancing the transparency of domestic public finance for the blue economy is at an early stage. For instance, countries like Indonesia have piloted a blue budget tagging system, and a refinement of its tagging exercise is envisioned as part of the 2026 work plan. In other countries, there is generally no dedicated tracking system.

In the case of private finance for the blue economy, comparable cross-country data are largely confined to philanthropic contributions. As shown in Figure 3.2, philanthropic contributions to Sustainable Development Goal (SDG) 14 in Southeast Asia show a fluctuating but generally positive trend between 2017 and 2023. The contributions reached their peak in 2019 (USD 13.8 million), declining thereafter to USD 3.2 million. The two largest philanthropic donors (between 2017 and 2023) to the region are Margaret A. Cargill Foundation (USD 15.5 million) and David and Lucile Packard Foundation (USD 14.1 million) – with the others providing less than USD 5 million each.

Figure 3.2. Philanthropic contributions to SDG14 in Southeast Asia



Source: Authors' calculations based on OECD (2025[9]), *Data Platform on Development Finance for the Sustainable Ocean Economy*, https://oecd-main.shinyapps.io/ocean/.

Data on purely private capital for the blue economy – whether from domestic or international sources – are largely incomplete. Efforts to enhance transparency in this area are burgeoning, with initiatives in Europe being most advanced. For instance, the European Union-funded BlueInvest initiative has carried out a study of blue economy investment activity in Europe. Focusing on blue economy deal flow (e.g. mergers and acquisitions, private equity and venture capital deals), the analysis has uncovered a significant increase in disclosed blue economy investments, reaching over USD 13 billion between 2018 and 2023 (BlueInvest, n.d.[10]). The analysis identifies blue renewable energy, blue technology and ocean observation, and aquaculture as the most dynamic sectors for blue investments in Europe. Digital platforms, such as the Hub Azul Dealroom, which function as a matchmaking platform for blue economy investments, also house data on private transactions related to the blue economy (Forum Oceano, 2025[11]). While the tool can conceivably have a global scope, its genesis and current coverage skew towards Europe and North America.

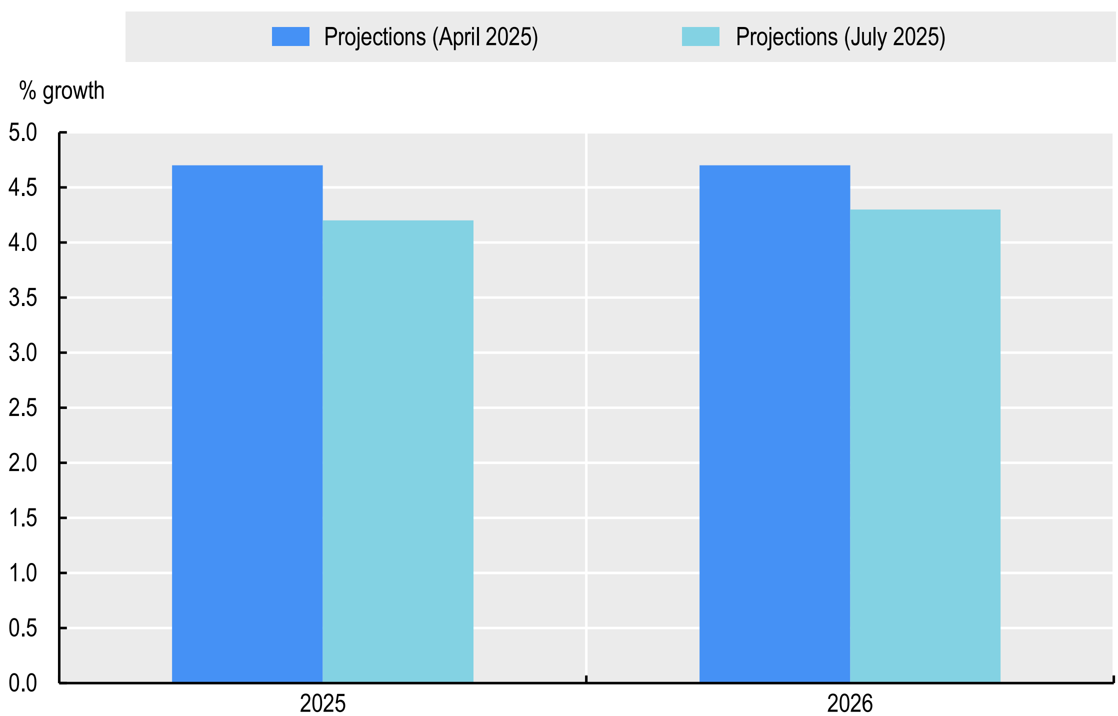
3.2. Macroeconomic conditions create uneven availability of capital for the blue economy across Southeast Asia

Blue economy finance does not exist in isolation from broader economic fundamentals (see Figure 2.3 in Chapter 2). The fiscal constraints, debt burdens, shallow financial markets and weak regulatory frameworks that limit overall capital availability in many Southeast Asian countries affect blue economy investments just as they constrain financing in any other sector.

Southeast Asia faces several economic headwinds, which may affect the availability of different capital types. For instance, external factors like escalating tariffs and trade uncertainty have hampered the region's near-term economic outlook, resulting in the Asian Development Bank downgrading projections for future growth (see Figure 3.3). More structural issues also plague the region: for example, while economic expansion over the past 40 years has been propped up by a growing working-age population, there is evidence that this is slowing in several countries.

But the vast heterogeneity of countries in the region means that limiting factors for blue economy finance and investment are most salient at the country level. The disparity in per capita output, an average of USD 6 840 in the ASEAN-5 countries (Indonesia, Malaysia, the Philippines, Singapore and Thailand) compared to less than USD 3 000 in countries like Cambodia, Lao PDR and Myanmar (IMF, 2025[12]), reflects fundamental differences in the availability of domestic public, as well as domestic and international private capital.

Figure 3.3. Downward adjustment of Southeast Asia's economic outlook



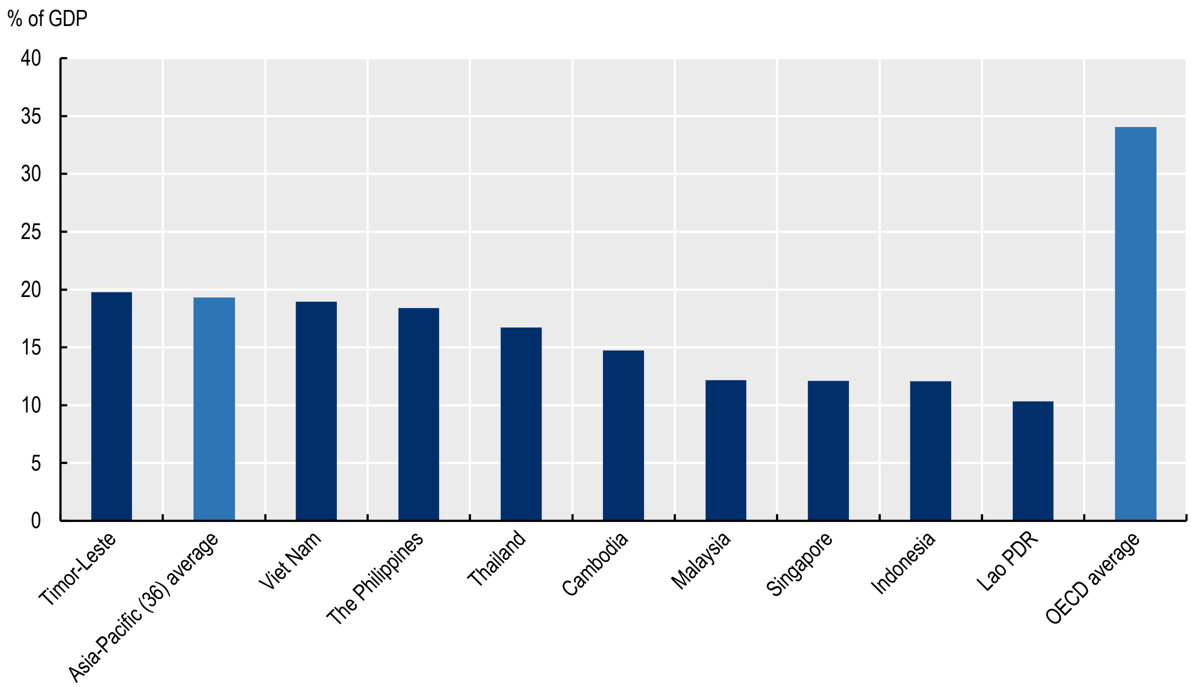
Source: (ADB, 2025[13]), Economic Forecasts for Asia and the Pacific, https://www.adb.org/outlook/editions/july-2025.

3.2.1. Fiscal constraints limit domestic public finance availability

The availability of domestic public finance for blue economy priorities depends fundamentally on three factors: revenue generation capacity (tax to gross domestic product (GDP) ratios), fiscal flexibility (debt sustainability and budget pressures), and public finance management quality (institutional capacity to allocate and track expenditures effectively). These disparities in fiscal capacity and public finance management have direct implications for blue economy finance. Countries with tight fiscal space are less able to allocate budget to non-revenue-generating but essential blue economy functions such as marine spatial planning, fisheries enforcement or coastal ecosystem restoration. Conversely, where tax capacity and public finance management are stronger, governments are better positioned both to fund these public goods directly and to provide the co-financing to crowd in private investment for the blue economy.

Southeast Asian countries demonstrate significant variation in revenue generation. In 2022, tax-to-GDP ratios ranged from 10.3% in Lao PDR to 19.0% in Viet Nam, with the regional average falling well below the OECD norm of over one-third of GDP (see Figure 3.4). Countries with relatively low ratios face greater potential to increase domestic revenues, while others lie closer to the observed regional maximums. The composition of tax revenues also varies. Indonesia balances corporate income (28.8%) and value-added tax (VAT) (28.2%), Malaysia relies heavily on income taxes (over two-thirds of revenue), while Cambodia, Lao PDR and Thailand depend more on VAT and trade taxes (OECD, 2024[14]). This creates uneven exposure to volatility – countries dependent on goods and services taxes face sharper revenue fluctuations tied to consumption patterns.

Figure 3.4. Tax-to-GDP ratio across Southeast Asian countries



Note: Brunei Darussalam and Myanmar are not included due to unavailable data.
Source: OECD (2024[14]), *Revenue Statistics in Asia and the Pacific 2024: Tax Revenue Buoyancy in Asia*, https://doi.org/10.1787/e4681bfa-en.

Debt sustainability issues are particularly acute in certain countries. Lao PDR faces debt distress with unsustainable public debt trajectories, requiring primary surpluses of approximately 17% of GDP annually between 2025-2029 to reduce debt to benchmark thresholds – a level neither politically feasible nor socially desirable without harming development prospects (see Box 3.2). Malaysia, the Philippines and Thailand face moderate sovereign stress, with Malaysia's debt servicing costs exceeding the 15% revenue threshold in 2024 despite ongoing reforms (IMF, 2024/25[15]). Indonesia, Singapore and Viet Nam demonstrate generally sound debt sustainability, though Viet Nam faces long-term pressures from aging-related expenditures and climate spending needs (IMF, 2024/25[15]). These debt positions risk crowding out domestic public spending on the blue economy while also increasing borrowing costs that constrain the resources available for the blue economy.

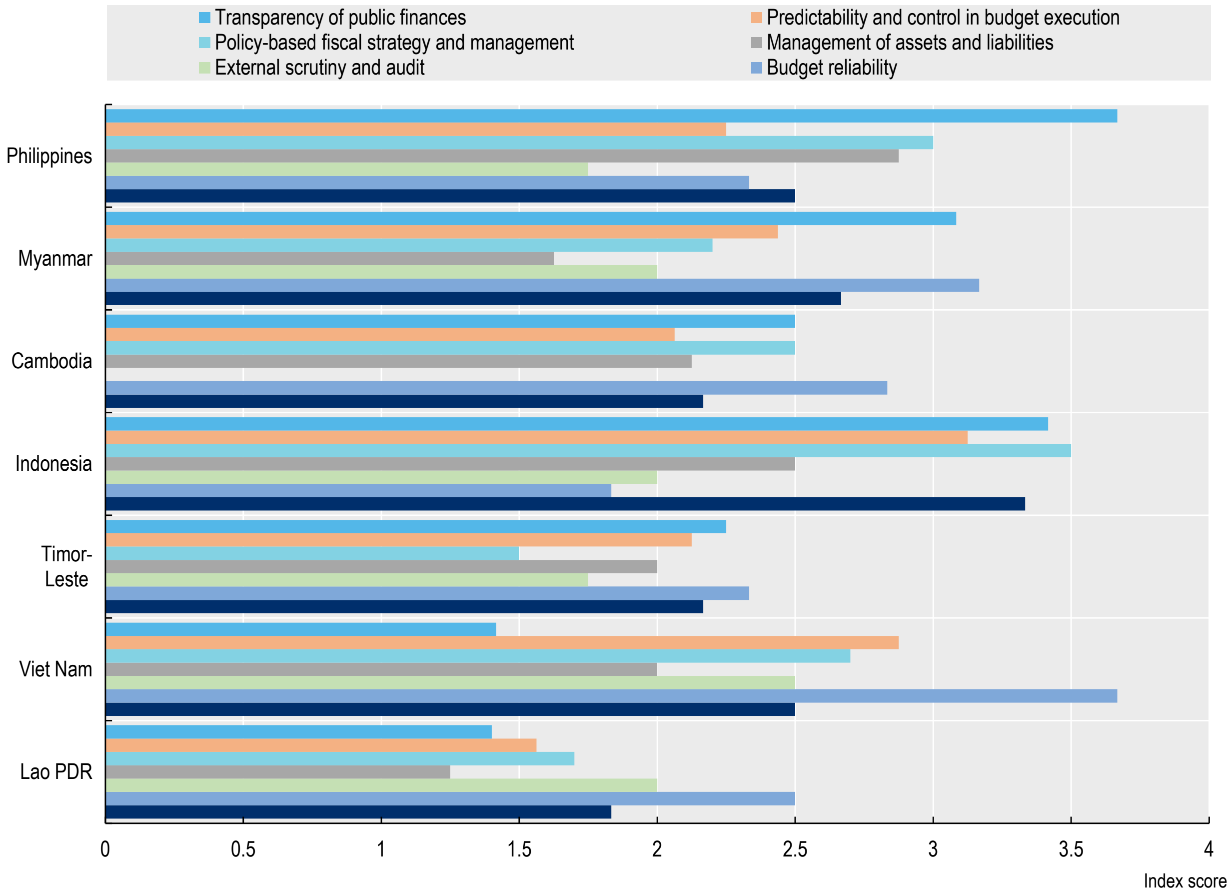
Box 3.2. Debt sustainability and Lao PDR

Lao PDR's debt distress stems from a confluence of structural weaknesses that have created an unsustainable fiscal trajectory. The country exhibits major economic vulnerabilities including significant financing needs, limited access to international markets and minimal international reserves. The economy continues to exhibit signs of stress, with persistent exchange rate depreciation and low foreign exchange reserves leading to persistently high domestic inflation while simultaneously worsening the public debt burden through currency valuation effects. The significant foreign currency exposure of Lao PDR's debt stock means that currency depreciation directly inflates the nominal value of foreign currency-denominated debt, creating a vicious cycle where macroeconomic instability begets further debt deterioration.

Source: IMF (2024[16]), Lao People's Democratic Republic, https://doi.org/10.5089/9798400293283.002.

Public finance management quality also varies substantially. World Bank Public Expenditure and Financial Accountability (PEFA) assessments reveal notable heterogeneity across Southeast Asia's seven pillars of public finance management, with transparency of public finances showing the starkest differences (see Figure 3.5) – some countries approach international good practice while others remain near basic performance levels. This matters for blue economy finance because, as discussed earlier, weak transparency, particularly for ocean-related revenue and expenditures, makes assessing whether available resources actually flow to blue priorities difficult. These disparities also constrain countries' capacity to design, implement and credibly manage revenue-raising fiscal tools such as tourism and shipping taxes or levies, even though such instruments illustrate how blue economy assets can themselves be harnessed to mobilise capital (OECD, 2020[17]).

Figure 3.5. Quality of public finance management according to the PEFA indices



Source: PEFA (2025[18]), PEFA Assessments, https://www.pefa.org/assessments.

It is also worth noting that reforming fiscal instruments so that they no longer incentivise harmful activities can not only support the transition to sustainability, but also create additional fiscal space. An illustrative example is harmful fisheries subsidies. OECD estimates indicate that, between 2020 and 2022, countries and territories accounting for 69% of global capture fisheries production spent USD 10.7 billion per year on public support for their fisheries, and that 65% of this support risks encouraging unsustainable fishing where effective management is lacking (OECD, 2025[19]). Repurposing these funds towards activities such as fish stock assessments, stronger fisheries management, or time bound income support for fishers adversely affected by crises would be a step in the right direction towards a more sustainable and resilient blue economy (OECD, 2025[19]).

3.2.2. International private capital concentrates in financially developed economies

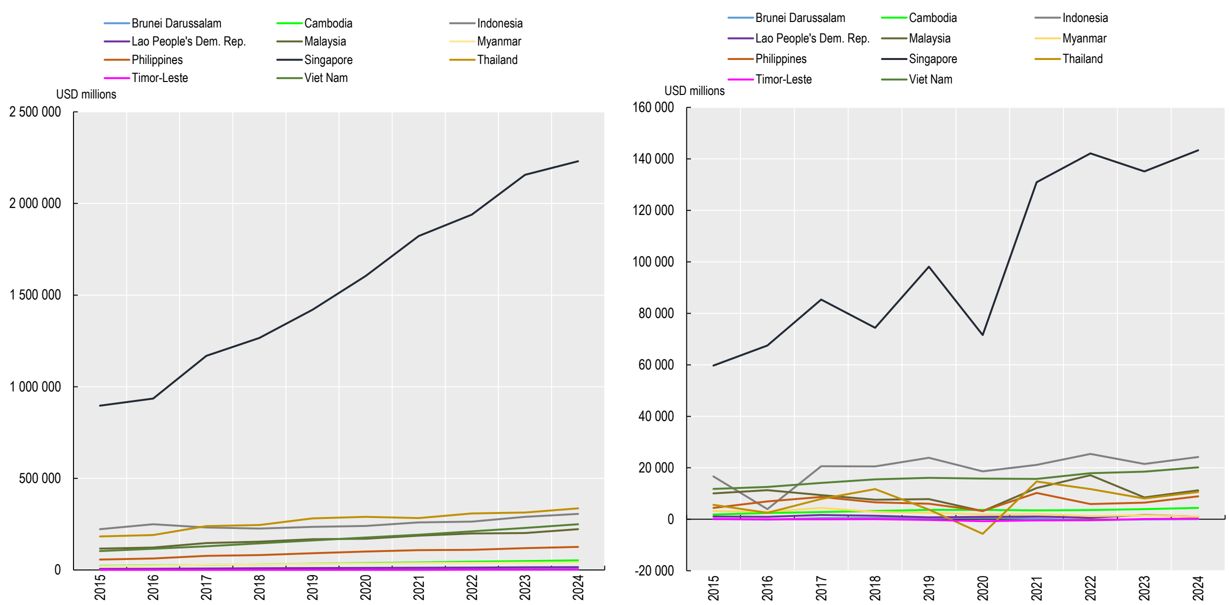
International private capital allocation – both flows and stocks – concentrates heavily in Southeast Asia's most financially developed economies, reflecting risk perceptions embedded in credit ratings and sovereign spreads, as well as varying degrees of financial market openness to foreign investment.

Foreign direct investment (FDI) concentrates heavily in Singapore, which consistently receives over USD 60 billion annually and held over USD 2.2 trillion in FDI stock by 2024 (Figure 3.6). Beyond this financial hub, emerging economies like Indonesia, Malaysia, Thailand and Viet Nam attract USD 10-25 billion per year with steady stock accumulation. Viet Nam recorded the fastest FDI stock growth, from USD 103 billion in 2015 to USD 249 billion in 2024, as firms relocated production from China. Smaller economies – Brunei Darussalam, Cambodia, Lao PDR, Myanmar, the Philippines and Timor-Leste – receive relatively modest inflows, reflecting early-stage integration into international investment networks. International credit follows similar patterns, with Singapore accounting for nearly 40% of total international private credit to Southeast Asia, followed by Indonesia and Malaysia (BIS, 2025[20]).

Creditworthiness explains part of this concentration. Singapore holds the highest credit ratings among Southeast Asian countries, while Indonesia, Malaysia, the Philippines and Thailand maintain investment-grade status (BBB-/Baa3 or above) (Trading Economics, 2026[21]), providing access to institutional investors legally mandated to hold only investment-grade securities. Cambodia, Lao PDR and Viet Nam hold sub-investment grade ratings (Trading Economics, 2026[21]), limiting access to narrower, more expensive investor bases. Sovereign spreads reinforce these patterns: Singapore and Thailand trade well below United States Treasuries, reflecting safe-haven status, while Indonesia and the Philippines show bond yields firmly above United States Treasuries, suggesting a risk premium (Chiang and Chang, 2025[22]).

Figure 3.6. FDI flows to and stocks in Southeast Asia

LHS: stocks; RHS: flows



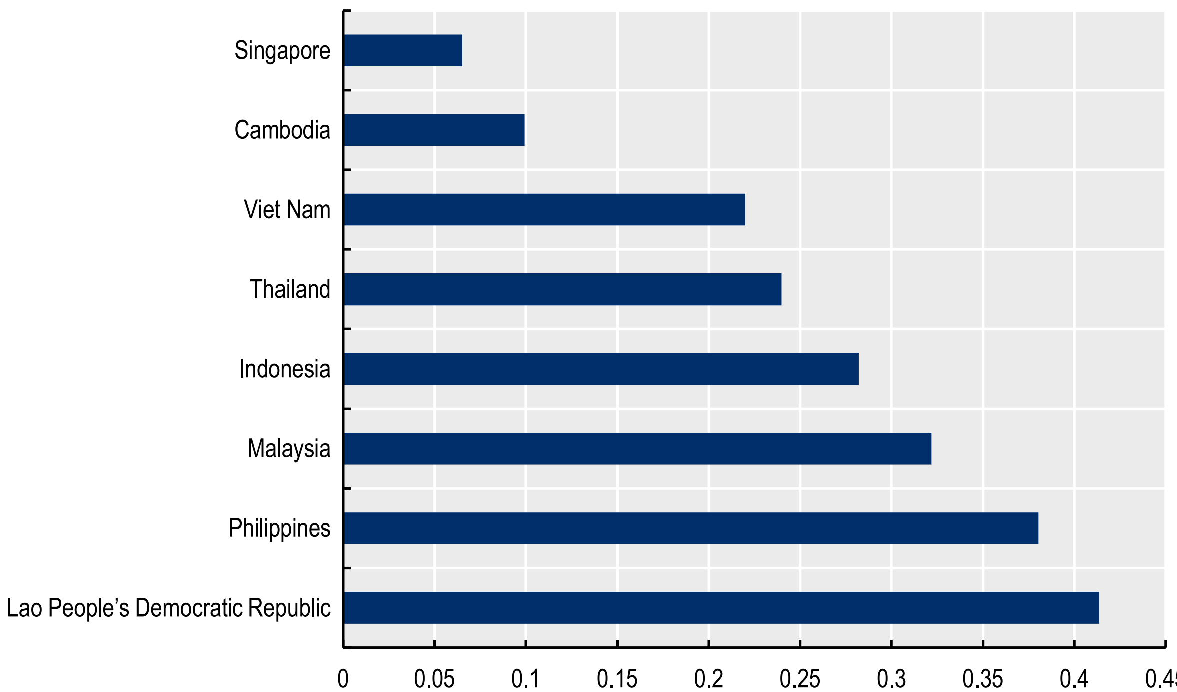
Source: Authors' calculation based on UNCTAD (2025[23]), Foreign direct investment: Inward and outward flows and stock, https://unctadstat.unctad.org/datacentre/dataviewer/US.FdiFlowsStock.

Financial liberalisation also shapes capital allocation significantly. The OECD FDI Restrictiveness Index reveals more than six-fold differences between most and least restrictive countries (see Figure 3.7): Lao PDR stands most restrictive (0.414), while Singapore demonstrates greatest openness (0.065). However, liberalisation itself generates considerable risks (e.g. of banking or currency crises) for lower-income countries (Hamdi and Jlassi, 2014[24]). Effective liberalisation requires preconditions – macroeconomic stability and robust domestic financial institutions (IMF, 2002[25]) – that developing economies may lack. Observed differences in restrictiveness may therefore reflect pragmatic policy choices, echoing underlying disparities in economic development and institutional capacity that themselves help explain the concentration of international private capital in certain markets.

From a blue economy perspective, this concentration implies that large-scale, commercially oriented blue investments – such as port infrastructure, offshore energy and industrial aquaculture – are far more likely to materialise in countries that already combine higher credit ratings, more open FDI regimes and deeper financial markets. This raises the risk that the availability of international private capital reinforces existing disparities in the blue economy, absent financial or policy intervention.

Figure 3.7. FDI Restrictiveness Index across Southeast Asian countries

Lower score implies less restrictive FDI policies



Source: OECD (2025[26]), FDI Regulatory Restrictiveness Index, https://www.oecd.org/en/topics/sub-issues/sustainable-investment/fdi-regulatory-restrictiveness-index.html.

Remittances and philanthropic contributions represent exceptions to the concentration of capital in the most financially developed countries, as they primarily serve household welfare and social objectives – including coastal resilience livelihoods and smaller-scale blue economy activities – rather than commercial returns. Both sources remain quite important in the ASEAN context. Remittances account for a notable share of GDP in several Southeast Asian countries: Timor-Leste (9.3%), the Philippines (8.9%) and Cambodia (6.6%) lead the region (World Bank, 2025[27]). Private philanthropies also channel significant funding to Southeast Asia, with Indonesia (USD 95.8 million) and the Philippines (USD 10.6 million) receiving the largest amounts in 2023 (OECD, 2025[28]). Though small in volume compared to other capital flows, these contributions were almost entirely grants, highlighting their value for activities like marine conservation and ocean science.

3.2.3. Domestic private capital availability is shaped by heterogenous financial market depth and inclusion

Domestic private capital availability is shaped by three interrelated factors: savings capacity (measured by gross domestic savings rates), financial market depth (captured by credit-to-GDP ratios and stock market capitalisation), and financial inclusion (access to banking and credit services across populations). Many factors affecting international capital allocation – particularly credit risk perceptions and policy predictability

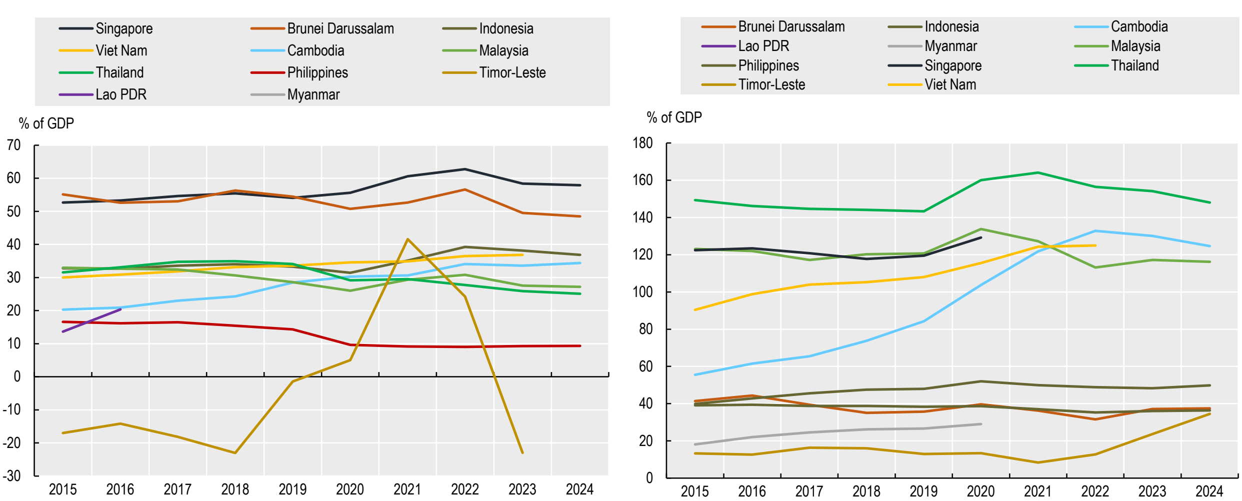
– also shape domestic capital availability, as local investors respond to similar country-level risk assessments.

Savings capacity varies significantly across countries and forms the foundation for domestic capital mobilisation. Brunei Darussalam and Singapore maintain high gross domestic savings rates (above 50% of GDP), indicating large and stable pools of domestic funds. At the other end of the spectrum, the Philippines exhibits persistently low savings (around 9%), constraining the resources available to its financial system. Cambodia and Viet Nam show upward trends in savings, suggesting improving capacity for domestic financial self-reliance.

In general, higher savings support greater domestic credit provision to the private sector, though the relationship is not strictly linear (see Figure 3.8). Malaysia, Singapore and Thailand illustrate how strong savings combined with effective financial intermediation translate into deep financial systems, with private credit often at or above 100% of GDP. By contrast, Myanmar and the Philippines maintain credit-to-GDP ratios below 50%, reflecting both lower savings and less developed banking sectors. Brunei Darussalam is a notable outlier: despite very high savings, its domestic financial system remains relatively shallow, indicating limited intermediation of available funds. Meanwhile, Cambodia and Viet Nam have experienced rapid increases in credit-to-GDP ratios, pointing to fast financial deepening. However, sustained credit growth above GDP can introduce financial stability risks – particularly if lending standards weaken or regulatory capacity does not keep pace.

Figure 3.8. Availability of domestic private capital

LHS: Domestic savings; RHS: Domestic credit to private sector



Source: World Bank (2025[29]), Gross savings (% of GDP), https://data.worldbank.org/indicator/NY.GNS.ICTR.ZS; World Bank (2025[30]), Domestic credit to private sector (% of GDP), https://data.worldbank.org/indicator/FS.AST.PRVT.GD.ZS.

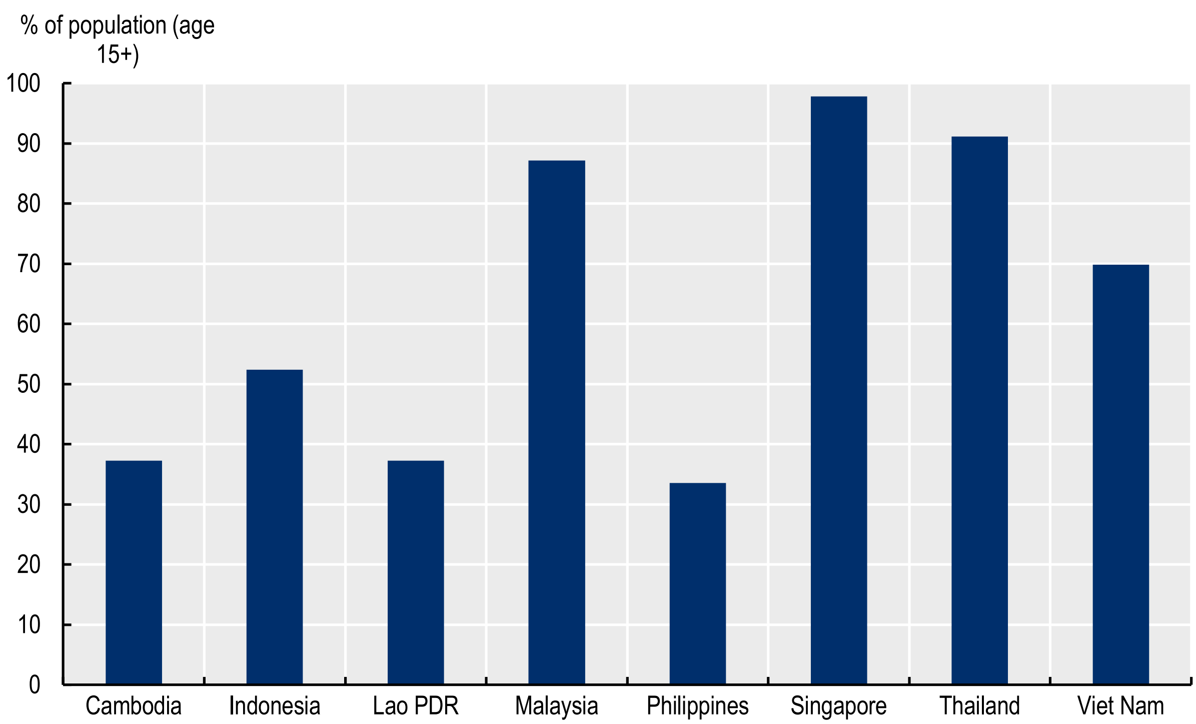
While credit-to-GDP ratios broadly capture the scale of debt-based – primarily bank-led – financing, stock market capitalisation reflects the role of equity-based, market financing. Across countries, the latter channel of capital allocation shows a similarly uneven pattern of development. Singapore and Malaysia possess highly capitalised equity markets (well over 100% of GDP) signalling strong equity financing capacity and well-developed capital markets (World Bank, 2025[31]). Indonesia and the Philippines show market capitalisation significantly higher than their domestic credit ratios, suggesting a relatively greater role for equity financing compared to bank lending (World Bank, 2025[31]; World Bank, 2025[30]). In contrast, in Thailand and Viet Nam, domestic credit exceeds market capitalisation, suggesting that bank lending plays a more prominent role in private sector financing. Cambodia's very small stock market capitalisation further underscores the uneven development of capital markets across the region.

Finally, financial inclusion determines whether available capital is broadly accessible. As shown in Figure 3.9, Malaysia, Singapore and Thailand couple deep financial markets with near-universal inclusion (over 87% banked adult population). Viet Nam, with around 70% inclusion, has relatively high access but still faces gaps. Inclusion levels are significantly lower in Indonesia (52%), Cambodia (37%) and the Philippines (34%), suggesting that large segments of the population remain underserved.

Differences in domestic private capital availability have direct implications for the blue economy. Where domestic credit markets are shallow and equity finance limited, even viable blue economy enterprises – such as sustainable aquaculture operations or eco-tourism businesses – may struggle to secure the capital needed to invest and scale. Separately, low financial inclusion means that coastal communities and micro-enterprises often lack access to basic financial services: small-scale fishers cannot obtain loans for improved gear or climate adaptation, and informal blue economy actors remain entirely outside formal credit systems. In both cases, the availability of domestic private capital for the blue economy is constrained.

Figure 3.9. Financial inclusion across Southeast Asia

Population with bank or similar financial institution account



Source: World Bank (2025[32]),Global Findex Database, https://databank.worldbank.org/source/global-findex-database.

3.3. Even when available, capital may not always flow to Southeast Asia's blue economy due to policy gaps

The overall envelope of capital available for the blue economy – and its sources – depends on generic economic factors. However, whether available capital actually flows to blue economy priorities is determined by whether these priorities represent viable and attractive investment propositions. Sound policies and regulations at the macro-level play a critical role by directing capital flows, reducing uncertainty and information asymmetries, and establishing a basis for investable projects (Halland et al., 2021[33]). Importantly, policy signals shape not only where capital should flow but also which activities should be excluded – whether for economic, environmental, or social reasons.

Singapore's Second Nationally Determined Contribution (NDC) illustrates this signalling function. The country explicitly states that marine, hydroelectric and conventional geothermal renewable energy options are unviable due to structural constraints including low wind speeds, narrow tidal range, a lack of fast-flowing river systems, and an absence of near-surface geothermal resources (Republic of Singapore, 2025[34]). By explicitly excluding these sectors, the policy statement deters misallocated investment and channels capital more efficiently towards feasible blue economy opportunities.

Yet such clarity in policy signalling is uneven across Southeast Asia. Nascent – albeit emerging – blue economy policy frameworks – and underdeveloped financial policy levers represent a constraint for the flow of available capital to the region's blue economy ambitions.

3.3.1. Certain Southeast Asian countries have advanced blue economy policies whilst others are in more nascent stages – though momentum is accelerating

Clear and integrated national blue economy policy frameworks are essential for channelling finance towards sustainable ocean development. At minimum, they signal national priorities to investors, donors and financial institutions. More detailed frameworks outline objectives, quantify financing requirements and articulate strategies that align public resources and co-ordinate donor funding. Marine spatial planning (see Box 4.2 in Chapter 4), for instance, helps identify and prioritise investments whilst mobilising donor finance. Whilst the ASEAN Blue Economy Framework provides regional direction, national-level clarity remains critical given substantial heterogeneity across Southeast Asian countries (see Chapters 1 and 3).

National policy framework development in Southeast Asia remains uneven (Table 3.2). Indonesia and Singapore have advanced furthest, with Indonesia explicitly positioning the blue economy as an engine for inclusive growth aligned with national development plans (Government of Indonesia, 2024[35]). Others – including Cambodia, Lao PDR and Timor-Leste – are currently developing the policies and institutions necessary for sound blue economy governance.

Table 3.2. Level of blue economy policy development across Southeast Asian countries

Country	Policy development
Brunei Darussalam	Solid institutional foundations and poised to accelerate progress
Cambodia	Laying the groundwork for long-term success by strengthening institutional capacity and building momentum
Indonesia	Trailblazer in the region, combining strong institutional capacity with rapid implementation progress
Lao PDR	Laying the groundwork for long-term success by strengthening institutional capacity and building momentum
Malaysia	Solid institutional foundations and poised to accelerate progress
Myanmar	Laying the groundwork for long-term success by strengthening institutional capacity and building momentum
The Philippines	Solid institutional foundations and poised to accelerate progress
Singapore	Trailblazer in the region, combining strong institutional capacity with rapid implementation progress
Thailand	Solid institutional foundations and poised to accelerate progress
Timor-Leste	Laying the groundwork for long-term success by strengthening institutional capacity and building momentum
Viet Nam	Solid institutional foundations and poised to accelerate progress

Source: Adapted from ASEAN Blue Wealth (2025[36]), *National Snapshots*, https://aseanbluewealth.com/asean-snapshots/.

Recent developments suggest momentum is building. Since first referencing the blue economy in 2020, Brunei Darussalam has integrated it into its Twelfth National Development Plan (ASEAN Blue Wealth, 2025[36]). The Philippines enacted the Blue Economy Act in 2024, integrating blue economy objectives into the Philippine Development Plan 2023-2028 (ASEAN Blue Wealth, 2025[36]). Malaysia incorporated the blue economy into the 12th Malaysia Plan (2021-2025) and elevated it to a key pillar in the 13th Malaysia Plan (2026-2030) (ASEAN Blue Wealth, 2025[36]).

The latest NDCs also reflect this uneven but accelerating policy attention. Following the first Global Stocktake, 61 of 66 updated NDCs globally included at least one ocean-based action (Khan et al., 2025[37]). Several Southeast Asian countries have incorporated ocean-based mitigation and adaptation measures into their 2025 NDCs (Table 3.3). Indonesia has adopted the most comprehensive approach – consistent with its advanced blue economy roadmap – with commitments spanning blue carbon ecosystems, strengthened marine spatial planning, expanded marine protected areas, ocean accounting and seagrass mapping (Republic of Indonesia, 2025[38]). Looking ahead, Singapore and Cambodia have joined the Blue NDC Challenge launched at the Third United Nations Ocean Conference, signalling growing regional commitment to placing ocean priorities at the heart of climate action (WRI, 2025[39]).

Table 3.3. Southeast Asian countries with ocean-related measures in their 2025 NDCs

	Countries
Mitigation measures	Indonesia, Malaysia, Singapore, Thailand, Cambodia
Adaptation measures	Indonesia, Malaysia, Singapore, Cambodia, Brunei Darussalam

Note: Only includes Southeast Asian countries who submitted their 2025 NDCs by 6 November 2025: Brunei Darussalam, Cambodia, Indonesia, Malaysia, Singapore and Thailand.
Source: Khan et al. (2025[37]), *The State of* Ocean-*Based Climate Action in 2025 Nationally Determined Contributions: A Preliminary Update*, https://doi.org/10.46830/wriwp.25.00004

3.3.2. Financial standards for blue economy investments and expenditures are underdeveloped

Financial guidelines and standards, alongside disclosure requirements, constitute critical mechanisms for channelling private capital towards policy objectives – particularly as blue economy sectors navigate different transitions (e.g. decarbonisation). Taxonomies operate by classifying economic activities and establishing performance thresholds that delineate sustainable from conventional investments, thereby reducing search costs for investors seeking eligible projects (Inderst and Opp, 2025[40]). Disclosure requirements complement this by mandating transparent reporting of sustainability information, which enhances market efficiency and mitigates "bluewashing"[1] risks that erode investor confidence (Inderst and Opp, 2025[40]). Together, these instruments ensure alignment between financial flows and substantive blue economy outcomes.

At the regional level, ASEAN has developed harmonised sustainable finance frameworks. The ASEAN Capital Markets Forum (ACMF) Action Plan 2026-2030 anchors this effort through two core instruments: the ASEAN Taxonomy for Sustainable Finance, which establishes regional classification criteria for sustainable investments, and the ASEAN Transition Finance Guidance, which provides credibility mechanisms for entities pursuing decarbonisation pathways to access capital markets (ASEAN Capital Markets Forum, 2025[41]). ASEAN has further standardised specific financing instruments through the ASEAN Green Bond Standards (2017), ASEAN Social Bond Standards (2018), ASEAN Sustainability Bond Standards (2018), and ASEAN Sustainability-Linked Bond Standards (2022) – all calibrated to international principles while adapted to regional market conditions (OECD, 2024[42]).

However, the blue economy is largely missing from these frameworks. The ACMF Action Plan 2026-2030 makes no explicit reference to the ASEAN Blue Economy Framework. It does, nonetheless, offer some entry points for strengthening support for the blue economy. The Plan includes the creation of the mARs Guide, a methodology for classifying climate change adaptation and resilience investments under the ASEAN Taxonomy. Such classification allows projects to access capital from labelled debt instruments (e.g. sustainable bonds) (ASEAN Capital Markets Forum, 2025[43]). Since much of ASEAN's adaptation needs are coastal, the Guide is well-placed to expand access to capital for coastal and marine projects, provided that its future iterations more explicitly recognise the blue economy.

Moreover, while the ASEAN Taxonomy for Sustainable Finance includes Environmental Objective 3 ("Protection of Healthy Ecosystems and Biodiversity"), it lacks sector-specific technical screening criteria for blue economy activities such as sustainable fisheries, aquaculture, or marine conservation (ASEAN Capital Markets Forum, 2024[44]). The ASEAN Transition Finance Guidance similarly omits blue economy-specific transition pathways (ASEAN Capital Markets Forum, 2024[45]). Regional bond standards provide unified frameworks aligned with international principles but do not explicitly delineate blue economy project categories. At the global level, tools such as the United Nations Environment Programme Finance Initiative (UNEP FI) Sustainable Blue Economy Finance Principles, a comprehensive framework for aligning financial institutions with SDG 14 (UNEP, 2018[46]), and the global practitioner's guide for bonds to finance the sustainable blue economy, developed jointly by the International Capital Markets Association, International Finance Corporation, Asian Development Bank, United Nations Global Compact, and UNEP FI (Asian Development Bank, 2023[47]), provide useful reference points for blue economy financing.

Disclosure requirements exhibit similar patterns. At the global level, climate and nature-related disclosure frameworks provide foundational architecture. The Task Force on Climate-related Financial Disclosures (TCFD) established the template for systematic climate risk reporting, which the Taskforce on Nature-related Financial Disclosures extended in September 2023 to encompass nature-related risks and dependencies, including marine and ocean ecosystems (Taskforce on Nature-related Financial Disclosures, 2023[48]). Within ASEAN, however, implementation is characterised by cross-country heterogeneity and minimal blue economy integration. Five member states (Indonesia, Malaysia, Singapore, Thailand, and Viet Nam) have mandatory requirements for corporate sustainability disclosure; three others (Cambodia, Lao PDR, and the Philippines) maintain softer regimes (OECD, 2024[42]). Implementation timelines diverge markedly. Indonesia phased in requirements between 2017 and 2025, while Thailand adopted its "One Report" standard in 2022. Reporting frameworks also vary. Malaysia and Singapore require climate-related disclosures consistent with TCFD recommendations, whereas Viet Nam's 2016 guidance references the Global Reporting Initiative's principles (OECD, 2024[42]). Across jurisdictions, disclosure focuses predominantly on climate change-related metrics (e.g. greenhouse gas emissions, energy consumption) (OECD, 2024[42]), with standardised approaches for blue economy-specific indicators – such as marine ecosystem health, sustainable fisheries performance, or coastal community impacts – remaining underdeveloped. As a result, policymakers and investors lack the information needed to consistently compare the sustainability and effectiveness of blue economy activities in the region.

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Note

[1] Bluewashing refers to firms leveraging sustainability initiatives to signal strong environmental, ethical or human rights performance, while making limited substantive changes to their actual practices (Berliner and Prakash, 2014[49]).

4 Financing solutions: Tools and lessons learned

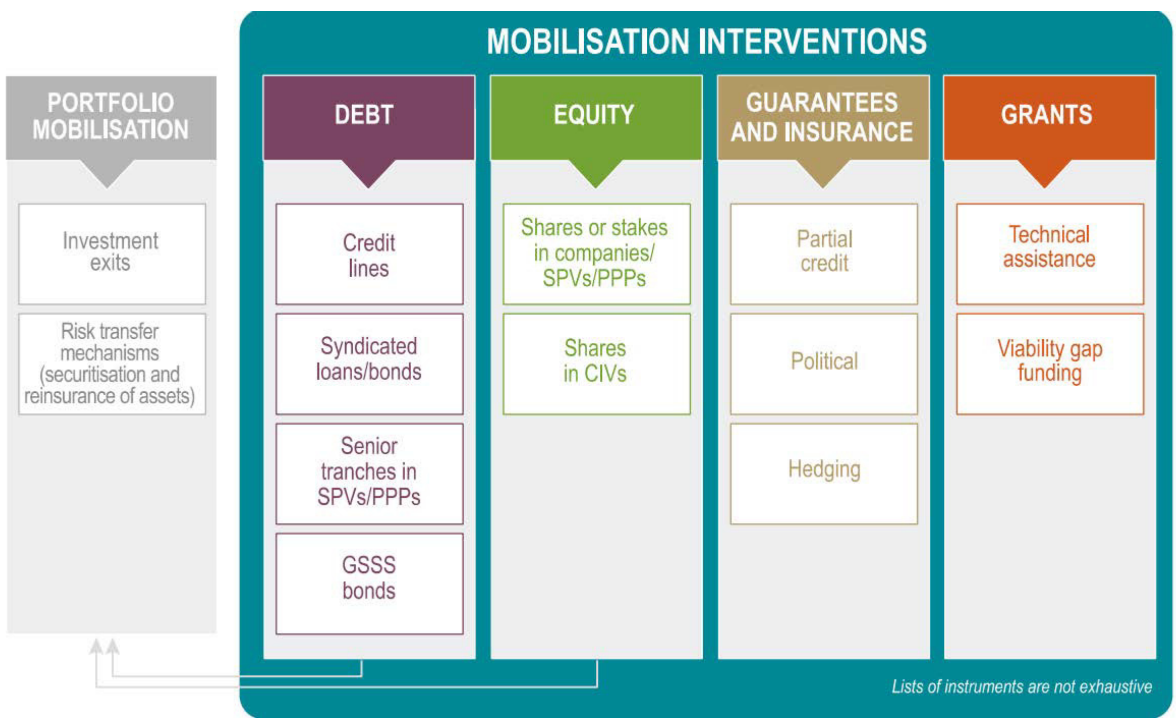
Macroeconomic conditions create uneven capital availability across Southeast Asia, while policy gaps impede capital flow to blue economy priorities. Addressing these challenges requires not only strengthening macroeconomic fundamentals and policy frameworks but also deploying financial mechanisms that can work within – and around – existing constraints. This chapter explores these mechanisms, starting with blended finance, before moving to the range of blue finance instruments. For each instrument, the analysis explores its theoretical use case, assesses its application in Southeast Asia to date and draws on global experience to identify critical success factors and persistent barriers – including their continued reliance on development assistance.

4.1. Blended finance can mitigate individual capital availability challenges but has achieved limited success

4.1.1. Blended finance can help de-risk blue economy investments by absorbing macroeconomic and project-level risks

Blended finance, defined by the OECD Development Assistance Committee as the strategic use of development finance for the mobilisation of additional finance towards sustainable development in developing countries (OECD, 2018[1]), employs various instruments to improve investment risk-return profiles. As shown in Figure 4.1, key mechanisms include guarantees that provide protection against losses; equity investments through structured funds with layered risk tranches; concessional loans and credit lines; and technical assistance to build project pipelines and enabling environments. By mitigating market failures and narrowing perceived risk gaps, these instruments lower barriers to investment. In particular, public finance absorbs risks that private investors are unable or unwilling to bear – such as political instability, currency volatility, regulatory uncertainty, or long payback periods.

Figure 4.1. Blended finance instruments



Source: OECD (2025[2]), *OECD DAC* Blended *Finance Guidance 2025*, https://doi.org/10.1787/e4a13d2c-en.

For the blue economy, blended finance offers particular relevance given the dual nature of financing barriers. At the macroeconomic level, many coastal developing countries face shallow capital markets, limited fiscal capacity and broader investment climate challenges that constrain all financing flows. At the project level, blue economy investments face specific barriers: complex infrastructure (e.g. ports, harbours) requiring large capital outlays; long investment horizons with uncertain revenue streams; nascent technologies with unproven business models; and difficulty establishing clear business cases despite strong development rationales. Blended finance can address both levels. For instance, guarantees and insurance can be used to mitigate country and currency risks; technical assistance can help to strengthen the enabling environment; structuring public-private partnerships with appropriate risk allocation can make individual projects bankable.

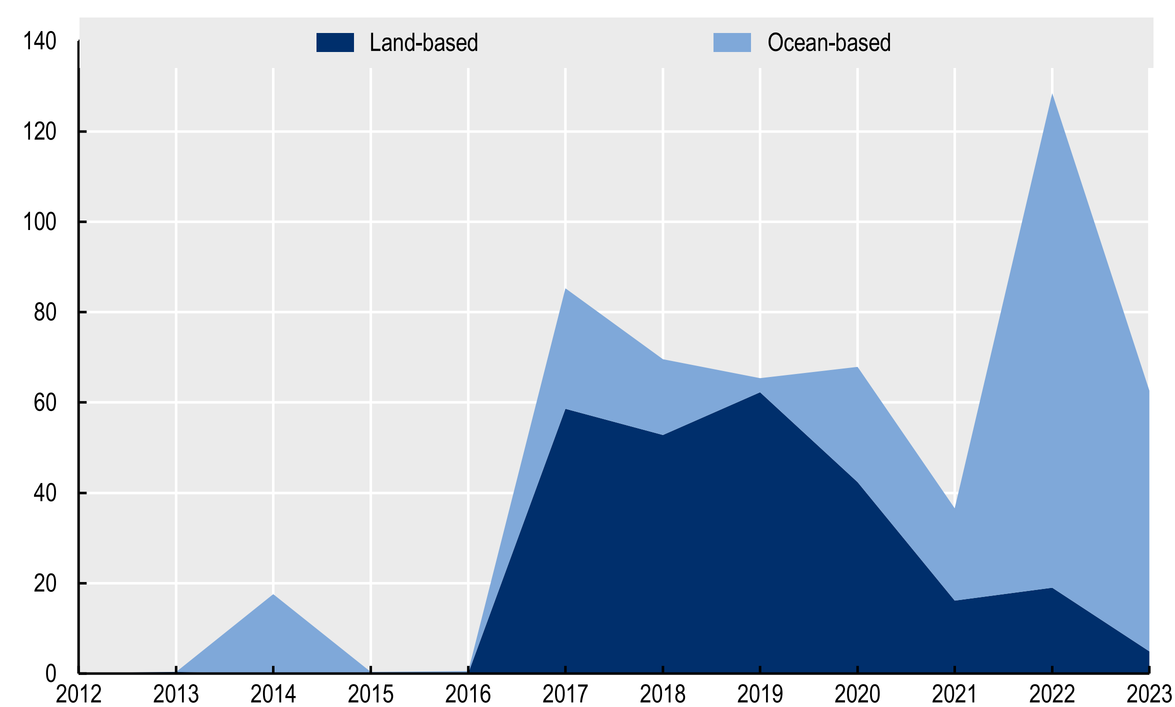
4.1.2. Blended finance deployment in Southeast Asia shows limited scale and is concentrated across a narrow set of geographies, sectors, and instruments

Despite renewed interest in the mobilisation agenda (UN, 2025[3]), blended finance has, globally, encountered challenges in mobilising adequate and effective capital for sustainable development. OECD analysis suggests that blended finance mobilised only USD 70 billion in 2023 and that the approach remains a "cottage" industry typified by fragmented and bespoke interventions (OECD, 2025[2]). Importantly, to date, blended finance has mobilised relatively low shares of private capital in challenging contexts including least developed countries and fragile situations, and in sectors like education and social services (OECD, 2025[2]).

The systematic challenges are, to some degree, apparent in the data on private finance mobilised for the blue economy in Southeast Asia. These data consist of two main indicators: private finance mobilised for ocean-based activities and that for land-based, but ocean-relevant, activities. Henceforth, the sum of these indicators is referred to as private finance mobilised for ocean-related activities. As shown in Figure 4.2, both streams began modestly before jumping significantly after 2017, suggesting growing interest. However, even at their peak, the total private finance mobilised by official development assistance (ODA) for ocean activities and land-based ocean-relevant activities in Southeast Asia amounted to only USD 128.4 million. Importantly, both indicators exhibit volatile and episodic tendencies. Private finance mobilised for ocean-based activities, for instance, spiked in 2022, but dropped the following year. Similarly, that for land-based ocean-relevant activities peaked in 2019, but dropped off significantly by 2021.

Figure 4.2. Private finance mobilised by ODA for ocean-related activities in Southeast Asia

USD millions

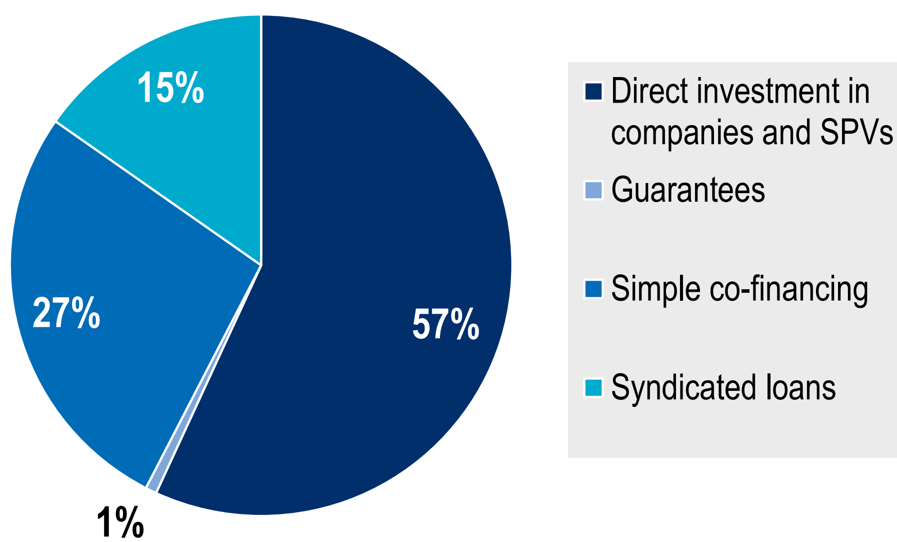


Source: Authors' calculations based on OECD (2025[4]), *Data Platform on Development Finance for the Sustainable Ocean Economy*, https://oecd-main.shinyapps.io/ocean/.

There is also notable geographical and sectoral concentration in private finance mobilised. Private finance mobilised for ocean-based activities skews heavily towards Indonesia and Thailand – which represent nearly 70% of the total amount mobilised between 2014 and 2023. Meanwhile, the private finance mobilised for ocean-relevant land-based activities skews principally to Indonesia and the Philippines. There is also heavy concentration of ocean-related private finance mobilised in infrastructure-oriented sectors such as transport and storage, and water supply and sanitation. Sectors like fisheries attract some investment, but marine protection and conservation remain marginal. While the precise factors underlying this skew are difficult to determine, it is plausible that the concentration reflects a common pattern of blended finance – namely, that it tends to be more easily deployable in sectors and contexts with fewer barriers to private investment. For example, the strong representation of Indonesia and the Philippines may indicate a bias toward countries with more developed blue economy policy frameworks and institutional capacity.

The use of financial instruments in mobilising private finance for the blue economy in Southeast Asia remains relatively narrow, dominated by direct investments (57%), simple co-financing (27%) and syndicated loans (15%) (see Figure 4.3). The limited use of guarantees and other risk-sharing mechanisms suggests that most mobilisation relies on conventional debt and equity structures, rather than the full spectrum of available tools.

Figure 4.3. Blended finance instruments for ocean-related activities in Southeast Asia



Source: Authors' calculations based on OECD (2025[4]), *Data* Platform *on Development Finance for the Sustainable Ocean Economy*, https://oecd-main.shinyapps.io/ocean/.

4.2. Innovative instruments could ease capital constraints, but adoption remains patchy in Southeast Asia

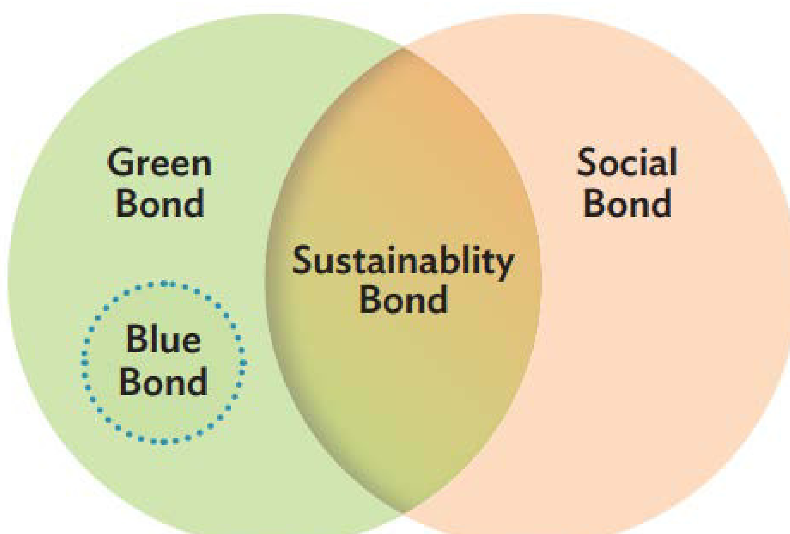
4.2.1. Blue debt instruments come in different forms, with blue use-of-proceed bonds being the most prominent in Southeast Asia

Traditional debt instruments can be adapted to support the blue economy

Blue debt instruments customise traditional bonds and loans to channel capital toward ocean- and water-related priorities (World Bank, 2025[5]). They come in two main forms, use-of-proceeds or performance-based, each addressing capital flow constraints identified in Chapter 3 in different ways.

Blue use-of-proceeds instruments (often called "thematic" or labelled debt) require that net proceeds of a bond or loan are exclusively allocated to a predetermined set of eligible blue economy projects, following strict criteria for project evaluation, selection, management of proceeds and reporting. "Blue bonds" are a prime example of this. These bonds mirror green bonds but confine the eligible basket of activities to ocean- or water-related activities (e.g. sustainable aquaculture, coastal resilience, marine pollution control, marine renewable energy) in line with frameworks such as the International Capital Market Association's Blue Bond guidance (see Figure 4.4) and the International Finance Corporation's (IFC) Blue Finance Guidance (IFC, 2025[6]).

Figure 4.4. Blue bonds and its relation to other use-of-proceed bonds



Source: ICMA (2023[7]), *Bonds to Finance the Sustainable Blue Economy*, https://www.icmagroup.org/assets/documents/Sustainable-finance/Bonds-to-Finance-the-Sustainable-Blue-Economy-a-Practitioners-Guide-September-2023.pdf.

Performance-based instruments shift the focus from how capital is spent to measurable results (outcomes). Unlike use-of-proceeds instruments, these structures focus on the actual achievement of impact. Within this category, two main structures exist:

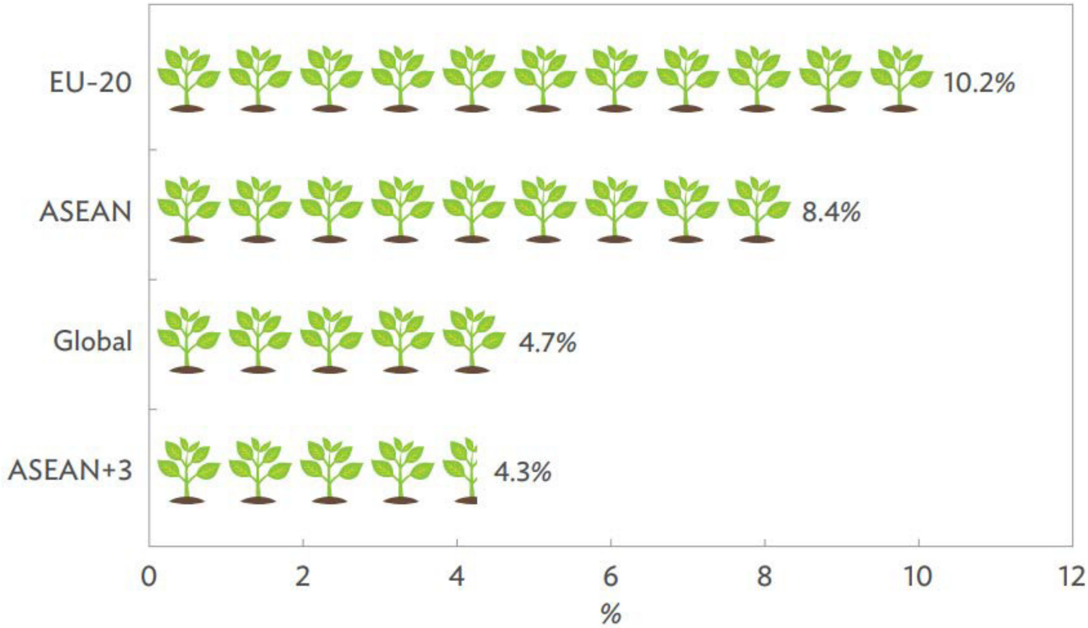
- **Sustainability-linked bonds (SLB):** SLBs are general-purpose debt instruments where the issuer commits to achieving predefined sustainability performance targets, measured via key performance indicators (ICMA, 2023[7]). If the issuer fails to hit those targets, financial costs rise – for example, via margin step-ups or coupon penalties – linking the cost of capital to the overall sustainability trajectory, but not to the outcomes of any ring-fenced projects. For instance, an SLB could penalise a shipping company for failing to meet its fleet-wide greenhouse gas reduction target.
- **Outcome-based:** These instruments link the repayment of principal and investor returns directly to the verified achievement of pre-agreed project outcomes, rather than the mere allocation of funds. In a blue economy context, an outcome-based bond might tie returns to the rate of marine ecosystem recovery or specific reductions in plastic pollution.

In theory, use-of-proceeds and performance-based instruments can address constraints in both the availability and flow of capital to the blue economy. On the availability side, blue bonds help tap investors seeking sustainable assets with transparent allocation, while performance-based instruments attract impact investors willing to accept outcome-linked returns in exchange for measurable environmental results. On the flow side, use-of-proceeds instruments channel capital through ring-fencing and accountability mechanisms tied to eligible activities, while performance-based instruments create ongoing incentives by linking financial terms to verified outcomes shifting monitoring responsibility to investors whose returns depend on target achievement (CGD, n.d.[8]). Blue debt instruments offer flexibility across issuers and structures. Sovereigns, sub-sovereigns, multilaterals, development banks, municipalities, and corporates (both financial and non-financial) can issue these instruments, and design choices (i.e. criteria for proceeds use or sustainability indicators) can accommodate diverse blue economy sectors.

Several early cases are emerging in Southeast Asia, paralleling the steady growth of the region's sustainable bond market

By mid-2025, Association of Southeast Asian Nations' (ASEAN) sustainable bond market was gaining a stronger presence within the broader Asia region. At the end of June, sustainable bonds outstanding across ASEAN+3 totalled USD 955.3 billion, representing 18.3% of the global sustainable bond market of USD 5.2 trillion, supported by robust issuance across most regional markets amid continued monetary easing. Within this total, ASEAN markets accounted for 10.5%, up from 8.3% in mid-2021. In Q2 2025, sustainable bonds made up 8.4% of total bond issuance in ASEAN countries, compared with 10.2% in the EU-20 and 4.7% globally (see Figure 4.5). While green bonds still dominate ASEAN+3's sustainable bond market, their share declined from 68.9% in June 2021 to 57.9% in June 2025 as other bond types continued to expand, led by social and transition bonds. The majority of sustainable bond issuance in ASEAN+3 comes from corporates, whereas in ASEAN itself, most issuance is from the public sector (ADB/ABO, 2025[9]).

Figure 4.5. Sustainable bond Issuance as Share of Total Bond Issuance in Q2 2025



Source: ADB/ABO (2025[9]), ASEAN+3 Sustainable Bonds Highlights, https://asianbondsonline.adb.org/newsletters/aboesg202509.pdf.

Building on this momentum, several case studies of use-of-proceeds and blue performance-based instruments have emerged in Southeast Asia. The Asian Development Bank's (ADB) blue bond issuance in 2021 provided a regional benchmark. ADB's inaugural dual-tranche deal, worth roughly USD 302 million, was issued under its *Green and Blue Bond Framework* to finance projects in marine ecosystem restoration, pollution control, sustainable aquaculture and marine renewable energy (ADB, 2021[10]). The issuance attracted major institutional investors such as Dai-ichi Life and Meiji Yasuda Life, with arrangers Citigroup Global Markets and Crédit Agricole. By 2025, ADB's cumulative blue bond issuance had exceeded USD 425 million, reinforcing its role as a cornerstone in building credible blue finance standards and investor confidence across Asia and the Pacific.

In the Philippines, BDO Unibank issued the country's first blue bond in 2022 – a USD 100 million instrument dedicated to reducing marine plastic pollution – with the IFC fully subscribing and providing technical guidance on eligible project selection (IFC, 2022[11]). In Thailand, the private sector followed suit: TMBThanachart Bank issued a domestic blue bond backed by IFC (IFC, 2022[12]), while Indorama Ventures secured a USD 300 million blue loan from a consortium led by IFC and ADB to expand their plastics recycling capacity and reduce marine plastic leakage. At the sovereign level, Indonesia broadened the market in 2023 with its first sovereign blue bond (JPY 20.7 billion), targeting investments in sustainable fisheries, coastal restoration and maritime-sector development (World Bank, 2025[5]).

Meanwhile, blue SLBs are generally nascent in Southeast Asia. For example, Thai Union Group issued a sustainability-linked bond in July 2021 for approximately THB 5 billion (around USD 155-170 million at the time). The bond is linked to performance targets related to ocean health, including enhanced monitoring and traceability of wild-caught tuna supply chains (Thai Union Group, 2021[13]). Though not specific to the blue economy, Thailand also issued a sovereign SLB in 2024 – marking the first sovereign SLB in the region (Thailand PMDO, 2025[14]). Initially targeting an issuance of THB 20 billion and ultimately increased to THB 30 billion in response to investor demand, the bond focuses on objectives such as greenhouse gas emissions reductions and clean mobility (Thailand PMDO, 2025[14]).

There are also early cases of blue outcome bonds in Southeast Asia. Indonesia is piloting a coral reef bond, with the aim of supporting marine protected areas (MPAs) (World Bank, n.d.[15]). The bond will focus on three MPAs covering approximately 1.9 million hectares, with the objective of improving coral reef health through indicators such as fish biomass, stakeholder engagement, sustainable fishing practices and the enforcement of zoning regulations. Investors will receive payments based on verified conservation outcomes, with funding sourced from the Global Environment Facility and BNP Paribas. The bond was proposed at the Third United Nations Ocean Conference and is expected to raise between USD 100 and 120 million (World Bank, n.d.[15]).

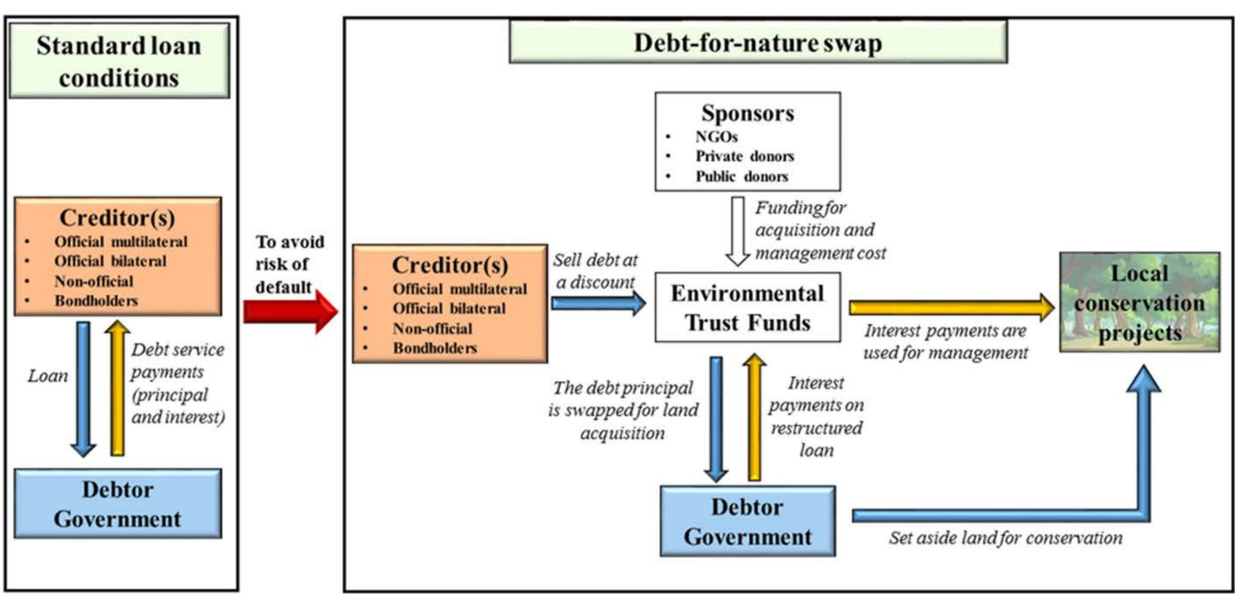
4.2.2. Debt-for-nature swaps can help ease domestic public finance constraints, but have had limited regional application

> *Debt-for-nature swaps serve a dual purpose of funding ocean conservation and reducing fiscal stress*

Debt-for-nature swaps (DFNS) represent one of the most sophisticated instruments in blue finance, simultaneously addressing sovereign fiscal stress and ocean conservation financing gaps. First pioneered in the 1980s for terrestrial conservation, these mechanisms have been adapted for marine environments over the past decade, with the blue variant emerging as a powerful tool for ocean-dependent nations facing dual pressures of debt burden and ecosystem degradation (Fedosova and Turner, 2023[16]).

The financial architecture (see Figure 4.6) typically involves buying back a portion of a country's existing sovereign debt at a discount, then refinancing the remaining obligation at lower interest rates through a new "blue bond" issuance. Credit enhancement from development finance institutions or non-governmental organisations (NGO), usually in the form of guarantees or political risk insurance, makes this refinancing viable by improving the credit profile and reducing borrowing costs (Albinet, Chekir and Kessler, 2024[17]). The fiscal savings generated from lower debt service are then channelled into long-term marine protection activities (Benzaken et al., 2022[18]).

Figure 4.6. Debt-for-nature swap architecture



Source: Nedopil, Yue and Hughes (2023[19]), “Are debt-for-nature swaps scalable: Which nature, how much debt, and who pays?”, https://doi.org/10.1007/s13280-023-01914-4.

The instrument's innovation lies in converting what appears to be a zero-sum trade-off – debt relief versus conservation spending – into a positive-sum transaction where both fiscal sustainability and environmental outcomes improve simultaneously. For heavily indebted nations facing binding constraints on domestic public finance due to debt distress, DFNS can provide immediate fiscal relief whilst maintaining market access and credit ratings – addressing the dual challenge of inadequate fiscal space and the risk of credit downgrades that would further restrict capital availability. At the same time, DFNS can help generate sustained funding for marine protection and sustainable livelihoods, addressing the chronic underfunding of ocean conservation, which typically relies on grant-based or public funds. Moreover, the long-term nature of conservation commitments (discussed below) – typically a feature of DFNS – can help strengthen the overall policy framework for ocean management, which can help attract capital from other sources.

Debt-for-nature-swaps are marginal in Southeast Asia.

No large-scale sovereign blue debt-for-nature swaps, comparable to the Caribbean and Latin American models, have been executed in Southeast Asia. However, the region has some experience with smaller-scale debt-for-nature mechanisms through the United States Tropical Forest and Coral Reef Conservation Act (TFCCA).

Indonesia represents the most significant case study for debt-for-nature mechanisms in Southeast Asia, having participated in four separate swaps with the United States totalling approximately USD 105 million since 2009. The earliest swap in 2009 redirected nearly USD 30 million of Indonesian debt over eight years toward protecting Sumatra's tropical forests – home to critically endangered Sumatran tigers, elephants, rhinos and orangutans – marking the largest TFCCA transaction at that time (US Department of Treasury, 2009[20]). Subsequent swaps in 2011 (US Department of State, 2011[21]) and 2014 (US Department of State, 2014[22]) generated an additional nearly USD 40 million for forest conservation, collectively demonstrating sustained bilateral commitment to this mechanism.

Most relevant to the blue economy, Indonesia, in 2024, signed a USD 35 million debt-for-nature swap specifically targeting coral reef conservation (US Department of Treasury, 2024[23]). Finalised in January 2025, this landmark transaction will direct funds over nine years to the Bird's Head and Lesser Sunda-Banda seascapes, two of the planet's most biodiverse reef systems within the Coral Triangle. The swap is made possible through contributions from the U.S. Government under TFCCA, with Conservation International and The Nature Conservancy providing additional support. Funds are managed through a Supervisory Committee led by Indonesia's Ministry of Maritime Affairs and Fisheries, consisting of the Ministry of Finance and participating NGOs, which will distribute grants to local organisations for coral reef protection and restoration, as well as community livelihood projects.

The broader Southeast Asian landscape for debt-for-nature swaps remains largely undeveloped beyond Indonesia's TFCCA experience. The Philippines participated in two earlier TFCCA swaps, but has had no recent activity (US Department of Treasury, 2024[23]). Thailand, Viet Nam and Malaysia, all with substantial coastal ecosystems and marine resources, have shown no engagement with debt-for-nature mechanisms. Their relatively strong credit ratings result in sovereign bonds trading near par, limiting the market-based buyback discount opportunities that characterise Caribbean swaps. Cambodia, Myanmar and Timor-Leste, together with Lao People's Democratic Republic – currently classified by the International Monetary Fund as in debt distress – have also not yet embarked on a DFNS.

4.2.3. There is momentum for monetising blue ecosystem services, but blue carbon schemes are not ubiquitous in the region

Blue carbon mechanisms can help create new revenue streams for marine and coastal protection

Monetising blue carbon – the carbon captured and stored in coastal and marine ecosystems like mangroves, seagrasses and tidal marshes[1] – represents an innovative market-based approach to environmental conservation (Locatelli et al., 2014[24]). These coastal ecosystems are exceptional carbon sinks, with mangroves storing up to four times more carbon per unit area than terrestrial forests while demonstrating remarkable resistance to natural hazards (Locatelli et al., 2014[24]). By assigning economic value to this carbon sequestration service, blue carbon initiatives create financial incentives for protecting and restoring coastal ecosystems that might otherwise be converted to other uses.

The fundamental mechanism is straightforward: scientists quantify the carbon stored in coastal ecosystems using established methods (see Table 4.1 for global estimates), and this carbon is then sold as credits that buyers can use to offset their emissions (Wylie, Sutton-Grier and Moore, 2016[25]). This approach transforms an intangible environmental benefit into a tradable commodity, effectively making coastal forests "worth more standing than cut down" (Salzman et al., 2018[26]). Blue carbon operates as a market-based form of payment for ecosystem services (PES) – a broader framework where landholders receive compensation for managing their land in ways that provide environmental benefits (Salzman et al., 2018[26]; Locatelli et al., 2014[24]). While PES programmes exist for various services like watershed protection and biodiversity conservation, blue carbon specifically targets the climate mitigation value of coastal carbon storage.

Table 4.1. Blue carbon ecosystems: global extent, rates of conversion, estimated carbon dioxide (CO_2) emissions due to human activities, and their estimated costs

Ecosystem	Global Extent (Mha)	Conversion Rate (% /year)	Organic Carbon in Biomass and Top Meter of Sediment (Mg CO_2 /ha)	Carbon Emissions (Pg CO_2 /year	Estimated Cost (Billion USD/ year)
Tidal Marsh	2.2–40 (5.1)	1.0–2.0 (1.5)	237–949 (593)	0.02–0.24 (0.06)	0.64–9.7 (2.6)
Mangrove	13.8–15.2 (14.5)	0.7–3.0 (1.9)	373–1 492 (933)	0.09–0.45 (0.24)	3.6–18.5 (9.8)
Seagrass	17.7–60 (30.0)	0.4–2.6 (2.5)	131–552 (326)	0.05–0.33 (0.15)	1.9–13.7 (6.1)
Total	33.7–115.2 (48.9)			0.15–1.02 (0.45)	6.1–41.9 (18.5)

Source: Ecosystem Marketplace (2024[27]), State of the Blue Carbon Market 2024, https://www.ecosystemmarketplace.com/wp-content/uploads/2024/10/State_of_the_Blue_Carbon_Market_final.pdf.

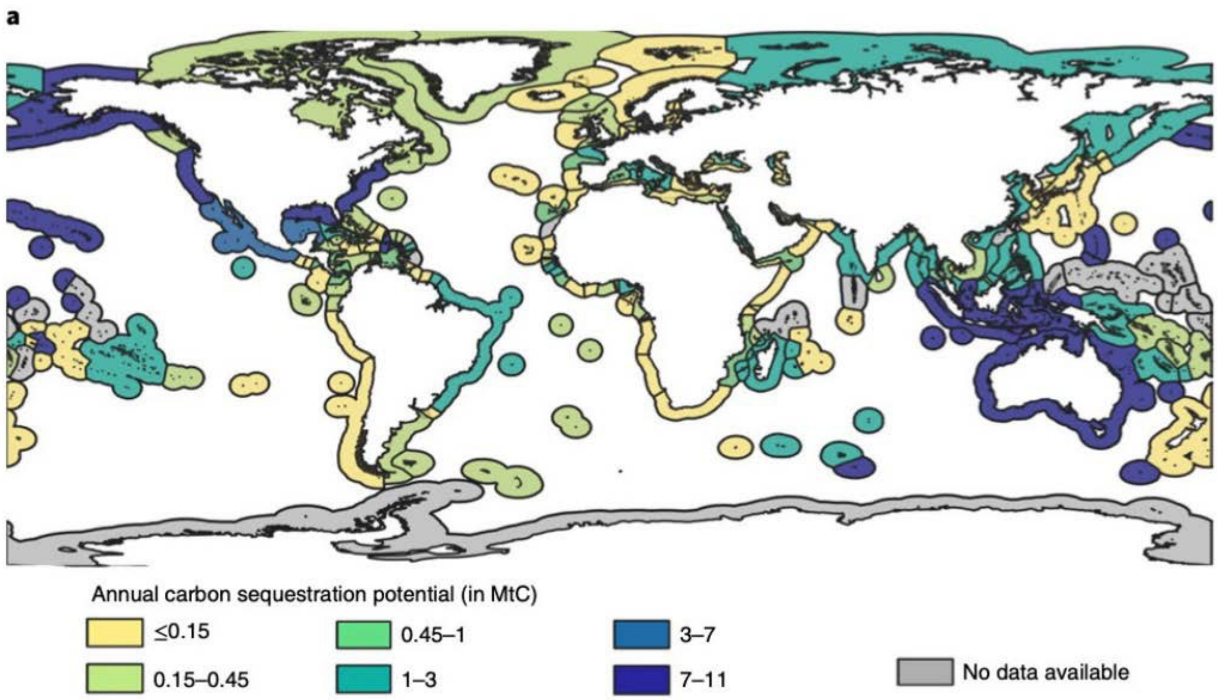
Two primary markets facilitate blue carbon transactions. In compliance markets, governments or companies purchase credits to meet mandatory emission reduction targets under national or regional systems (Wylie, Sutton-Grier and Moore, 2016[25]) and increasingly, mechanisms established under the Paris Agreement. The voluntary market allows organisations to buy credits voluntarily to demonstrate environmental leadership or achieve corporate sustainability goals. For blue carbon projects, the voluntary market has proven more accessible, with credits verified under standards like Verra Verified Carbon Standard or American Carbon Registry (Wylie, Sutton-Grier and Moore, 2016[25]).

Blue carbon mechanisms address capital availability and flow constraints by monetising the carbon sequestration services of coastal ecosystems – mangroves, seagrasses and salt marshes – thereby transforming previously non-monetised ecosystem services into tradable assets that generate revenue. By establishing clear revenue streams tied to measurable environmental outcomes, blue carbon mechanisms can attract diverse capital sources and are a conceivable way to reduce reliance on scarce public funds for conservation.

Southeast Asia shows growing but uneven adoption of blue carbon initiatives

Southeast Asia has become a focal region for blue carbon development, driven by its extensive mangrove coverage and hence blue carbon potential (Figure 4.7), as well as concerning rates of coastal habitat loss (Miller and Taylor, 2023[28]). Yet despite growing policy interest, the translation from concept to implementation remains uneven, with progress varying across countries and projects.

Figure 4.7. Mean annual carbon sequestration potentials



Source: Bertram et al. (2021[29]), The blue carbon wealth of nations, https://doi.org/10.1038/s41558-021-01089-4.

Indonesia leads Southeast Asia in terms of blue carbon projects. Mangrove restoration initiatives across provinces including Aceh, Banten, East Kalimantan and North Sulawesi involve diverse partnerships spanning government agencies, international research institutions, local NGOs and funding organisations (Thomas, 2014[30]). Early projects targeted between 3 800 and 105 000 tonnes of CO_2 equivalent annually over 20-30-year periods, with some pursuing validation under voluntary carbon standards (Thomas, 2014[30]). The country has also established a national carbon trading framework while continuing to develop institutional arrangements needed to incorporate blue carbon into carbon market mechanisms (Paledung, Amahoru and Latupeirissa, 2026[31]).

Viet Nam demonstrates an alternative pathway through its "Markets and Mangroves" project in Ca Mau province. Rather than relying on carbon credit sales, this initiative achieves carbon sequestration goals through sustainable aquaculture certification (Wylie, Sutton-Grier and Moore, 2016[25]). By certifying 1 150 shrimp farmers under international organic standards while requiring 50% mangrove cover, the project prevents deforestation and maintains carbon stocks[2] (Wylie, Sutton-Grier and Moore, 2016[25]). Project developers deliberately bypassed the more complex mechanisms of the United Nations Framework Convention on Climate Change, finding that linking conservation to premium market access for organic shrimp achieved similar environmental outcomes more efficiently (Wylie, Sutton-Grier and Moore, 2016[25]).

This case illustrates how blue carbon objectives can sometimes be met through creative market mechanisms beyond traditional carbon credit sales.

The Philippines has engaged with blue carbon concepts at the policy level, with national discourse evolving "from conversion to conservation to carbon" in recent decades (Song et al., 2021[32]). However, translating policy interest into functional projects has proven challenging, with research highlighting difficulties around governance co-ordination, procedural fairness and equitable benefit distribution (Thompson, Primavera and Friess, 2017[33]).

As elaborated in more detail below, substantial barriers constrain blue carbon implementation across Southeast Asia. While some projects are now operational and generating credits, many remain in planning or validation stages. Key obstacles include the absence of regulatory drivers for carbon payments, contested land tenure on communal and state-owned coastal lands, high upfront development costs, expensive measurement and verification requirements and technical challenges in quantifying soil carbon – which represents the majority of blue carbon stocks (Thomas, 2014[30]; Locatelli et al., 2014[24]). Small-scale community projects particularly struggle to achieve the economies of scale needed to cover transaction costs and compete in global carbon markets (Wylie, Sutton-Grier and Moore, 2016[25]).

4.2.4. Parametric insurance can de-risk blue economy investments but lacks regional precedents

Insurance innovations – such as through parametric triggers – can reduce the risks of investing in the blue economy

Parametric insurance provides automatic payouts based on predefined environmental triggers – wind speed thresholds, sea surface temperature anomalies, rainfall levels or earthquake magnitude (OECD, 2021[34]). When these measurable parameters reach specified thresholds, payouts trigger immediately, regardless of actual damage incurred (OECD, 2021[34]).

This differs fundamentally from traditional indemnity insurance, which requires post-event damage assessments and lengthy claims verification (Broberg, 2019[35]). Parametric insurance eliminates individual loss verification and on-site damage assessment, reducing transaction costs substantially (OECD, 2021[34]). Disbursements occur within days rather than months (Maltby et al., 2022[36]). Transparent trigger mechanisms reduce disputes between insurers and policyholders.

The relevance to blue economy finance is direct. Parametric insurance de-risks investments in climate-vulnerable coastal regions (World Bank, 2025[5]). Blue economy activities – sustainable fisheries, marine conservation, coastal tourism and aquaculture – face mounting financial risk from tropical cyclones, storm surges and extreme weather. Uninsured risks from tropical cyclones, storm surges and extreme weather constrain private capital availability – investors cannot adequately price uncertain losses, leading to capital rationing or market withdrawal from climate-vulnerable coastal regions (Kousky, Treuer and Mach, 2024[37]). Parametric insurance resolves this by using predefined triggers (e.g. wind speed thresholds, wave heights) to provide rapid and transparent payouts. This not only helps safeguard coastal livelihoods, but also makes investments bankable, enabling private capital to flow to ocean-based activities that would otherwise appear too volatile.

While evidence of uptake in Southeast Asia remains limited, parametric models have been implemented elsewhere in the blue economy space

Parametric insurance schemes with blue economy applications remain relatively under-documented in Southeast Asia. However, experiences from the Caribbean and Mesoamerican region provide valuable insights into the design and operation of such mechanisms (see Table 4.2).

The Mesoamerican Reef Insurance Programme demonstrates parametric insurance supporting marine conservation (World Bank, 2025[5]). Spanning Belize, Guatemala, Honduras and Mexico, this mechanism provides immediate financial resources for post-hurricane reef restoration. Wind speed thresholds activate payouts based on tropical cyclone model outputs, using data from independent meteorological agencies. The programme uses a "cat in nested circles" approach – payout amounts increase based on wind speed and proximity to reef sites (World Bank, 2025[5]). The Mesoamerican Reef Fund purchases the parametric policy, with AXA Climate and Munich Re providing coverage and the InsuResilience Solutions Fund covering initial premium costs (World Bank, 2025[5]). Following Hurricane Lisa in 2022, the first activation enabled swift coral reattachment, debris removal and reef stabilisation by pre-trained local NGOs and conservation organisations (World Bank, 2025[5]).

The Caribbean Catastrophe Risk Insurance Facility Segregated Portfolio Company (CCRIF SPC) represents the first multi-country parametric risk pool, formed in 2007 to provide parametric catastrophe insurance to governments across Caribbean and Central American countries. CCRIF SPC covers earthquakes, tropical cyclones and excess rainfall (Charles, 2023[38]). By 2019, CCRIF SPC made over 36 payouts to 13 member governments totalling USD 130.5 million (Charles, 2023[38]). The facility operates at the sovereign level, with governments purchasing policies to close liquidity gaps when responding to natural catastrophes (Peterson, 2020[39]).

Building on CCRIF SPC's institutional foundation, the Caribbean Oceans and Aquaculture Sustainability Facility (COAST) applies parametric insurance to the fisheries sector (World Bank, 2025[5]). Launched in 2019 and developed by CCRIF SPC and the World Bank, COAST provides rapid financial protection to small-scale fishers and value chain actors in Grenada and Saint Lucia (World Bank, 2025[5]). Payouts trigger when modelled annual losses from rainfall and waves exceed thresholds, with CCRIF SPC transferring funds to national governments within 14 days for distribution to pre-registered beneficiaries (World Bank, 2025[5]).

Table 4.2. Parametric insurance programmes in the Caribbean

Programme	Region / Scope	Type / Purpose	Trigger
Mesoamerican Reef Insurance Programme	Belize, Guatemala, Honduras, Mexico	Parametric insurance for reef restoration	Wind speed thresholds based on tropical cyclone models
Caribbean Catastrophe Risk Insurance Facility	Caribbean & Central America	Regional parametric insurance for governments	Modelled hazard intensity (earthquakes, cyclones, rainfall)
Caribbean Ocean and Aquaculture Sustainability Facility	Grenada, Saint Lucia	Parametric insurance for fisheries	Modelled rainfall and wave losses exceeding set thresholds

Source: Authors' based on World Bank (2025[5]), "Accelerating Blue Finance: Instruments, Case Studies, and Pathways to Scale", https://www.worldbank.org/en/topic/environment/publication/accelerating-blue-finance-instruments-case-studies-and-pathways-to-scale.

4.3. Viability of these instruments is not guaranteed – making development support and hence additionality essential

4.3.1. Economic viability remains an underlying constraint

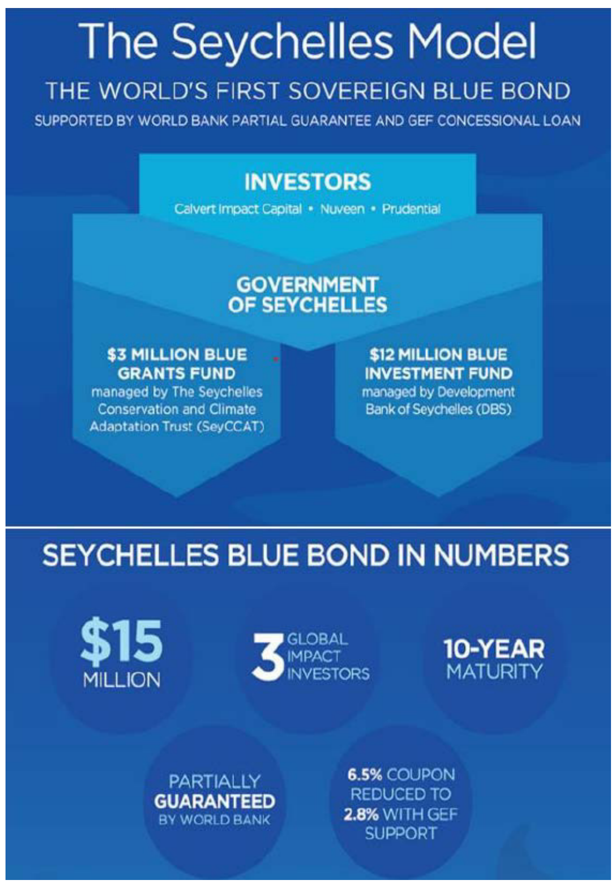
The viability of innovative financial instruments depends on the same factors that constrain capital availability initially – macroeconomic stability and project-level economics. Development support is therefore often needed to overcome these bottlenecks, even for instruments intended to reduce aid dependence.

Blue debt instruments and debt-for-nature swaps still require credit enhancements

Creditworthiness is an important consideration – especially for blue debt instruments. Sovereign and corporate creditworthiness fundamentally shapes investor participation and borrowing costs, meaning that bond issuances in sub-investment-grade ASEAN countries remain out of reach or prohibitively expensive. This limits the instrument's immediate applicability to the region's more creditworthy economies.

A solution to this is the strategic use of development finance. In fact, early blue bond transactions have depended heavily on development finance institutions to provide guarantees to make issuances viable for countries with weaker ratings. For example, the Seychelles' sovereign blue bond (Figure 4.8) was backed by a USD 5 million partial guarantee from the World Bank to enhance the issuance's creditworthiness, alongside a USD 5 million concessional loan from the Global Environment Facility that further reduced net financing costs (World Bank, 2025[5]).

Figure 4.8. The Seychelles sovereign blue bond



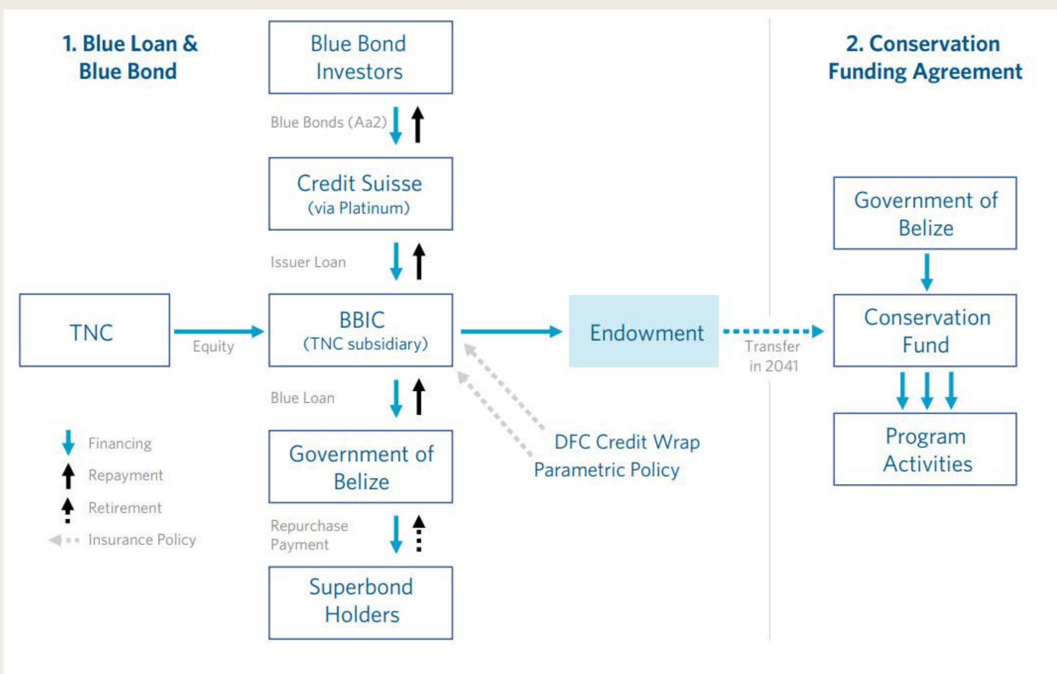
Source: Adapted from World Bank Group (2018[40]), *Innovative Financing for Healthy Oceans*, https://www.worldbank.org/en/news/infographic/2018/10/25/innovative-financing-for-healthy-oceans.

Despite their promise to alleviate fiscal stress, debt-for-nature swaps also depend on credit enhancement from development finance institutions in order to reduce borrowing costs and enable investment-grade ratings for restructured debt. The United States International Development Finance Corporation (DFC) and Inter-American Development Bank (IDB) have provided political risk insurance and partial guarantees reducing interest rates by 200-400 basis points (Albinet, Chekir and Kessler, 2024[17]). Belize combined DFC insurance with Nature Conservancy guarantees to achieve 5% interest versus 9-12% on existing distressed debt (Albinet, Chekir and Kessler, 2024[17]). Ecuador leveraged DFC and IDB enhancement despite political turmoil during negotiations (Albinet, Chekir and Kessler, 2024[17]). Credit enhancement comes with standards: DFC mandates environmental/social requirements, independent verification and debt transparency protocols. These standards mean that credit enhancement can, in addition to helping access institutional investors with minimum rating thresholds, signal credibility and create accountability.

Box 4.1. Belize debt-for-nature-swap

The Belize debt-for-nature initiative combined two integrated components: the debt conversion mechanism and the conservation funding agreement (see Figure 4.9). The debt conversion mechanism involved Belize repurchasing a portion of its outstanding external commercial debt at a discounted rate, thereby reducing its total debt burden and easing long-term fiscal pressure. A special-purpose vehicle provided the upfront funding to cover the discounted debt repayment, and the arrangement included financial safeguards, such as insurance against political and natural disaster risks, as well as reserve accounts to ensure Belize could meet its obligations. This structure enabled the country to convert a portion of its debt service into funding for environmental objectives rather than traditional debt repayment.

Figure 4.9. Belize blue bond transaction structure



Source: The Nature Conservancy (n.d.[41]), Case Study: Blue Bonds for Ocean Conservation, https://www.nature.org/content/dam/tnc/nature/en/documents/TNC-Belize-Debt-Conversion-Case-Study.pdf.

Parametric insurance in climate-vulnerable countries is costly, requiring risk-absorption mechanisms

Parametric insurance premiums can be prohibitively expensive for climate-vulnerable countries and communities. As a result, these schemes often rely on blended structures – combining concessional capital from development institutions with market-based insurance – to improve affordability (World Bank, 2025[5]). The Mesoamerican programme illustrates this approach. The InsuResilience Solutions Fund provided premium financing, effectively subsidising the cost of coverage and making reef insurance accessible to regional environmental trust funds (World Bank, 2025[5]).

Beyond improving affordability, development co-operation can also help overcome broader market barriers by funding early-stage research, supporting risk modelling and providing credit enhancements that reduce perceived risks for private insurers and investors (OECD, 2025[42]). By absorbing part of the upfront costs and uncertainties associated with new insurance products, concessional support can prompt private insurers and capital markets to participate in markets that might otherwise be deemed commercially unviable (World Bank, 2025[5]). As the market matures and risk models improve, the need for concessional support can decline.

The financial viability of blue carbon schemes is not assured – making alternative revenue streams necessary safety nets

Financial viability remains uncertain for many blue carbon projects, particularly those relying solely on carbon revenue. Blue carbon requires willing buyers, yet without regulatory mandates, demand remains uncertain (Thomas, 2014[30]). Project economics vary dramatically by context. In Thailand, carbon payments would need to range from USD 3.14 to USD 156 per tonne CO_2 equivalent to replace income from shrimp farming – illustrating substantial variation in opportunity costs (Thomas, 2014[30]). Blue carbon projects face higher upfront costs and more expensive monitoring than terrestrial forest projects. Evidence from developing economies suggests that projects can struggle to achieve financial viability from carbon credits alone, even with optimistic price projections (Macreadie et al., 2022[43]).

This financial challenge has led some projects to bundle carbon credits with other revenue streams – sustainable fisheries, ecotourism, coastal protection payments or premium agricultural certification (Macreadie et al., 2022[43]). The Viet Nam Markets and Mangroves case demonstrates that achieving blue carbon conservation goals through alternative market mechanisms (like organic certification) may sometimes prove more practical than navigating complex carbon credit systems (Wylie, Sutton-Grier and Moore, 2016[25]). However, quantifying and verifying non-carbon co-benefits like biodiversity enhancement or fisheries productivity remains technically challenging, limiting the scalability of bundled approaches (Macreadie et al., 2022[43]).

4.3.2. System-wide capacity needs constrain implementation

The complexity of innovative instruments creates binding capacity constraints in data management, governance and fund deployment. Addressing these gaps through capacity development determines whether instruments can function as intended.

Data demands are generally high across all instruments

Robust verification and monitoring systems are essential to ensure that proceeds from debt instruments broadly – including those specific to the blue economy – achieve their intended environmental or social outcomes (OECD, 2025[42]). For use-of-proceeds blue bonds, this involves reporting on fund allocation, adherence to eligibility criteria and tracking environmental outputs such as hectares of restored mangroves or reductions in marine plastic pollution. Sustainability-linked bonds and outcome-based bonds require verification of predefined sustainability performance targets or project outcomes, as investor returns are contingent on meeting these benchmarks. Yet such verification is time- and expertise-intensive: outcomes like marine ecosystem recovery may only be observable after several years and demand specialised technical skills. Countries with limited capacity therefore face significant implementation hurdles, making solutions such as third-party verification and standardised monitoring frameworks critical to sustaining investor confidence and market credibility.

For blue carbon projects, establishing accurate baseline blue carbon inventories is a prerequisite. This is a technically intensive exercise. Quantifying carbon stocks in coastal ecosystems requires measuring both above-ground biomass and soil carbon, with soil representing the majority of blue carbon storage (Locatelli et al., 2014[24]). However, measuring soil carbon remains technically complex and expensive, requiring substantial additional scientific work compared to above-ground measurements (Wylie, Sutton-Grier and Moore, 2016[25]; Locatelli et al., 2014[24]). Many projects exclude soil carbon from their accounting due to these measurement difficulties, leaving much of the climate mitigation potential untapped (Wylie, Sutton-Grier and Moore, 2016[25]). Additional complications arise from measuring ecosystem degradation versus complete loss, accounting for climate change impacts like sea-level rise and distinguishing changes in ecosystem health from changes in spatial extent (Locatelli et al., 2014[24]).

Meanwhile, the central challenge in parametric insurance is basis risk[3] – the mismatch between when payouts occur and when actual losses arise (Peterson, 2020[39]). The key is strengthening the index-loss relationship – how closely environmental triggers (like wind speed or rainfall) correlate with actual economic losses (Peterson, 2020[39]). When this relationship is precise, parametric products become more attractive because stakeholders trust that payouts will align with genuine needs. Building this precision requires substantial investment in data infrastructure and risk modelling. The Pacific Catastrophe Risk Assessment and Financing Initiative, a World Bank programme, exemplifies this approach through its development of comprehensive hazard models and risk assessment tools for Pacific Island countries (Peterson, 2020[39]). Experience, particularly from CCRIF implementation, also highlights the importance of country-specific data and local input, with models improving when local actors defined parameters for their own contexts and contributed relevant data (Peterson, 2020[39]).

Sound governance is another prerequisite for the success of blue finance instruments

Prior experiences show that conservation effectiveness in debt-for-nature swaps depends on robust governance, which ensures funds are channelled to their intended purposes over multi-decade timeframes and remain insulated from political interference. A vehicle for administering funds from the country's DFNS, the Seychelles Conservation and Climate Adaptation Trust exemplifies good practice: an independent institution with a legally protected endowment, a multi-stakeholder board, competitive grant processes and transparent reporting subject to external audit.

Marine spatial plans (MSPs) play an important role in realising the conservation benefits of DFNS. Conceptually, MSPs serve several functions in operationalising conservation objectives, such as translating high-level percentage targets into specific geographic commitments that can be monitored or creating concrete implementation roadmaps with specific timelines, enforcement provisions and so on (see Box 4.2). Previous DFNS have included MSPs as a prerequisite or co-developed commitment. Seychelles completed MSP before deploying proceeds, using a participatory process to zone its 1.4 million km² exclusive economic zone, balancing 30% no-take zones with sustainable use for fisheries, tourism and resources. Belize committed to MSP alongside the 30% protection target, recognising that spatial planning is needed to determine which areas to protect and in what sequence, given limited enforcement capacity, while minimising conflicts with fishing communities and tourism operators.

Box 4.2. What is marine spatial planning

Marine spatial planning (MSP) is a comprehensive management approach designed to analyse and allocate the distribution of human activities across marine areas to achieve ecological, economic and social objectives through structured political processes. MSP serves as a critical component of integrated ocean management, working to achieve goals established through political decision-making. This planning framework creates an integrated system that accommodates both existing and emerging marine uses while minimising conflicts between different activities and protecting ecosystem integrity and essential ecosystem services. The approach recognises that ocean spaces face increasing pressure from diverse sectors including fishing, shipping, energy development, tourism and conservation, necessitating careful co-ordination to prevent degradation of marine environments.

By systematically identifying which ocean spaces are most suitable for various uses and activities, MSP balances development demands with marine ecosystem protection while pursuing social and economic goals through transparent and structured processes. The planning process typically involves multiple stakeholders – including government agencies, industry representatives, conservation organisations and local communities – working together to map current and future ocean uses, assess environmental sensitivities and develop spatial plans that reduce user conflicts. This collaborative approach helps ensure that decisions about ocean space allocation consider both immediate economic needs and long-term sustainability, creating a framework for managing increasingly crowded and contested marine environments in ways that maintain healthy ocean ecosystems for future generations.

Source: Frazão Santos et al. (2019[44]), Marine Spatial Planning, https://doi.org/10.1016/B978-0-12-805052-1.00033-4.

For blue carbon projects, secure land tenure is foundational, though challenges vary by context. Because mangroves predominantly exist on communal or state-owned land in developing countries, unclear property rights pose fundamental challenges (Locatelli et al., 2014[24]). Without secure tenure, communities and landholders may lack incentive to invest in long-term conservation, and uncertainty about who legally owns carbon rights – communities, individuals or governments – can complicate project development (Locatelli et al., 2014[24]).

Community engagement and procedural justice also emerge as critical conditions for blue carbon project sustainability. Evidence suggests that successful blue carbon initiatives tend to integrate three dimensions of justice: fair distribution of benefits, transparent decision-making processes and recognition of local knowledge and rights (Locatelli et al., 2014[24]). The Mikoko Pamoja project in Kenya exemplifies this approach. Its success appears to stem from genuine community participation through transparent processes, strong scientific foundation and, crucially, facilitation by trusted individuals who spent 20 years building relationships among community members, researchers and government agencies (Wylie, Sutton-Grier and Moore, 2016[25]).

The readiness to use generated funds from blue finance instruments is another bottleneck

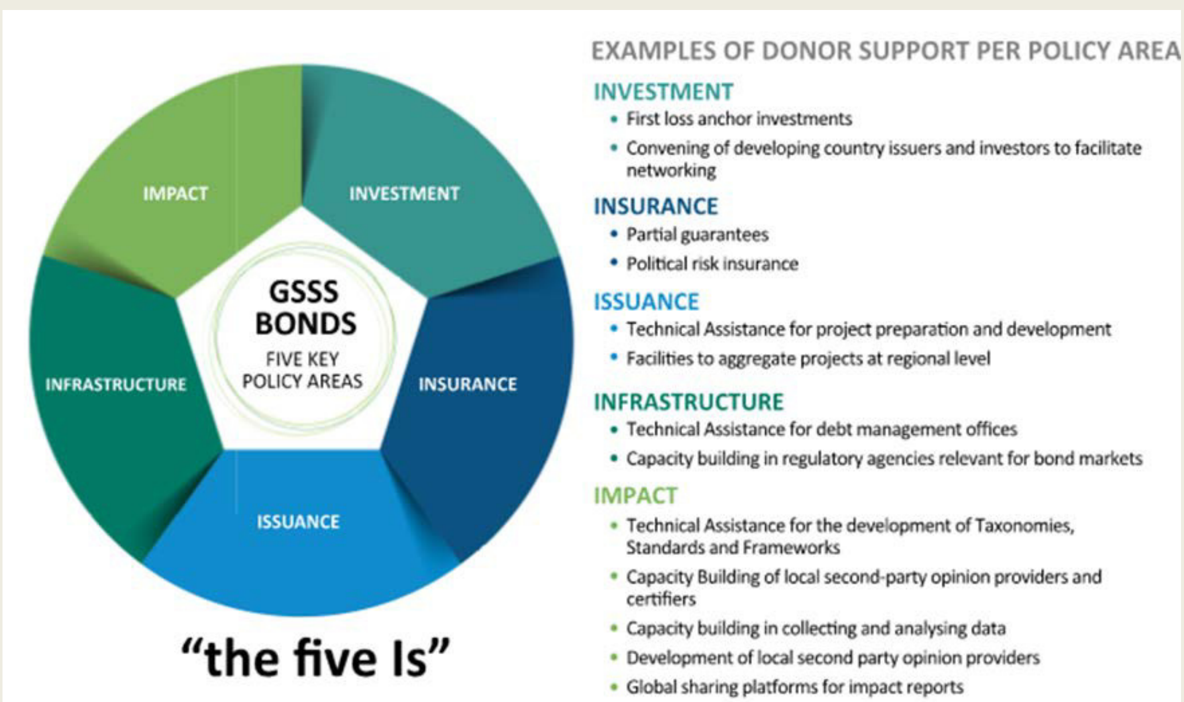
In the case of blue debt instruments, a structured pipeline of bankable projects is necessary to ensure that capital raised is deployed efficiently and produces tangible outcomes. Without a pipeline, funds risk remaining idle. The Asian Development Bank's Blue Southeast Asia Finance Hub, which aimed to develop bankable blue projects worth USD 300 million by 2024 (ADB, n.d.[45]), exemplifies how development finance institutions can address pipeline constraints through technical assistance and project preparation support – catalytic functions particularly important for lower-capacity ASEAN countries (see Box 4.3 for a broader elaboration of different areas of development co-operation support for the issuance of sustainability and sustainability-linked bonds).

Box 4.3. The “Five Is” Framework: Donor support to green, social, sustainability and sustainability-linked (GSSS) bond issuances

Green, social, sustainability and sustainability-linked (GSSS) bonds hold potential for financing sustainable development in developing countries, yet issuances remain limited. Issuers face numerous, often interrelated challenges. They can be technical – relating for example to the lack of capacity needed to structure and issue bonds, to missing pipelines of bankable projects (for use-of-proceeds bonds), or to challenges in choosing appropriate yet feasible targets (in the case of sustainability-linked instruments). They can also relate to the broader enabling environment – for example, a low level of capital market development.

The OECD “Five Is” Framework identifies five major policy areas that donors can target to comprehensively support GSSS bond issuances: investment, insurance, (market)-infrastructure, issuance and impact (Figure 4.10). Although many donors are already supporting issuances, at present they tend to act separately. Donor co-ordination across the five policy areas is therefore at the core of the Framework, and of successful scaling issuances.

Figure 4.10. The “Five Is” framework: Policy areas where donors can support GSSS bond issuances



Source: Adapted from OECD (2023[46]), *Green, social and sustainability bonds in developing countries: the case for increased donor co-ordination*, https://doi.org/10.1787/1cce4551-en.

For parametric insurance schemes, implementation capacity is key: rapid payouts are ineffective if no one is prepared to use the funds effectively. Parametric insurance's speed advantage – disbursing funds within days – only translates into real impact when implementation frameworks are established in advance (World Bank, 2025[5]). The Mesoamerican programme demonstrates this principle. Pre-trained reef response brigades – comprising local NGOs and conservation organisations – were identified, trained and equipped before any hurricane struck (World Bank, 2025[5]). When Hurricane Lisa triggered the insurance in 2022, these brigades could immediately begin coral reattachment, debris removal and reef stabilisation because they knew exactly what to do and had the necessary skills and equipment (World Bank, 2025[5]). Without this advance preparation, funds would have sat unused whilst organisations scrambled to mobilise resources and expertise. Similarly, COAST's success depends on pre-registration of beneficiaries (World Bank, 2025[5]). Fishers and value chain actors register before disasters occur, providing verified bank details and documentation. When parametric triggers activate, CCRIF SPC can transfer funds to governments within 14 days, and governments can immediately disburse to known beneficiaries (World Bank, 2025[5]). This pre-registration eliminates the chaotic post-disaster process of identifying affected parties, verifying claims and establishing payment mechanisms – steps that typically delay traditional insurance payouts by months.

4.3.3. Additionality concerns require careful scrutiny

A critical question, especially if blue finance instruments rely on development assistance, is whether they deliver genuine "additionality". Additionality manifests in two ways.

- **Financial additionality**: mobilising capital unavailable through conventional instruments, often reflected in pricing advantages or expanded investor access
- **Outcome additionality:** delivering outcomes that genuinely exceed what would have occurred absent the blue label

Reliable data on the additionality delivered by blue debt instruments remains limited because proving additionality requires demonstrating a counterfactual – what would have happened without the blue labelled instrument – which is inherently difficult. Evidence from blue bond yields is inconclusive in terms of financial additionality. While comparative analysis between blue and conventional bonds suggests higher yields for blue bonds, the results lack statistical significance (Erixon and Sidstedt, 2024[47]). Meanwhile, research on green bonds, the more mature predecessor, reveals that many labelled instruments simply refinance existing debt or fund already-planned projects: a comprehensive U.S. study found only 2% of green bond proceeds financed genuinely novel environmental projects (Lam and Wurgler, 2024[48]).

Similarly, the case for debt-for-nature swaps rests on demonstrating genuine additionality – that they generate conservation funding and debt relief beyond what countries could achieve independently. This requirement is highly context-dependent. In debt-distressed situations where sovereign bonds trade at significant discounts, the logic is straightforward: debt buybacks create fiscal space that would not otherwise exist. Belize's 2021 transaction exemplifies this – bonds trading below 50 cents on the dollar enabled restructuring USD 550 million (30% of gross domestic product) and achieving 12% debt reduction (Albinet, Chekir and Kessler, 2024[17]). But in non-distressed contexts where bonds trade near par, the additionality question becomes acute. Swaps cannot reduce debt stock; they can only improve debt profiles through lower interest rates or extended maturities. The case of Gabon illustrates this challenge: with bonds near par and maintained investment-grade market access, whether marginal debt profile improvements justified the swap's transaction costs remained unclear (Albinet, Chekir and Kessler, 2024[17]). Even Southeast Asia's relatively nascent experience with debt-for-nature mechanisms raises additionality concerns. Academic analysis of Indonesia's 2009 TFCCA swap questioned whether genuine additionality existed: the USD 30 million redirected represented already-obligated debt service, creating no new fiscal capacity, while evidence remained ambiguous whether conservation grants supplemented or simply substituted domestic budgets (Cassimon, Prowse and Essers, 2011[49]). Put simply, if swaps merely redirect existing financial flows rather than mobilising new resources, transaction costs become deadweight losses relative to direct grants.

Blue carbon mechanisms' credibility[4] also requires demonstrating additionality – that carbon sequestration results from the project rather than business-as-usual coastal management. Experience from terrestrial PES programmes shows that paying for conservation does not automatically target areas that would otherwise be destroyed rather than rewarding ecosystems already secure (Salzman et al., 2018[25]; Alix-Garcia and Wolff, 2014[50]). The importance of blue carbon to climate action – and hence institutional structures like the Paris Agreement – also introduces complications for additionality. Whilst carbon markets nominally create tradable assets from coastal ecosystem services, countries must simultaneously meet their Nationally Determined Contributions (NDCs), which restricts the volume of credits that can be exported internationally. If a country relies on mangrove or seagrass restoration to achieve its own NDC targets, those same carbon reductions cannot credibly be sold to foreign buyers without risking double-counting (Schneider et al., 2019[51]) – the same tonne of CO_2 sequestration cannot count towards both the host country's climate commitments and that of the purchasing entity.

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Notes

[1] In addition to these three, there are other blue carbon ecosystems, such as peatlands and microalgae forests (Ecosystem Marketplace, 2024[27]).

[2] The 50% threshold can be met by either conservation or by planting new forests (Wylie, Sutton-Grier and Moore, 2016[25]). In the case of the latter, there is a time lag for carbon sequestration.

[3] Basis risk manifests in two ways: payouts may trigger when no real damage occurs, or damage may occur without triggering a payout. Minimising this mismatch requires robust data collection and accurate parametric modelling (Peterson, 2020[39]).

[4] The need for such credibility and additionality is reflected in the Core Carbon Principles of the Integrity Council for the Voluntary Carbon Market, which set a global benchmark for high-integrity carbon credits and require that emission reductions or removals be real, additional and permanent (UNDO, 2024[52]).

5 Synthesis of recommendations

The preceding analysis reveals a challenging financing landscape for Southeast Asia's blue economy. Official development assistance faces declining trajectories, among other limitations. Macroeconomic conditions and policy gaps constrain access to alternative capital. While blended finance and innovative instruments offer potential pathways, each requires enabling conditions many countries lack. This suggests that no single capital source can bridge the USD 2.1 trillion financing gap through 2030. Success instead depends on co-ordinated action across three levels: national governments building foundational conditions for blue finance; Association of Southeast Asian Nations (ASEAN) translating regional commitments into co-ordinated finance actions; and development partners maximising the value of scarce concessional resources. This chapter synthesises key findings into actionable recommendations for each actor.

The preceding chapters have documented three interconnected realities that shape Southeast Asia's blue economy financing landscape – each with implications for how the region can mobilise the USD 2.1 trillion needed by 2030.

First, substantial heterogeneity exists across ASEAN member states in macroeconomic conditions, institutional capacity and ocean dependency. This variation means national contexts fundamentally impact what financing strategies are viable. Yet as a regional institution with an explicit mandate to narrow development gaps, ASEAN has a distinctive role – particularly because the Blue Economy Framework has established a regional benchmark that member states have collectively endorsed.

Second, underlying enabling conditions influence whether capital flows to blue economy priorities. The emergence of innovative financing instruments – blue bonds, debt-for-nature swaps, blue carbon markets, parametric insurance – represents meaningful progress. However, the analysis shows these mechanisms depend on macroeconomic stability, policy frameworks and institutional capacity that they do not inherently create. Without addressing these fundamentals, even sophisticated instruments fail to mobilise capital effectively.

Third, development finance remains operationally necessary even as strategic diversification becomes imperative. Declining official development assistance (ODA) budgets and the broader development trajectory both create pressure to reduce reliance on development assistance. Yet the evidence reveals that pathways to financial autonomy frequently require development support as their foundation – whether through credit enhancement, capacity building or de-risking interventions. The challenge is therefore how to deploy ODA more effectively, i.e. directly funding activities that markets systematically undervalue while simultaneously strengthening the conditions that enable broader resource mobilisation.

These findings generate distinct but complementary recommendations for three stakeholder groups: national governments (reflecting country-level heterogeneity), ASEAN as a regional institution (given its co-ordination and harmonisation functions) and development partners (recognising development co-operation's continued role in addressing resource constraints and building enabling conditions).

5.1. National level: Build the foundation for blue economy finance

5.1.1. Develop clear national priorities, institutional set-up and assess financing needs

Effective capital mobilisation begins with sound policy and institutional architecture. For countries still developing blue economy frameworks, this represents the foundational step. Where countries opt for standalone blue economy strategies, integration with existing planning instruments – development plans, nationally determined contributions, sectoral strategies – determines whether priorities translate into budgetary allocations and implementation rather than remaining aspirational.

Once policy frameworks are established, countries need to quantify financing needs across priority areas to enable targeted resource mobilisation. Implementing monitoring mechanisms such as blue budget tagging allows tracking of expenditure against stated commitments. Developing effective financing strategies requires mapping different capital sources to specific needs based on their respective functions and constraints – recognising that domestic public finance, international private capital and domestic private capital serve distinct purposes and face different deployment barriers.

A critical dimension of financing strategy development is assessing whether existing mechanisms can meet identified needs before pursuing new instruments. Given the imperative to demonstrate additionality – that new instruments mobilise resources beyond what alternatives would achieve – established financing pathways often prove more efficient than complex innovations that generate substantial transaction costs without commensurate resource mobilisation. This assessment criterion helps ensure that instrument selection serves strategic objectives rather than innovation for its own sake.

5.1.2. Leverage existing models, guidelines and platforms

Countries need not develop approaches from scratch. Regional experience provides actionable models: Indonesia has established detailed blue economy policy frameworks, blue budgeting processes and operational experience with multiple instruments including outcome-based bonds, bilateral debt-for-nature swaps and blue carbon mechanisms.

International tools offer structured methodologies. The World Bank's Blue Public Expenditure Review provides an analytical framework for assessing current spending patterns and identifying inefficiencies (World Bank, 2021[1]). The United Nations Development Programme's Biodiversity Finance Initiative supports evidence-based finance planning through systematic assessment of expenditures, needs and solutions (UNDP, 2025[2]). For specific instruments, the International Capital Market Association and Asian Development Bank have developed standardised guidance for blue bond issuance that enhances investor confidence while reducing transaction costs (ADB, 2023[3]).

International knowledge exchange platforms complement these technical resources by facilitating peer learning across countries. In addition to ASEAN-specific platforms, there exists a range of initiatives relevant to different ASEAN member states that function as spaces where policymakers can compare approaches, identify good practices and adapt lessons to national contexts. Some examples include the Commonwealth Blue Charter (The Commonwealth, 2026[4]), for Southeast Asian countries who are members of the Commonwealth, the Sustainable Ocean Initiative, for parties of the Convention on Biodiversity (CBD, 2025[5]), and the Coral Triangle Initiative (Coral Triangle Initiative, 2026[6]) – for Coral Triangle countries.

5.1.3. Integrate blue economy into aid co-ordination

For aid-reliant countries in Southeast Asia, development assistance remains necessary even as diversification becomes imperative. Integrating blue economy priorities into aid co-ordination mechanisms is essential. Development finance can fund public goods that markets undervalue, provide technical assistance that strengthens institutions, and supply credit enhancements that make innovative instruments viable. Systematic aid co-ordination enables countries to determine the appropriate support modality for specific needs: direct project funding, sectoral budget support or assistance establishing financing mechanisms such as blue bonds or blended finance structures.

Effective integration requires cross-ministerial engagement to ensure blue economy priorities receive consideration in donor programming alongside competing sectoral demands, rather than operating as isolated initiatives unable to access the development support necessary for building long-term financing capacity.

5.2. ASEAN-level: Translate regional policies to co-ordinated action

5.2.1. Establish a mechanism for knowledge exchange and agenda setting on blue finance

In the short term, ASEAN is encouraged to prioritise knowledge exchange and peer learning between member states. Establishing an ASEAN Blue Finance Working Group as a subset of the ASEAN Co-ordinating Taskforce for the Blue Economy would facilitate regular, focused discussions on the technical aspects of blue economy finance.

Critically, these meetings require cross-ministerial representation – bringing together officials from finance ministries, environment ministries and economic planning agencies – because blue economy finance sits at the intersection of fiscal policy, environmental management and sectoral line ministries. Establishing national-level co-ordinating bodies on blue finance would facilitate this, with focal points from these bodies representing their countries in the regional working group.

Beyond knowledge sharing, the working group would set the agenda for longer-term regional blue finance priorities and co-ordinate blue finance actions. This may, for instance, entail developing common monitoring, reporting and verification frameworks – alongside impact measurement standards – for blue finance instruments.

5.2.2. Mainstream blue economy in regional integration efforts

ASEAN's regional financial architecture offers multiple entry points for integrating blue economy priorities.

The ASEAN Taxonomy Group could explicitly incorporate blue economy criteria. This would standardise definitions across member states and ensure blue investments are recognised within regional sustainability frameworks. Similarly, the Regional Investment Promotion Action Plan, which co-ordinates investment promotion to position ASEAN as a unified destination, could be leveraged to showcase bankable blue economy projects and reduce transaction costs for investors seeking marine and coastal opportunities.

Recent capital market initiatives suggest growing scope for alignment. The ASEAN Capital Markets Forum Action Plan 2026-2030 commits to scaling up financing for clean energy, resilient infrastructure and climate projects through increased adoption of the ASEAN Taxonomy and Transition Finance Guidance (ASEAN Capital Markets Forum, 2025[7]). Whilst not blue economy-specific, these instruments provide technical foundations that could channel capital toward sustainable blue sectors. Corporate accountability mechanisms also present opportunities: planned enhancements to the ASEAN Corporate Governance Scorecard would elevate sustainability performance indicators (ASEAN Capital Markets Forum, 2017[8]), while the ASEAN Simplified Environmental, Social and Governance (ESG) Disclosure Guide (ASEAN Capital Markets Forum, 2025[9]) aims to help small and medium-sized enterprises demonstrate credible ESG performance and access capital markets.

These examples are illustrative rather than exhaustive, but they demonstrate that mainstreaming blue economy finance does not require wholesale institutional redesign. It requires deliberate alignment of existing regional tools with blue economy objectives.

5.2.3. Explore a regional blue finance facility, learning from other regional efforts

In the medium to long term, ASEAN could consider establishing a regional blue finance facility, drawing on global models such as Europe's BlueInvest initiative (European Commission, n.d.[10]) and the emerging Unlocking Blue Pacific Prosperity (Pacific Community, 2024[11]). However, any such facility should leverage existing regional architecture rather than duplicating it. The Asian Development Bank's Healthy Ocean Action Plan, for instance, already operates the Blue Southeast Asia Finance Hub (ADB, n.d.[12]). Hence, engagement and collaboration with the Asian Development Bank (ADB), in a similar vein to the ASEAN Infrastructure Fund (administered with the ADB), would be fitting (ADB, n.d.[13]).

Key design elements for a regional facility could include: 1) ensuring value beyond the national level through resource pooling and leverage; 2) supporting both supply (capital availability) and demand (project preparation and technical assistance); and 3) establishing mechanisms to accommodate different project sizes, from community-based coastal management initiatives to large transboundary infrastructure investments.

5.3. Development partners: Maximise value in a constrained environment

The OECD's guidance on development co-operation for a sustainable ocean economy provides a global framework for the application of development finance in supporting partner countries (OECD, 2025[14]). It emphasises the need for an integrated approach – promoting four goals concurrently (economic development and resilience, equity, climate action and healthy ecosystems). It calls for improving the delivery and use of ODA, leveraging partnerships to unlock capital from multiple sources and directing support to specific mechanisms bearing in mind their use cases and challenges. This guidance applies generally to Southeast Asia.

Within this broader framework, and in the context of Southeast Asia, two priorities emerge as particularly critical given current constraints on development assistance.

5.3.1. Prioritise strategically in a challenging aid environment

With ODA projected to decline significantly in coming years, development partners face difficult prioritisation decisions. For Southeast Asia's blue economy, this means:

- Channelling direct funding where markets systematically undervalue impact: Concessional resources can be concentrated on marine conservation, blue economy policy development, institutional capacity building and other public goods that private capital will typically not finance. These investments can even generate broader enabling environment improvements that unlock capital through enhanced governance, clearer property rights and stronger regulatory frameworks. For instance, support for marine spatial planning or ocean accounting can create the information infrastructure that catalyses subsequent private investment.
- Using guarantees and blended finance strategically and with development impact in mind: Available guarantees and risk-sharing instruments can be deployed strategically to enable countries to access other sources of capital, particularly for commercially viable blue economy projects that face excessive risk premiums due to weak enabling environments. The European Fund for Sustainable Development Plus is one example of this approach. However, the core principles of effective blended finance – prioritising development impact and public value, focusing on mobilising commercial finance, tailoring efforts to the local context, strengthening partnerships and monitoring for transparency and results – all apply.
- Supporting the establishment of blue finance instruments when aligned with country priorities and demonstrating clear additionality: Blue finance instruments can form part of the broader financing solution for Southeast Asia's blue economy. While they may represent a pathway towards financial autonomy, they often remain contingent on development partner support. Channelling development assistance towards these instruments may be justified in certain contexts, but this requires clear alignment with recipient countries' priorities and needs, as well as demonstrable additionality beyond what traditional instruments can achieve.

5.3.2. Seek co-benefits across blue economy and broader sustainable development goals

Effectiveness in a resource-constrained environment requires strategic integration at project and policy levels. Development partners can actively pursue co-benefits across ocean health, climate, biodiversity and development objectives. While these are often treated as separate issue areas with distinct financing streams and institutional mandates, they are fundamentally interconnected. Climate finance can support blue carbon ecosystems like mangroves and seagrass beds that simultaneously sequester carbon, protect coastlines from storms and provide nursery habitat for commercially important fish species. Biodiversity finance can strengthen coastal resilience and support livelihoods of fishing communities. Blue economy investments can advance multiple Sustainable Development Goals (SDG) simultaneously – SDG 14 on oceans, SDG 13 on climate, SDG 1 on poverty and SDG 8 on economic growth.

This integrated approach maximises the impact of scarce resources by designing interventions that generate multiple benefits rather than optimising for single objectives.

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Financing Southeast Asia’s Blue Economy

Development Assistance and Beyond

Southeast Asia’s development trajectory is closely intertwined with its blue economy – a diverse range of economic activities reliant on and non-market ecosystem services provided by the region’s rich coastal, marine, and freshwater resources. However, as highlighted in the ASEAN Blue Economy Framework, the overarching policy guiding Southeast Asia’s blue economy priorities, adequate and effective finance and investment are essential for the region to realise the potential of the blue economy for sustainable development. A key consideration is the need to move beyond excessive reliance on development assistance towards longer-term financing solutions, particularly in light of increasing pressures on international public finance. To support this shift and the implementation of the ASEAN Blue Economy Framework, this report assesses the current landscape of development assistance for the blue economy in Southeast Asia and examines the range of financial sources and instruments that can be leveraged to advance sustainable blue economy financing in the region.

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